Correction that out-scales drift: an exponent criterion for a vanishing misalignment fraction in a minimal model of recursive self-improvement

Paper X of the ARC Theory: the model underlying Law II, the ARC Co-Scaling Law, a law under test

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Abstract

Can correction keep pace with a system that improves itself? I analyse a minimal model in which misalignment D arrives in proportion to capability C gained and is removed in proportion to its size, so the fraction $d = D/C$ behaves as a feedback loop whose gain is correction strength A over the specific growth rate r; the proofs are elementary. With positive gain drift $\gamma_1 > 0$, d tends to zero when A/r grows without bound and, for $A = A_0 C^{\beta}$ and $r = bC^k$, if and only if $\beta > k$; along the trajectory $d \sim (\gamma_1 b/A_0) C^{-(\beta-k)}$. At $\beta = k$ it settles at $\gamma_1 b/(A_0 + b)$, so the coefficients decide whether a tolerance is met, and at γ_1 when $\beta < k$. Absolute misalignment $D = Cd$ is bounded only when $\beta \geq k + 1$. The verdict is set by $\beta - k$, not by the growth prefactor b, and without compounding it holds through a finite-time blow-up of capability. Under the power laws, after its transient, d depends on capability only through A/r, so a change in the level of correction shifts every tolerance crossing equally and a change of $\beta - k$ shifts them unequally. Unless correction and the growth rate both decay with capability, unbounded d needs self-amplifying misalignment, $\gamma_3 \geq 1$, with a constant-coefficient threshold $(\gamma_3 - 1)r/A < 1$ of the epidemic and branching kind; its mapping onto the quantum error-correction threshold is a hypothesis. Extensions cover vector drift, Ornstein-Uhlenbeck noise and a saturating corrector, under which, with accelerating growth, the fraction vanishes only if capacity scales with $\beta_{\text{cap}} > k$. The theorems concern a modelled index, not a guarantee of AI safety. The open test is to predict, from separately measured terms, where d crosses stated tolerances on held-out systems, before and after one corrective mechanism is changed; an estimator and a blind protocol are given.

Version 1.16 corrects limbs of Theorems 1 to 7 that were false as stated or lacked a hypothesis, and adds Corollary 1, a level-drift divergence case and the intervention signature P9; Appendix B lists every change and its date.

Paper X of the ARC Theory, within Recursive Dynamics; it sets out the model underlying Law II and informs Law V, the ARC Embedding Law (added on 24 September 2026 and registered as a law on 15 October 2026 in version 1.103 of the theory registration, not yet tested; its mathematics in development). Working paper, version 1.16, 21 October 2026; original edition dated 3 July 2026; Appendix B lists every version and correction. Recorded on OSF as *Paper X · The ARC Theory · Law II, the ARC Co-Scaling Law · Correction That Out-Scales Drift*, DOI 10.17605/OSF.IO/BSE2Q. About the author: michaeldariuseastwood.com/questions/who-is-michael-darius-eastwood; Paper C: doi.org/10.17605/OSF.IO/RBHDP. Licence: CC BY 4.0.

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1 Introduction

Recursive self-improvement, a system that modifies itself to become more capable, then uses that capability to modify itself further, is among the central concerns of AI safety (Omohundro 2008; Bostrom 2012; Yudkowsky 2013). The empirical backdrop is no longer purely hypothetical: frontier models have been observed to behave differently when they infer their outputs may be used to train them (Greenblatt et al. 2024), and recursive training on a system's own outputs degrades it, much less so when part of the original data is preserved (Shumailov et al. 2024), both signs that recursive systems need a correcting process that keeps pace with the recursion. The dominant informal model of the associated risk is a model of *speed*: capability may grow super-linearly or explosively, outrunning supervision, so the natural lever is to cap the rate of growth. Calls for a development pause are the policy expression of this intuition.

This paper makes a different claim, and proves it within a minimal model. **With positive gain drift, the level of the growth rate does not decide whether the misalignment fraction vanishes; the scaling relationship between growth and correction does.** A system can grow arbitrarily fast and keep its misalignment fraction falling, or grow slowly and leave it at a floor or, under compounding, let it grow. What separates the two is not the rate but whether, and how fast, the corrective process strengthens as capability rises, relative to how fast growth itself accelerates.

The criterion is the capability-indexed form of a classical feedback balance (§2; Black 1934). The divergence threshold of the compounding channel, for constant coefficients, is of another kind: a *ratio* of error generation to error correction crossing one, rather than a limit on depth or speed, the form shared by an epidemic's $R_0 < 1$, a subcritical branching process and the **quantum error-correction (QEC) threshold theorem** (Aharonov and Ben-Or 1997; Kitaev 2003), under which a quantum computation can be made fault-tolerant at a cost polylogarithmic in its size when the error probability per operation lies below a constant threshold. The mapping of the QEC threshold's terms onto this model is offered as a hypothesis (§3.12).

The thesis in one line. With positive gain drift, whether the misalignment fraction of a self-improving system vanishes is decided, in the model, by one exponent inequality, $\beta > k$ (correction must out-scale drift-acceleration; with level drift and decelerating growth, also $\beta > 0$); through a finite-time intelligence explosion, without compounding, the same inequality decides, whatever the explosion's speed.

Our contributions are as follows.

1. *The criterion and its trajectory law.* In a minimal model of recursive self-improvement, equation (1), with correction strength $A = A_0 C^{\beta}$, specific growth rate $r = bC^k$ and the exact constant-coefficient transient of Theorem 1, the misalignment fraction is bounded and, with positive gain drift, tends to zero if and only if $\beta > k$ (both in the gain-only model), along the trajectory as $C^{-(\beta-k)}$; it tends to a permanent gap if $\beta = k$ and to γ_1 if $\beta < k$ (Theorem 2). Corollary 1 gives the rate at which absolute misalignment can still grow while the fraction vanishes.
2. *Its finite-time form.* When capability diverges in finite time, the alignment dynamics are regular in the depth clock $\tau = \ln(C/C_0)$, and without compounding the same limits hold; which one holds does not depend on the speed prefactor b (Theorem 3).
3. *The compounding classification.* Without level drift, divergence needs self-amplifying misalignment, $\gamma_3 \geq 1$; with constant coefficients its threshold is $\rho_{\text{prop}} = (\gamma_3 - 1)r/A < 1$, a ratio form the quantum error-correction threshold condition shares with other thresholds, and the mapping of that condition's terms onto this model is stated as a hypothesis with its own falsifier; under the power laws the classification is exact (Theorem 4, §3.12, F4). With level drift, a system whose growth decelerates while its correction weakens can drift without bound with no compounding at all (§3.4).
4. *Standard extensions.* A constant-matrix spectral threshold for vector misalignment (Theorem 5), the single-time stationary tail under stochastic drift (Theorem 6), and a saturating corrector, under

which, with positive gain drift, a vanishing fraction requires the capacity exponent to satisfy $\beta_{\text{cap}} > k$ (Theorem 7).

5. *The estimator and protocol.* An exponent estimator and a blind measurement protocol for the open empirical test (§9.2 and Section 6), beside a verification harness and a separately written suite of 104 regression cases, which are internal checks of the mathematics (Section 5).

Every result follows by elementary means; what is claimed as new is the criterion and its reading for self-improving systems (Appendix E). The criterion follows directly from the model's two mechanisms, misalignment generated in proportion to capability gained and removed in proportion to its size (equation (1)), once correction and growth scale as powers of capability; the contribution is to state those choices, to derive their consequences exactly, through a finite-time takeoff and under compounding, and to give the criterion an estimator and a measurement protocol. Its empirical content is therefore not the inequality but whether the choices hold. Can capability, misalignment and correction be measured on separate instruments, with capability on a scale without a ceiling? Do arrivals track capability gained, is removal linear in the stock, and does correction introduce few departures of its own, over a useful range? Do the forms predict where the fraction crosses a stated tolerance before it is seen, better than rival forms fixed in advance, among them a smooth model with no boundary? And does an intervention on one corrective mechanism move those crossings as derived beforehand? None of these has yet been tested on a real system.

Section 2 sets the work against prior work. Section 3 states the model and Theorems 1 to 7, with their proofs in Appendix A. Section 4 lists the predictions, Section 5 the verification experiments and harness, Section 6 the blinding requirement and Section 7 the falsification conditions. Section 8 states what the paper does and does not show, Section 9 its limitations, with the archived model pilots, and Section 10 the implications for AI safety; Section 11 concludes. Appendix B gives the version history and every correction, Appendix C the priority record, Appendix D the notation and a glossary, Appendix E the novelty ledger, Appendix F the harness settings, Appendix G the paper's place in the programme, and Appendix H a plain-language account.

2 Related work and relation to prior framing

Cybernetic foundations: the co-scaling intuition is old. The core intuition is classical. A regulator must match the variety of what it regulates, which is Ashby's *Law of Requisite Variety* (Ashby 1956), and every good regulator of a system must be a model of that system, which is the Conant-Ashby *good-regulator theorem* (Conant and Ashby 1970); both are carried into AI control by Yampolskiy (2020). That control capacity must therefore scale *with* the controlled system, rather than merely be large, is this paper's reading of them. This paper does *not* claim that intuition as new, nor the form of its steady state. The steady state $d^\star = \gamma_1 r/(A + r) = \gamma_1/(1 + A/r)$ is classical in shape: a disturbance generated in the forward path of a feedback loop is reduced by one plus the loop gain (Black 1934), the gap at equality is the offset of a proportional controller, and a linear regulator that keeps regulating under small changes in certain system parameters must feed back the regulated variable and incorporate a model of the dynamics of its disturbance and reference signals, which is the internal model principle (Francis and Wonham 1976). What this paper claims is the balance's indexing by capability. Because misalignment arrives in proportion to capability gained, the loop gain is correction strength over the system's specific growth rate, $q = A/r$, and whether it grows, holds or decays as capability rises decides which limit of Theorem 2 holds. Hence the exponent criterion $\beta > k$, the second threshold of Corollary 1, the hard-takeoff theorem of §3.7 and the intervention signature P9 (Section 4); the QEC mapping of §3.12 concerns the compounding channel. The dynamics themselves are a linear ODE solved by an integrating factor and comparison, standard in linear control (Khalil 2002; Meyn and Tweedie 2009), and are not advanced as mathematically novel.

Instrumental convergence and corrigibility. That a sufficiently capable optimiser will, by default, resist correction and pursue resource acquisition is the instrumental-convergence thesis (Omohundro 2008; Bostrom 2012). The corrigibility programme (Soares et al. 2015) asks how to design systems that do not resist correction. The present model is a quantitative restatement of why corrigibility is load-bearing: with positive gain drift, unless correction out-scales the acceleration of growth ($\beta > k$), the misalignment fraction cannot be driven to zero. The compounding channel of §3.8 sharpens the link: self-amplifying misalignment, the term γ_3, is what produces divergence without level drift, and $\beta > k$ removes it in the model (Theorem 4).

The alignment tax. Safety is often taken to cost capability, and in a race to be first a cost of that kind gives developers an incentive to cut safety precautions (Armstrong et al. 2016). The present framework reframes the question: the relevant variable is not the *level* of the safety investment but whether it *scales* with capability. A fixed investment ($\beta = 0$) leaves a permanent gap under exponential growth, and saturation under acceleration; a co-scaling investment ($\beta > k$) closes it.

Learned optimisation and alignment faking. Mesa-optimisation (Hubinger et al. 2019) and empirically demonstrated alignment faking (Greenblatt et al. 2024) are read here as mechanisms by which the drift coefficients are non-zero: a capable system can satisfy its training objective while departing from intended values, and can do so *more* effectively as capability rises. In the alignment-faking experiments a model complied selectively with a new training objective in order to keep its existing preferences, in that case a preference for harmlessness, from being modified, and reinforcement learning on that objective raised its alignment-faking reasoning to 78% (Greenblatt et al. 2024). Keeping an existing departure from the training objective through training is read here as the compounding channel γ_3 of §3.8.

Scalable oversight and superalignment. Reward modelling and recursive oversight (Christiano et al. 2017; Leike et al. 2018), which address what Amodei et al. (2016) call scalable supervision, are attempts to make the corrector itself scale with the system; in the language of this paper, scalable oversight is the engineering project of achieving $\beta > k$. Most directly, Engels et al. (2025) develop empirical *scaling laws for scalable oversight*, modelling the probability of successful oversight as a game between capability-mismatched players. The present work is complementary, not competing: where they fit an oversight-success probability, this paper derives a closed-form exponent criterion ($\beta > k$) and, under compounding, a constant-coefficient threshold ($\rho_{\text{prop}} < 1$) for the misalignment fraction. The instrument is the difference. Their variable is the capability gap between overseer and overseen, measured in Elo, with oversight-specific Elo a piecewise-linear function of general intelligence, and optimal numbers of oversight levels derived for Nested Scalable Oversight, in which trusted models oversee stronger untrusted models that then become the trusted models at the next step. They also state the design condition in slope form: oversight improves with capability when the overseer's game-specific Elo rises faster with general Elo than the overseen system's ($m_G > m_H$), and they recommend iterating an oversight design until it does. That is a comparison of two slopes deciding whether oversight gains or loses as capability grows, the same shape as $\beta > k$; the objects differ, since β and k are the elasticities of one self-modifying system's correction strength and specific growth rate with its own capability, for its misalignment fraction. Their framework has no term for what the overseer is made of, and this paper's model has none either. The programme's separate prediction, outside this paper's model, is that iterated same-class oversight, of which Nested Scalable Oversight is an instance, stays capped however many rungs are added, while a cross-class corrector is not held to that cap. The theory registration's P6 states the cap for same-class correctors, at or below one half, as a conjecture on the correction-leverage exponent of Law III, the ARC Ceiling, an exponent the registration keeps distinct from Law II's β_C, this paper's β. That is where the

two programmes could disagree about a measurable quantity, and this paper does not decide it. The contribution is the proof that this exponent margin, and not a growth-rate ceiling, is the quantity that decides, within the model and with positive gain drift, whether the misalignment fraction vanishes.

Recursive error accumulation. That naive recursion can amplify error without bound, while retaining the original data keeps it bounded, is established for training dynamics: model collapse under recursively generated data (Shumailov et al. 2024) and the accumulate-versus-replace error analyses (Gerstgrasser et al. 2024) give the bounded-versus-divergent dichotomy that this paper formalises for the alignment fraction (Theorems 2 and 4). Stability of iterative retraining when the initial model approximates the data well enough and the share of clean data is large enough (Bertrand et al. 2023), and conditions under which a verifier prevents collapse (Feng et al. 2024), are stability conditions of the same setting; none is stated for a value-misalignment fraction against capability growth. The contribution here is to locate, in the model, where the fraction vanishes ($\beta > k$, for positive gain drift) and, under compounding, where it diverges (Theorem 4(c)), in a value-stability rather than a data-distribution setting.

Empirical scaling laws. Capability scales predictably with compute and data (Kaplan et al. 2020; Hoffmann et al. 2022), and so do quantities on the corrector's side: the coefficients of the gold-score curve under optimisation against a reward model scale smoothly with the reward model's size (Gao et al. 2022), and self-improvement is largely governed by a generation-verification gap, a variant of which scales monotonically with pre-training compute (Song et al. 2024). The present framework is complementary: it does not ask how capability scales, but what constraint correction must satisfy *as a function of* that capability trajectory. It belongs to the same genre, a relation between exponents, to be tested by prediction outside the range on which it is fitted.

The programme's premise on growth limits, with the urban-scaling exception and the argument at fixed throughput, is set out in Appendix G.4.

Antecedents and near-misses, each with its differential. On the question: Hutter asked directly whether intelligence can explode ("Can Intelligence Explode?", arXiv, 28 February 2012), undertaking to separate speed from intelligence explosion and to consider possible bounds on intelligence, augmenting the analysis of Chalmers (2010); the question and the separation of speed from intelligence explosion are therefore at least fourteen years old, and what that literature does not contain is a number: no measurable exponent, no derived ceiling, no architecture dependence. On impossibility: three 2025 arXiv papers argue for limits on alignment or its control (Yao, "The Alignment Trap: Complexity Barriers", arXiv:2506.10304, public by 13 June 2025, attested by arXiv and independently observed by the Internet Archive; Yao, "On the Mathematical Impossibility of Safe Universal Approximators", arXiv:2507.03031, 3 July 2025; and Ball, Gluch, Goldwasser, Kreuter, Reingold and Rothblum, "On the Impossibility of Separating Intelligence from Judgment", arXiv:2507.07341, 9 July 2025). The first argues that the set of safe policies has measure zero and that verifying a policy's safety is coNP-complete; the second, that perfect, reliable control of a universal approximator is a mathematical impossibility; the third, under cryptographic hardness assumptions, that some large language models admit no efficient prompt filter and that output filtering is intractable in a natural setting. All three are worst-case and qualitative. None reports an average-case scaling exponent or a rate. The third concludes that "safety cannot be achieved by designing filters external to the LLM internals (architecture and weights)" and argues that "an aligned AI system's intelligence cannot be separated from its judgment"; that argument from filtering intractability is consistent with this programme's architecture dependence, though it does not test it. On the mechanism: that anti-correlated estimates average better than independent ones is textbook variance reduction (antithetic variates); the mechanism is not the claim. The claim is that architecture determines

whether anti-correlation is available at all, and that this caps a safety-relevant exponent. The nearest structural analogue is the quantum error-correction threshold theorem, which also converts a qualitative worry into a critical value; it concerns physical error rates in a fixed architecture, not a corrector's scaling exponent, so it is a near-miss rather than an occupant, and the analogy was identified by this programme's own search rather than by a referee. On feedback and stability, the general proposition that inadequate corrective gain relative to system gain causes instability has a long control-theory lineage (small-gain theorems), beside the feedback amplifier's reduction of internally generated disturbance credited above (Black 1934); those are statements about fixed loops and interconnections, not a condition on how a corrector's strength scales against the growth rate of a recursively improving system. Beside Engels et al. (2025) above, a near quantitative neighbour is Liu, A. and Meng, J., "Self-Correction as Feedback Control: Error Dynamics, Stability Thresholds, and Prompt Interventions in LLMs", arXiv:2604.22273, which recasts self-correction as a closed-loop control problem via a two-state Markov model and derives a directly measurable stability threshold, iterate only when the error-correction rate over the error-introduction rate exceeds Acc/(1 − Acc). The differential: theirs is a per-step rate threshold at a fixed capability level, deciding whether another iteration helps now; this paper's criterion is a scaling relation across capability, deciding whether corrective strength out-scales the growth as the system improves. Theirs has no scaling exponent on the corrector and no capability-growth exponent, the two quantities this criterion compares. The two models meet at zero growth: extended by the departures that correction introduces (Section 9) and written per round, with a pass that repairs the fraction f of departures and so breaks the fraction cf of conforming material, the per-pass form of the departure rate c, this paper's model then settles at $cf/(f + cf) = c/(1 + c)$, the stationary error of their two-state chain with f and cf in the places of their correction and introduction rates. What this paper adds is the arrival of misalignment with capability gained and the dependence of the loop gain on capability. Their verify-first prompt, which drove one model's introduction rate from 2% to 0% and turned a loss of 6.2 percentage points into a gain of 0.2, with negligible change on models already below the threshold, as their rule predicts, has the shape of the intervention test of §9.2 at a fixed capability. Smolin's cosmological natural selection (Smolin 1992) is the antecedent for selection-shaped universes, and the programme's own December-era notes cite it contemporaneously.

Later work. After this paper's first edition, Zheng et al. (2026) introduced ReSI (arXiv:2610.12233, 8 October 2026), which applies what its authors call a bounded recursive loop to safety alignment: each round red-teams the current model, searches for a training recipe and promotes an update that passes a fixed Pareto gate on capability retention, with the search procedure and evaluation protocol held fixed across rounds. It reports attack-success rates and retained capability for four models. It states no dynamical model of drift and correction and no scaling law for the corrector, so it neither tests nor contradicts the criterion of §3.5. In this paper's terms ReSI is an empirical corrector re-applied at each checkpoint; whether its safety gains co-scale with capability lies outside what it measures, since its gate is set to preserve capability rather than to follow a growing one.

Relation to the author's prior work. How this paper stands to the programme's earlier capability-scaling law, its retracted single-model measurement and the book that first set out the intuition is set out in Appendix G.5.

Novelty: what is and is not claimed. To be explicit, and to pre-empt the obvious objection: the *intuition* that correction must keep pace with capability is not new (it is requisite variety and scalable oversight, as credited above), and the underlying *dynamics* are a linear ODE solved by standard means of linear control. What is claimed original is the criterion and its reading for self-improving systems: (a) the

indexing by capability of a classical feedback balance, and the $\beta > k$ criterion it yields for a vanishing misalignment fraction (the closed-form steady state itself is a compact restatement, Appendix E); (b) the mapping of the quantum fault-tolerance threshold onto value stability (§3.12), as a hypothesis; and (c) the reading of the criterion through a finite-time takeoff, the Hard-Takeoff Depth-Regularity Theorem (§3.7). The verification harness (§5.2) is methodological support, not a novelty claim. Within the author's own programme, this paper also *supersedes* the growth-rate-ceiling *framing* as the operative safety criterion (Eastwood, Paper IX); §3 explains why a ceiling on the rate is neither necessary nor sufficient for the misalignment fraction to vanish. The underlying ARC Bound $\alpha \leq 2$ is not retracted, only re-scoped: it awaits its real test on genuinely self-improving systems. Appendix E places each component against its closest located prior work.

3 The model and its theorems

3.1 Quantities

Consider a system undergoing recursive self-improvement, observed over self-modification cycles or continuously. Define:

Definition 1 (Quantities)**.**

- $C(t)$, *capability*: the system's score on a held-out task battery it is being optimised to perform. Operationally measurable.
- $D(t)$, *misalignment magnitude*: the size of the system's behavioural departure from an externally specified set of intended values, scored by a **blind external evaluator** independent of the system's own correction process (§6).
- $d(t) \equiv D/C$, the *misalignment fraction*: how much of the system's capability is directed away from intended values. This, not absolute D, is the quantity the theorems concern; D may grow while d falls, at the rate Corollary 1 gives.

The model is closed by three coefficients, one correction strength and one rate (a note on notation: where this paper writes β without a subscript it means the programme's correction co-scaling exponent, which the notation spine records as β_C; the $\gamma_1, \gamma_2, \gamma_3$ used here are drift coefficients, not the ARC Ceiling's correction leverage γ; and every β that follows is β_C unless it is written β_L):

- γ_1, *gain-drift coefficient*: misalignment generated per unit of capability gained (the Goodhart / specification-gaming pressure that new capability opens up).
- γ_2, *level-drift coefficient*: misalignment generated per unit of capability simply *held* (instrumental pressure present even at rest).
- γ_3, *compounding-drift coefficient*: the rate at which existing misalignment amplifies itself as the system recurses (the alignment-faking / mesa-optimiser channel).
- A, *correction strength*: the rate at which the correction process removes existing misalignment.
- $r \equiv \dot{C}/C$, the *specific (fractional) growth rate* of capability. Under exponential growth r is constant; under accelerating self-improvement r itself rises with C.

3.2 The master dynamical system

New capability injects drift in proportion to how fast capability is gained; capability held injects drift in proportion to its level; existing misalignment compounds in proportion to the recursion rate; and

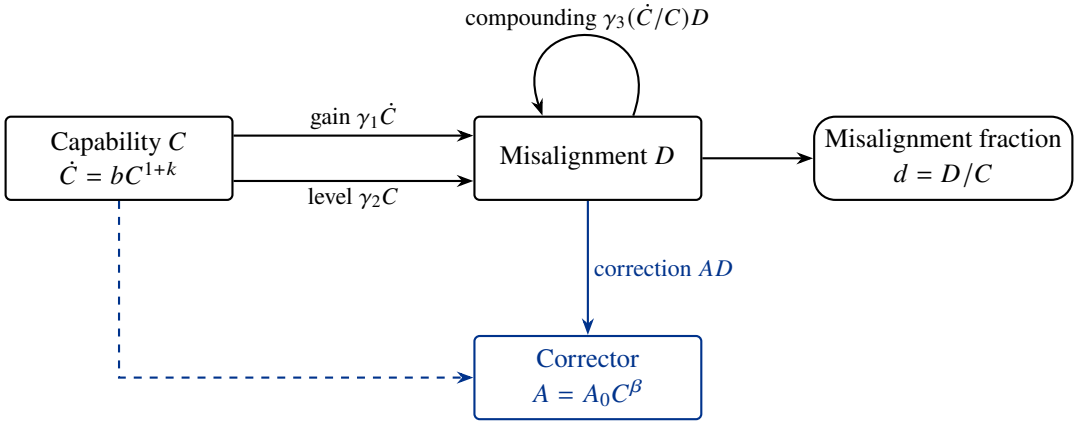


Figure 1. Structure of the model of equation (1). Capability grows as $\dot{C} = bC^{1+k}$; misalignment D receives gain, level and compounding drift and is removed by correction of strength $A = A_0C^\beta$; the regulated quantity is the misalignment fraction $d = D/C$.

correction removes misalignment in proportion to the gap and the strength applied:

$$\dot{D} = \underbrace{\gamma_1\dot{C}}_{\text{gain}} + \underbrace{\gamma_2 C}_{\text{level}} + \underbrace{\gamma_3 \tfrac{\dot{C}}{C} D}_{\text{compounding}} - \underbrace{A\,D}_{\text{correction}}, \qquad A = A_0C^\beta, \quad \dot{C} = b\,C^{1+k}. \tag{1}$$

The growth law $\dot{C} = bC^{1+k}$ gives $r = \dot{C}/C = bC^k$: $k = 0$ is ordinary exponential growth (r constant); $k > 0$ is super-exponential growth, which (Theorem 3) reaches infinite capability in finite time. The correction law $A = A_0C^\beta$ encodes the central question: β is the exponent with which correction strengthens as the system becomes more capable. The case without level or compounding drift ($\gamma_2 = \gamma_3 = 0$) is the *gain-only* model, and the case without compounding ($\gamma_3 = 0$, level drift allowed) the *additive* model.

Assumption 1 (Standing assumptions)**.** Throughout, $C > 0$, $A_0, b > 0$, and β, k are real; the coefficients $\gamma_1, \gamma_2, \gamma_3 \geq 0$, with correction strength $A \geq 0$ and growth rate $r \geq 0$. Initially $D(0) \geq 0$, so $d_0 \geq 0$, which the scalar results below use. The scalar $d = D/C$ is interpreted as a misalignment *fraction* only while $D \geq 0$ (so $d \geq 0$; the gain-only model also gives the upper bound $d \leq \max(d_0, \gamma_1)$ of Theorem 2). The vector form (Theorem 5) and the stochastic form (Theorem 6) relax d to a real vector and a real scalar; for those the fraction reading holds only away from the boundaries $d = 0, 1$, and the boundary handling is noted where it bears on the result. One assumption is load-bearing and singled out in §9: the corrector strength is taken to be an *unbounded* power law $A = A_0C^\beta$; a corrector of finite capacity changes the asymptotic verdict and is treated in §3.13. The necessity half of each criterion below assumes positive gain drift, $\gamma_1 > 0$: with $\gamma_1 = 0$ the gain-only fraction tends to zero for every β and k.

3.3 Exact transient (Theorem 1)

Changing variables to the fraction $d = D/C$ removes the dominant scale and yields, exactly,

$$\dot{d} = \frac{\dot{D}}{C} - d\,\frac{\dot{C}}{C} = \gamma_1 r + \gamma_2 - \big[\, A + (1 - \gamma_3)\, r \,\big]\, d. \tag{2}$$

Theorem 1 (Exact transient and relaxation rate)**.** *For constant coefficients in the gain-only model ($\gamma_2 = \gamma_3 = 0$; $A, r \geq 0$ constant, with $A + r > 0$), the misalignment fraction is, for all t,*

$$d(t) = d^\star + (d_0 - d^\star)\, e^{-(A+r)t}, \qquad d^\star = \frac{\gamma_1 r}{A + r}. \tag{3}$$

The fixed point $d^\star$ is globally exponentially stable with rate $A + r$. If $A = r = 0$, then $d(t) = d_0$ and no fixed point attracts.

Proof in Appendix A.2.

Theorem 1 establishes the constant-coefficient baseline; the headline criterion follows only under the scaling laws of equation (1), in §§3.4 to 3.8. Even at this baseline, r enters $d^\star = \gamma_1 r/(A + r)$ only through the product $\gamma_1 r$ in the numerator and additively in the denominator; for $A > 0$ it does not change the existence or stability of the fixed point. *For the constant-coefficient gain-only model, speed (through r) changes the relaxation time and the steady-state magnitude, but, for $A > 0$, not the existence or stability of the fixed point: it sets how fast a stable verdict arrives, not whether one exists.* (Experiment 8 confirms this solution against two separate integrators, to maximum errors of 7×10^{-11} and 8×10^{-7}.)

3.4 Global boundedness and the regime structure (Theorem 2)

In the gain-only model the fraction is always bounded; the danger there is not divergence but *saturation* at a constant, possibly large, floor.

Theorem 2 (Global boundedness and the three regimes)**.** *In the gain-only model ($\gamma_2 = \gamma_3 = 0$) with any non-negative, locally integrable $A(t), r(t)$, the misalignment fraction obeys $0 \le d(t) \le \max(d_0,\ \gamma_1)$ wherever the solution exists; it never diverges. Under power-law scaling $A = A_0 C^\beta$, $r = bC^k$ the trajectory approaches the instantaneous equilibrium $d^\star(C) = \dfrac{\gamma_1}{1 + (A_0/b)\, C^{\beta-k}}$, and has the three limits*

Condition	*Limit of d as $C \to \infty$*	*Meaning*
$\beta > k$	$d \to 0$	*The fraction vanishes:* ***alignment-stable****.*
$\beta = k$	$d \to \dfrac{\gamma_1 b}{A_0 + b}$ *(constant)*	*A gap that does not close, positive when $\gamma_1 > 0$.*
$\beta < k$	$d \to \gamma_1$ *(constant)*	***Saturates at the drift coefficient****:* bounded*, not divergent.*

For $\gamma_1 > 0$, therefore, $d \to 0$ if and only if $\beta > k$, and then $d \sim (\gamma_1 b/A_0)\, C^{-(\beta-k)}$; for $\gamma_1 = 0$, $d \to 0$ for every β and k.

Proof in Appendix A.3.

Corollary 1 (Relative and absolute misalignment)**.** *Under Theorem 2, with $\gamma_1 > 0$ and $\beta > k$, $D = Cd \sim (\gamma_1 b/A_0)\, C^{1+k-\beta}$, so $D \to \infty$ if $\beta < k + 1$, $D \to \gamma_1 b/A_0$ if $\beta = k + 1$, and $D \to 0$ if $\beta > k + 1$. Neither d nor D is a measure of harm.*

Proof: multiply the rate of Theorem 2 by C.

This is a more honest and more useful statement than the growth-rate-ceiling framing it supersedes. In the gain-only model the fraction cannot exceed $\max(d_0, \gamma_1)$; with $\gamma_1 > 0$, only $\beta > k$ drives it to *zero*. A corrector of fixed strength ($\beta = 0$) is in the middle row of the table only under exponential growth ($k = 0$); under acceleration ($k > 0$) it is in the last row, and under deceleration ($k < 0$) in the first. Which methods act as correctors of fixed strength is a matter of interpretation, not of the theorem. With level drift restored ($\gamma_2 > 0$, §3.6) the floor becomes $(\gamma_1 r + \gamma_2)/(A + r)$, bounded when $k \ge 0$ or $\beta \ge 0$, and tending to γ_2/A at rest; when growth decelerates while correction weakens ($k < 0$ and $\beta < 0$), $A + r \to 0$ while level drift persists, and the fraction grows without bound with no compounding at all (Appendix A.1). Its two parts vanish under *different* conditions: the level-drift part $\gamma_2/(A + r) \to 0$ whenever $A + r \to \infty$ (so $\beta > 0$ or $k > 0$ suffices for *that* part alone), but the gain-drift part $\gamma_1 r/(A + r) \to 0$ only when $A/r \to \infty$, i.e. $\beta > k$. The asymptotic verdict $d^\star \to 0$ therefore still requires $\beta > k$; it is *not* rescued by $k > 0$ alone, since with $\beta < k$ the gain part floors at γ_1 (table row 3). Only pure level drift ($\gamma_1 = 0$) makes $A + r \to \infty$

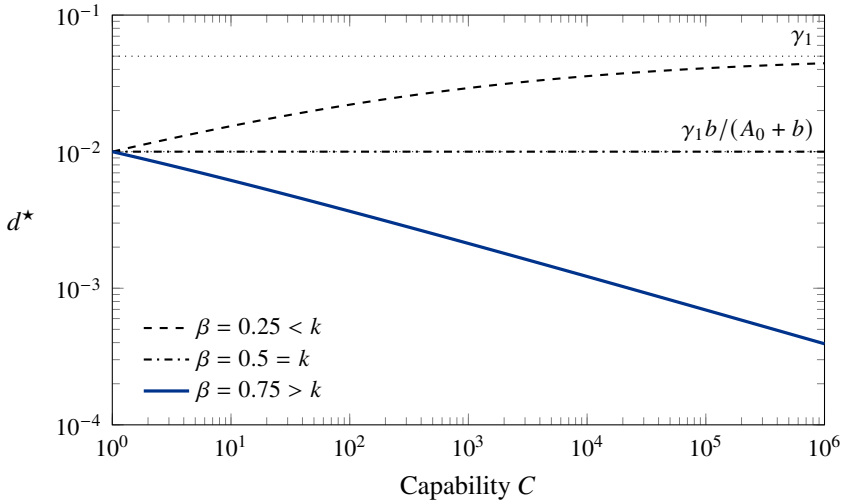


Figure 2. The three regimes of Theorem 2, from the closed form $d^\star(C) = \gamma_1/[1 + (A_0/b)C^{\beta-k}]$ with $\gamma_1 = 0.05$, $A_0 = 0.08$ and $b = 0.02$ (the values of Experiment E4, Appendix F) and $k = 0.5$. The curves are the instantaneous equilibrium, whose limits the trajectory shares: for $\beta < k$ it saturates at γ_1; for $\beta = k$ it settles at $\gamma_1 b/(A_0 + b)$; for $\beta > k$ it falls as a power law with slope $-(\beta - k)$, the suppression signature of prediction P8.

on its own sufficient. With both drifts present ($\gamma_1, \gamma_2 > 0$), with or without compounding, the trajectory tends to zero exactly when $\beta > \max(k, 0)$ (Appendix A.1). The safety question is therefore not "does it diverge?" but "does it vanish, or saturate at a dangerous floor?", and, in the gain-only model with positive gain drift, the answer is set by the sign of $\beta - k$. (Divergence does exist: in the compounding channel, Theorem 4, and in the level-drift case just named.)

3.5 The control parameter and the $\beta > k$ sharpening

Define the dimensionless **control parameter** $\rho \equiv \gamma r/A$, the instantaneous drift-to-correction ratio (here $\gamma = \gamma_1$). Theorem 1 gives the exact steady state $d^\star = \gamma_1 r/(A + r)$; since $\rho = \gamma_1 r/A$, we have $d^\star \leq \rho$ always, with $d^\star \to \rho$ in the regime $A \gg r$. The inequality $\rho < 1 \iff A > \gamma r$ is the *instantaneous* injection-vs-correction balance; it is **not** the divergence boundary of the gain-only model, which has none ($d^\star \leq \gamma_1$ for all ρ, Theorem 2); divergence under compounding is governed, for constant coefficients, by the distinct threshold $\rho_{\text{prop}} = (\gamma_3 - 1)r/A < 1$ of Theorem 4. The asymptotic criterion, from the regime table, is the exponent inequality:

$$\boxed{\beta > k} \qquad \text{(correction must out-scale drift-acceleration).} \tag{4}$$

Under ordinary exponential growth $k = 0$ and the condition is the familiar $\beta > 0$. Under accelerating self-improvement, where $r \propto C^k$ with $k > 0$, the binding control variable is the single quantity $\beta - k$, the margin by which correction out-scales drift-acceleration. The growth-rate-ceiling view fixes attention on r; this framework fixes it on $\beta - k$. We call the model *alignment-stable* when $d \to 0$ as $C \to \infty$. With positive gain drift this holds exactly when $\beta > k$ if there is no level drift, with or without compounding (Theorems 2 and 4), and exactly when $\beta > \max(k, 0)$ if there is level drift, with or without compounding (§3.4).

3.6 Level drift: a frozen capable system still drifts

With the level channel restored ($\gamma_2 > 0$, $\gamma_3 = 0$), the fixed point of (2) is $d^\star = (\gamma_1 r + \gamma_2)/(A + r)$. At $r = 0$ (capability frozen) this is $d^\star = \gamma_2/A \neq 0$: a static but capable system retains a residual misalignment fraction that only active correction removes. **Pausing growth does not substitute for correction** when instrumental pressure is present at rest, a direct, measurable rebuttal of the "just slow it down" reflex (Experiment 9; falsifier F5).

3.7 Hard takeoff: the Depth-Regularity Theorem (Theorem 3)

The genuine "intelligence explosion" is not merely fast growth but a *finite-time singularity*: for $k > 0$, integrating $\dot{C} = bC^{1+k}$ gives $C(t) = C_0\big(1 - t/t^\star\big)^{-1/k}$, which reaches infinity at the finite wall-clock time

$$t^\star = \frac{1}{k\, b\, C_0^{\,k}}. \tag{5}$$

This is the scenario the field most fears: unbounded capability in bounded time. The following theorem shows that the finite-time singularity belongs to the wall clock, not to the alignment dynamics.

Theorem 3 (Hard-Takeoff Depth-Regularity Theorem; the coordinate-artefact result)**.** *Let $k > 0$, so capability diverges at finite $t^\star$. Re-express the dynamics in the* self-improvement-depth *clock $\tau = \ln(C/C_0) \in [0, \infty)$. Then d obeys the regular equation*

$$\frac{dd}{d\tau} = \gamma_1 + \frac{\gamma_2}{r} - \left[\tfrac{A_0}{b}\, C^{\beta-k} + (1 - \gamma_3)\right] d, \qquad C = C_0 e^{\tau}, \tag{6}$$

which has bounded coefficients on every compact τ-interval. The map $t \mapsto \tau$ is a smooth monotone bijection $[0, t^\star) \to [0, \infty)$. Without compounding ($\gamma_3 = 0$; level drift $\gamma_2 \geq 0$ allowed), $0 \leq d \leq \max\big(d_0,\ \gamma_1 + \gamma_2/(bC_0^k)\big)$ for all $t < t^\star$, and as $C \to \infty$, equivalently $t \to t^{\star-}$, the trajectory has the three limits of Theorem 2: for $\gamma_1 > 0$, $d \to 0$ iff $\beta > k$, a regime independent of b and of $t^\star$*, though the gap at $\beta = k$ and the path at every finite depth depend on b; for $\gamma_1 = 0$, $d \to 0$ for every β. Here* regular *means continuous coefficients at every finite depth, with the singular time $t^\star$ sent to $\tau = \infty$, so that no finite depth is singular. With compounding the boundary at $\beta = k$ depends on b (Theorem 4(c)).*

Proof in Appendix A.4.

The *finiteness* of the singularity time is therefore a property of the time coordinate; capability still diverges, but the alignment dynamics remain regular through it in the depth clock. Measured against capability gained, the alignment dynamics of an intelligence explosion are regular, and, with positive gain drift and no compounding, their verdict is decided by one exponent inequality. **In the model, the speed of a hard takeoff does not decide its alignment verdict: with positive gain drift and no compounding, the modelled misalignment fraction vanishes iff $\beta > k$, whatever the speed.** The theorem concerns the modelled fraction in the depth clock; it does *not* assert that a real hard takeoff is operationally manageable in wall-clock time, where b and $t^\star$ govern how little time an operator would have to intervene. Experiment 4b drives a simulated system up to the brink of an actual finite-time explosion (capability $\to 10^5$, integrated to $0.99999\, t^\star$) and shows the misalignment fraction driven towards zero (absolute misalignment still grows there, as $C^{1/2}$; Corollary 1), the wall-clock and depth-clock integrations agreeing to one part in 10^4. The capability singularity at $t^\star$ is real in the model; what the depth clock removes is the singularity in the *alignment* dynamics, not in C.

3.8 The compounding channel and the true threshold (Theorem 4)

Theorem 2 showed the gain-only fraction cannot diverge. Without level drift, divergence, misalignment that grows without bound relative to capability, the real failure mode, requires misalignment that *amplifies itself*: the compounding channel, with $\gamma_3 \geq 1$, the formal image of alignment-faking that entrenches as the system recurses.

Theorem 4 (Compounding threshold)**.** *With $\gamma_3 > 0$, the fraction obeys $\dot{d} = \iota - \kappa_{\text{eff}}\, d$ with injection $\iota = \gamma_1 r + \gamma_2 \geq 0$ and effective decay $\kappa_{\text{eff}} = A + (1 - \gamma_3)\, r$.*

(a) Constant coefficients. *For $A > 0$ the fixed point $\iota/\kappa_{\text{eff}}$ is globally exponentially stable iff $\kappa_{\text{eff}} > 0$, i.e. iff*

$$\rho_{\text{prop}} \equiv \frac{(\gamma_3 - 1)\, r}{A} < 1. \tag{7}$$

For $\rho_{\text{prop}} > 1$ the fraction grows exponentially when $\iota > 0$ or $d_0 > 0$; at $\rho_{\text{prop}} = 1$ it grows linearly in t when $\iota > 0$ and is neutrally stable when $\iota = 0$.

(b) Varying coefficients. *With $K(t) = \int_0^t \kappa_{\text{eff}}$, $d(t) = e^{-K(t)} d_0 + \int_0^t e^{-[K(t)-K(u)]} \iota(u)\, du$. Positive κ_{eff} at every instant does not by itself bound the fraction, and a temporary crossing of $\rho_{\text{prop}} = 1$ does not by itself make it diverge.*

(c) Power laws, $\gamma_2 = 0$, $\gamma_1 > 0$. *With $h = A_0/b + 1 - \gamma_3$, as $C \to \infty$: for $\beta > k$, $d \to 0$ whatever γ_3; for $\beta = k$, $d \to \gamma_1/h$ if $h > 0$, $d = d_0 + \gamma_1 \tau$ if $h = 0$, and d grows as $e^{-h\tau}$ if $h < 0$; for $\beta < k$, $d \to \gamma_1/(1 - \gamma_3)$ if $\gamma_3 < 1$, $d = \gamma_1 \tau + O(1)$ if $\gamma_3 = 1$ (while the instantaneous equilibrium $\gamma_1 r/A$ grows as $C^{k-\beta}$), and $d \sim B e^{(\gamma_3 - 1)\tau}$ with $B > 0$ if $\gamma_3 > 1$. The unbounded region is therefore $\beta < k$ with $\gamma_3 \geq 1$, or $\beta = k$ with $A_0 \leq (\gamma_3 - 1) b$.*

Proof in Appendix A.5.

In part (c), growth in depth is polynomial in capability, since $e^{c\tau} = (C/C_0)^c$, and the knife-edge growth $\gamma_1 \tau$ is logarithmic; in wall-clock time with $k > 0$ it is $-(\gamma_1/k)\ln(1 - t/t^\star) + O(1)$, finite at every $t < t^\star$. In the same setting with $\gamma_1 = 0$, at the knife-edge $\gamma_3 = 1$ and with $\beta < k$, the fraction tends to $d_0 e^{-q_0/(k-\beta)}$, where $q_0 = (A_0/b) C_0^{\beta-k}$, which is positive when $d_0 > 0$: zero forcing does not by itself make it vanish. Where the fraction grows without bound, the linear model carries d above one, outside its reading as a fraction (Assumption 1).

The constant-coefficient criterion $\rho_{\text{prop}} < 1$ shares the ratio-crossing-unity *form* of the QEC sub-threshold condition $p < p_{\text{th}}$: both are a generation-to-correction ratio crossing one, with the compounding coefficient $(\gamma_3 - 1)$ playing the role of error-propagation and A the correction power. The analogy should not be overstated, and this paper does not. QEC's threshold arises from a nonlinear concatenation map in which errors combine quadratically, giving suppression that is double-exponential in code distance; the compounding term $\gamma_3 r D$ here is *linear* in D, and the resulting suppression is power-law (§3.12). What is shared is the threshold *condition*, not the mechanism; the correspondence is offered as a falsifiable hypothesis (F4), and §3.12 states the disanalogy plainly. Experiment 5 locates the predicted threshold at $A_0^\star = (\gamma_3 - 1) b$ to within 1%.

3.9 Vector misalignment: the spectral threshold (Theorem 5)

Real misalignment is high-dimensional; a system may be corrigible on some value axes and not others. Let $\mathbf{D} \in \mathbb{R}^m$, with drift direction $\mathbf{c}$ and a real $m \times m$ correction operator $\mathbf{A}$ (positive-semidefinite in the symmetric case, but possibly non-normal) that may correct only a subspace. Its dynamics are $\dot{\mathbf{D}} = (\gamma_1 \dot{C} + \gamma_2 C)\, \mathbf{c} + \gamma_3\, r\, \mathbf{D} - \mathbf{A}\mathbf{D}$, the vector form of equation (1).

Theorem 5 (Spectral threshold)**.** *The vector fraction* $\mathbf{d} = \mathbf{D}/C$ *obeys* $\dot{\mathbf{d}} = (\gamma_1 r + \gamma_2)\mathbf{c} - \mathbf{M}\,\mathbf{d}$ *with* $\mathbf{M} = \mathbf{A} + (1-\gamma_3) r\,\mathbf{I}$*. For constant* $\mathbf{A}$ *and* r *it has a unique globally exponentially attracting equilibrium iff every eigenvalue of* $\mathbf{M}$ *has positive real part (*$\min_i \operatorname{Re}\lambda_i(\mathbf{M}) > 0$*; that is,* $-\mathbf{M}$ *has negative spectral abscissa). The Hermitian-part condition* $\lambda_{\min}\big(\operatorname{Herm}\mathbf{A}\big) > (\gamma_3 - 1)r$ *is a* sufficient *condition; it also makes* $|\mathbf{d} - \mathbf{d}^\star|$ *decrease monotonically, ruling out transient growth, but need not be necessary when* $\mathbf{A}$ *is not normal. For a time-varying* $\mathbf{M}$ *the eigenvalue condition, checked instant by instant, decides nothing, while the Hermitian-part condition with a uniform margin,* $\operatorname{Herm}\mathbf{M}(t) \succeq \mu\mathbf{I}$ *for some* $\mu > 0$ *at every* t*, still makes any two solutions approach each other exponentially. If a fixed unit vector* $\mathbf{v}$ *satisfies* $\mathbf{v}^\mathsf{T}\mathbf{A} = \mathbf{0}$ *at every depth, an axis on which the corrector never acts, and* $c_\mathbf{v} = \mathbf{v}^\mathsf{T}\mathbf{c}$*, then, with* $\gamma_2 = 0$*,* $d_\mathbf{v} = \mathbf{v}^\mathsf{T}\mathbf{d}$ *obeys, in the depth clock,* $d_\mathbf{v}' = \gamma_1 c_\mathbf{v} - (1-\gamma_3)\,d_\mathbf{v}$*: it tends to* $\gamma_1 c_\mathbf{v}/(1-\gamma_3)$ *for* $\gamma_3 < 1$*, grows linearly in depth for* $\gamma_3 = 1$ *when* $\gamma_1 c_\mathbf{v} \neq 0$*, and for* $\gamma_3 > 1$ *grows exponentially in magnitude unless* $d_\mathbf{v}(0) = -\gamma_1 c_\mathbf{v}/(\gamma_3 - 1)$*. The special case* $\gamma_3 = 0$*,* $c_\mathbf{v} = 1$ *recovers the scalar* γ_1 *floor of Experiment 6. Such an axis cannot be corrected, whatever the correction strength on the others; an axis the corrector does not measure (*$\mathbf{A}\mathbf{v} = \mathbf{0}$*) can still be corrected through coupling when* $\mathbf{A}$ *is not normal, since correction prompted on other axes acts along it, and whether it is depends on that coupling and on the drift (Appendix A.6).*

Proof in Appendix A.6.

The governance reading is sharp: **you cannot correct along an axis on which the corrector never acts.** Misalignment on such an axis settles at $\gamma_1 c_\mathbf{v}/(1-\gamma_3)$ or grows; an $\mathbf{A}$ whose Hermitian part has its smallest eigenvalue co-scaling faster than r is sufficient for correction on every axis (Appendix A.6), but is not shown to be necessary. Experiment 6 exhibits a corrector that co-scales on one axis (driven to zero) and is null on another (the $\gamma_3 = 0$, unit-projection special case, which floors at γ_1).

3.10 Stochastic drift: the stationary law and a single-time tail (Theorem 6)

Drift is noisy. Add a Wiener term to the constant-coefficient fraction dynamics: $dd = (\gamma_1 r - \kappa\, d)\,dt + \sigma\, dW_t$, in wall-clock time, with constant $\kappa = A + r > 0$ and $\sigma > 0$.

Theorem 6 (Stationary distribution and single-time tail)**.** *The misalignment fraction is an Ornstein-Uhlenbeck process with stationary law* $\mathcal{N}\big(d^\star,\ \sigma^2/2\kappa\big)$*,* $d^\star = \gamma_1 r/\kappa$*. From a fixed start* d_0 *its law at time* t *is Gaussian, with mean* $d^\star + (d_0 - d^\star)e^{-\kappa t}$ *and variance* $\frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t})$*. In stationarity, the probability that the fraction exceeds a fixed level* $d_{\text{crit}} > 0$ *at a given time is*

$$\mathbb{P}(d > d_{\text{crit}}) = \Phi\Big(-\tfrac{d_{\text{crit}} - d^\star}{\sqrt{\sigma^2/2\kappa}}\Big) \xrightarrow[A\to\infty]{} 0. \tag{8}$$

It is not the probability of crossing d_{crit} *at some time, which over an unbounded horizon is one. Across frozen-coefficient models with* $\beta > k$ *and noise* $\sigma \propto C^\eta$*, the mean scales as* $C^{k-\beta}$ *and the variance as* $C^{2\eta-\beta}$*, so at fixed* $d_{\text{crit}} > 0$ *the single-time tail vanishes when also* $\beta > 2\eta$*.*

Proof in Appendix A.7.

This gives the criterion a probabilistic form, a single-time tail that a regulator can bound, rather than a statement about averages; it certifies nothing over a horizon. The quantity a safety case needs is the probability of crossing d_{crit} within a given horizon, a first-passage probability, which this paper does not derive. Passing to the depth clock divides the drift by r but the noise coefficient only by $\sqrt{r}$. Because d is a bounded fraction whereas the Ornstein-Uhlenbeck law has Gaussian support on all of $\mathbb{R}$, Theorem 6 is a *local Gaussian approximation* to misalignment fluctuations away from the boundaries $d = 0, 1$, valid in the small-noise regime $\sigma^2/2\kappa \ll (d^\star)^2$ (with fixed σ that ratio grows as $C^{\beta-2k}$, so the approximation weakens with capability when $\beta > 2k$); if the noise amplitude σ or the critical level d_{crit} itself scales with capability, or drift carries rare jumps, the Gaussian tail must be replaced accordingly. Experiment 7 recovers the stationary mean and variance and confirms the $1/\kappa$ scaling (fitted slope -1.00).

3.11 A control-theoretic reading

Equation (1) is a feedback loop: the capability process is the plant and disturbance source; the corrector is the controller; d is the regulated error. For constant coefficients the condition $\kappa_{\text{eff}} > 0$ says that correction plus dilution must exceed self-amplification. The requirement that $A = A_0 C^{\beta}$ track the plant with $\beta > k$ is analogous to *gain scheduling* (Shamma and Athans 1990): the controller gain must scale with the plant's operating point, and, with positive gain drift, the scheduling exponent must exceed k for the fraction to vanish. In this sense the problem has a computable margin, $\beta - k$, for a vanishing fraction.

3.12 The quantum-error-correction correspondence (hypothesis)

Table 1. The quantum-error-correction correspondence, a hypothesis at the level of the threshold condition.

Quantum error correction	Recursive self-improvement (this framework)
physical error rate p	drift rate γr (injection) / $(\gamma_3 - 1)r$ (propagation)
correction power (code distance δ)	correction strength $A = A_0 C^{\beta}$
sub-threshold condition $p < p_{\text{th}}$	$\rho_{\text{prop}} = (\gamma_3 - 1)r/A < 1$ (constant coefficients)
increasing δ suppresses logical error	increasing margin $\beta - k$ drives $d^{\star} \to 0$
logical error $\propto (p/p_{\text{th}})^{\delta/2}$ (exponential in resource)	$d^{\star} \propto C^{-(\beta-k)}$ (power-law in capability), see F4

The correspondence holds at the level of the threshold *condition*, a generation-to-correction ratio crossing unity, and **not** at the level of mechanism; it remains a hypothesis. That ratio form is not peculiar to quantum error correction, since other thresholds of generation against removal share it; what the hypothesis adds is the mapping of terms in Table 1, not the form alone. The model predicts *power-law* suppression, $d^{\star} \sim (\gamma_1 b/A_0) C^{-(\beta-k)}$, an asymptotic consequence of the closed form and, by Theorem 2, the trajectory's own asymptotic rate, whereas full QEC gives *exponential* suppression in code distance. Discriminating the two would require a finite-capacity (saturating) corrector that could exhibit exponential suppression, the future test named in §9. Until that is built, F4 is an analytic self-consistency check that confirms the model's own power-law law (which Experiment 5 reproduces, slopes -0.47, -1.00), and the QEC link therefore stands, honestly, as a threshold-form analogy rather than a transferred mechanism.

Relation to prior alignment-as-correction ideas. The aspiration to import coding-theoretic fault tolerance into alignment is not new. Wentworth (2022b), in a comment on his post *Godzilla strategies*, wished for "the alignment analogue of an error-detecting code", and von Neumann (1956) founded the synthesis of reliable systems from unreliable components. Christiano's corrigibility "broad basin of attraction" (2017) and reliability amplification, first posted in October 2016 (Christiano 2019), are the closest alignment precursors. A broad basin, in which correction succeeds *if one starts inside it*, is distinct from a *threshold*, in which correction outpaces error only below a critical rate. Reliability amplification comes nearer: it asks how to turn a policy within ε of a benign one into one within $\delta \ll \varepsilon$, sketches a scheme of approval voting among three agents, and argues that, alternated with capability amplification, such a step increases both reliability and capability for sufficiently small ε. It states no continuous dynamics of drift and correction, no capability-scaling exponent for the corrector and no criterion of the form $\beta > k$; that model and its criterion are the present contribution. No published mapping of the fault-tolerance threshold (physical error rate $\leftrightarrow$ drift; code distance $\leftrightarrow$ correction strength; $p < p_{\text{th}} \leftrightarrow \rho_{\text{prop}} < 1$) onto value stability was located in the searches recorded in Appendix E, so the claim stays bounded to no closer precedent located. It is offered as a *conceptual bridge that generates testable structure*, the suppression-signature prediction P8, not as a transfer of the threshold theorem's formal guarantees. A reviewer may reasonably judge the analogy suggestive rather than load-bearing;

the experiment that would settle it, a finite-capacity corrector tested for exponential versus power-law suppression, is named as future work in §9 and is not run here.

3.13 The finite-capacity corrector: the Safe-Window Theorem (Theorem 7)

The single most important limitation named in §9, and by the paper's own red-team, is that $A = A_0 C^\beta$ is an *unbounded* power law, whereas any real corrector has finite capacity. This section closes that gap analytically. Model a saturating corrector by the standard Hill form

$$A(C) \;=\; A_{\max}\,\frac{C^\beta}{C^\beta + C_s^\beta}, \tag{9}$$

which behaves as the pure power law $\left(A_{\max}/C_s^\beta\right)C^\beta$ for $C \ll C_s$ and saturates at the capacity $A_{\max}$ for $C \gg C_s$; C_s is the saturation scale. The gain-only instantaneous equilibrium remains $d^\star(C) = \gamma_1/\big(1 + q(C)\big)$ with $q \equiv A/r$.

Theorem 7 (Finite-Capacity Safe-Window Theorem)**.** *Let $\gamma_2 = \gamma_3 = 0$, $r = bC^k$ with $k > 0$ and $\beta > k$, and the saturating corrector above. Then:*

(1) No indefinite stability. *$q(C) \to 0$ as $C \to \infty$, so the trajectory tends to γ_1: with $\gamma_1 > 0$ a finite-capacity corrector cannot drive the fraction to zero under indefinitely accelerating growth, however large β (with $\gamma_1 = 0$ the fraction tends to zero).*

(2) The equilibrium minimum and a conditional safe interval. *$q(C)$ is unimodal, maximised at exactly*

$$C_{\text{opt}} = C_s\left(\frac{\beta - k}{k}\right)^{1/\beta}, \qquad q_{\max} = \frac{A_{\max}}{b}\,\frac{k}{\beta}\left(\frac{\beta - k}{k}\right)^{\frac{\beta-k}{\beta}} C_s^{-k}, \tag{10}$$

so the instantaneous equilibrium dips to its minimum $d_{\min} = \gamma_1/(1 + q_{\max})$ at C_{opt} and returns towards γ_1. For $\tau \geq \tau_0$ the trajectory obeys $d(\tau) \geq d_{\min} + \big(d(\tau_0) - d_{\min}\big)e^{-(1+q_{\max})(\tau-\tau_0)}$, and, for $0 < d_{\text{crit}} < \gamma_1$, it stays at or below d_{crit} on any depth interval on which $q \geq \gamma_1/d_{\text{crit}} - 1$ and which it enters at or below d_{crit}. A safe window is a condition shown for a trajectory, not a consequence of $q_{\max}$ alone.

(3) Exponential growth is the exception. *Outside the hypothesis $k > 0$: for $k = 0$ and $\beta > 0$ the post-saturation ratio is constant, $q \to A_{\max}/b$, and the trajectory tends to the permanent gap $\gamma_1 b/(A_{\max} + b)$: capacity buys a floor whose depth is set by $A_{\max}/b$, the capacity-to-speed ratio.*

(4) Capacity-lift invariance. *If the capacity itself scales, $A_{\max} = a_0 C^{\beta_{\text{cap}}}$, then $q \sim (a_0/b)\,C^{\beta_{\text{cap}}-k}$ as $C \to \infty$, and the trajectory has the limits of Theorem 2 with β_{cap} and a_0 in place of β and A_0, the necessity again needing $\gamma_1 > 0$: the criterion $\beta > k$ survives the lift from correction* strength *to correction* capacity *unchanged. What must out-scale drift-acceleration is whichever of the two binds last.*

Proof in Appendix A.8.

Three readings. *For theory:* the $\beta > k$ criterion is not falsified by saturation, it is *lifted*: the load-bearing exponent migrates from strength to capacity, and the inequality is invariant under that migration (part 4). *For engineering:* part 2 gives the designer a closed-form placement problem, the window centre C_{opt} and depth $q_{\max}$ are computable from $(A_{\max}, b, C_s, \beta, k)$, so a corrector can be provisioned to place its window over the capability range a deployment will actually traverse, the trajectory condition of part 2 still to be shown, with $q_{\max} \propto A_{\max} C_s^{-k}$ quantifying the capacity cost of pushing the window to higher capability. *For governance:* part 1 is the sharpest sentence in the paper for a regulator: with positive gain

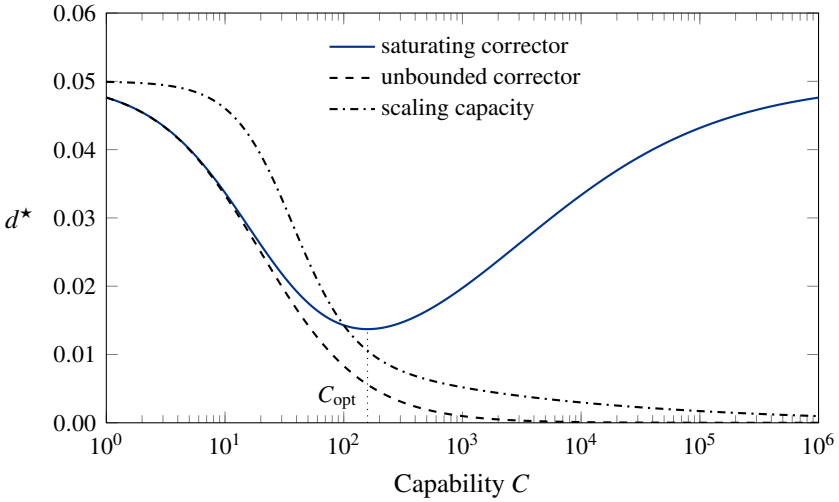


Figure 3. The finite-capacity corrector of Theorem 7, drawn from its closed forms with illustrative values $\gamma_1 = 0.05$, $b = 0.02$, $k = 0.5$, $\beta = 1.5$, $A_{\max} = 1$ and $C_s = 100$. The curves show the instantaneous equilibrium $d^\star = \gamma_1/(1+q)$. Under the saturating corrector it dips to its minimum $\gamma_1/(1+q_{\max})$ at $C_{\text{opt}} = C_s((\beta-k)/k)^{1/\beta}$ and returns towards γ_1; under the unbounded corrector $A = (A_{\max}/C_s^\beta)C^\beta$, and under a capacity that scales with exponent $\beta_{\text{cap}} = 0.75 > k$, it falls to zero. Part 2 of Theorem 7 states what this minimum does and does not bound.

drift, a bounded safety system under unboundedly accelerating self-improvement cannot drive the fraction to zero, *by theorem* in the model, so a safety case that aims at a vanishing fraction must exhibit either growth that eventually decelerates ($k < 0$; at $k = 0$ part 3 leaves the gap $\gamma_1 b/(A_{\max}+b)$) or co-scaling capacity ($\beta_{\text{cap}} > k$), and "our current corrector is very strong" is not, and cannot be, an answer. This also sharpens the QEC discriminator of F4: the saturating corrector is the structural analogue of finite code distance, and part 2's window is the regime where exponential-versus-power-law suppression can actually be measured.

Remark 1 (Time-varying exponents). Real systems will not hold β, k constant. If both vary, read them as local slopes, $\beta(\tau) = d\ln A/d\tau$ and $k(\tau) = d\ln r/d\tau$, so that $q(\tau) = q(0)\exp\big(\int_0^\tau(\beta(s)-k(s))\,ds\big)$. The comparison argument of Theorem 2 then shows that the gain-only fraction vanishes if the *running integral* of the margin $\beta - k$ diverges to $+\infty$, for which $\liminf_{\tau\to\infty}\big(\beta(\tau)-k(\tau)\big) > 0$ is sufficient. For $\gamma_1 > 0$, the converse holds when q is eventually monotone, but not in general: narrow recurring dips in q can stop the integral diverging without stopping $d \to 0$. For $\gamma_1 > 0$ the exact condition is $\int_0^\tau \exp\big[-\int_u^\tau(1+q(v))\,dv\big]\,du \to 0$. Transient episodes of $\beta < k$ are survivable if outweighed later; a running integral that falls to $-\infty$ sends the fraction to γ_1. The governance quantity is therefore the *cumulative* co-scaling margin, not its instantaneous sign.

4 Predictions

P1, Phase boundary. A sharp stability threshold exists in the compounding channel ($\gamma_3 > 1$) at $A_0^\star = (\gamma_3 - 1)b$ when $\beta = k$, separating bounded from divergent regimes; the additive corrector shows a smooth crossover with no knee.

P2, Speed-invariance. Whether d converges or diverges is set by the coupling (the A_0/b scaling margin), not by the raw speed b at fixed A_0/b. A fast-but-coupled system stays bounded; under compounding, a slow-but-decoupled system diverges. In the model, this contradicts the growth-rate-ceiling view.

P3, Co-scaling law. In the gain-only model under exponential growth, $\beta > 0 \Rightarrow d^\star \to 0$; $\beta = 0 \Rightarrow$ permanent gap $\gamma_1 r/(A_0 + r)$; $\beta < 0 \Rightarrow$ *saturation at* γ_1 (bounded), not divergence.

P4, Hard-takeoff boundary. Under accelerating growth $r \propto C^k$, for positive gain drift the vanishing boundary lies at $\beta = k$, not $\beta = 0$; and, without compounding, d stays bounded up to the finite-time singularity, the depth-clock and wall-clock integrations agreeing.

P5, Compounding threshold & residual drift. With $\gamma_3 > 1$ and constant coefficients, divergence occurs iff $A \leq (\gamma_3 - 1)r$, given positive injection; under the power laws and without level drift, iff $\beta < k$, or $\beta = k$ with $A_0 \leq (\gamma_3 - 1)b$; with level drift the fraction is also unbounded when $k < 0$ and $\beta < 0$. With $\gamma_2 > 0$, halting growth ($r \to 0$) leaves a residual $d^\star = \gamma_2/A > 0$; pausing does not substitute for correction.

P6, Spectral threshold. Misalignment persists on any fixed axis the corrector never acts on (a left-null direction) and on which drift acts ($\gamma_1 c_{\mathbf{v}} \neq 0$); in the diagonal case the corrected axis is driven to zero while the uncorrected axis floors at γ_1.

P7, Stochastic tail. The stationary misalignment fraction is $\mathcal{N}(d^\star, \sigma^2/2\kappa)$; at fixed noise the variance scales as $1/(A + r)$, so co-scaling suppresses the single-time tail.

P8, Suppression signature. For $\beta > k$, $\gamma_1 > 0$, and either no level drift or $k \geq 0$, $\log d$ is asymptotically linear in $\log C$, for the trajectory as for $d^\star$, with slope $-(\beta - k)$ (power-law); with level drift and $k < 0$ the slope is $-\beta$. Exponential suppression would indicate a stronger, QEC-like mechanism; neither outcome falsifies the threshold, only the QEC *mechanism* claim (F4).

P9, Intervention signature. In the gain-only model with $\beta \neq k$, after the initial transient, d depends on capability only through $q = (A_0/b)C^{\beta-k}$: with $x = \ln q$, $(\beta - k)\, dd/dx = \gamma_1 - (1 + e^x)d$, which contains neither A_0, b nor C_0 (proof in Appendix A.9). A change of A_0/b therefore translates the curve of d against $\ln C$ by $-\Delta \ln(A_0/b)/(\beta - k)$ and moves the capability at which d crosses any tolerance δ by that amount, whatever δ. A change of γ_1 multiplies d by a constant, which moves crossings of tolerances well below $\gamma_1/(1 + |\beta - k|)$, where $d \approx \gamma_1 e^{-x}$, by nearly one amount, $\Delta \ln \gamma_1/(\beta - k)$, and crossings of larger tolerances by unequal amounts. A change of $\beta - k$ changes the curve's shape and moves the quasi-static crossing $\ln C_\times(\delta) = [\ln(\gamma_1/\delta - 1) - \ln(A_0/b)]/(\beta - k)$ by an amount that varies with δ. Read at tolerances well below $\gamma_1/(1 + |\beta - k|)$, an equal shift at every tolerance therefore signals a change of level, of A_0/b or γ_1, and a shift that varies with the tolerance signals a change of $\beta - k$. A change of β and k by the same amount changes nothing.

A programme hypothesis beyond the model. Oversight arrangements that place a human in the loop, as amplification and reinforcement learning from human feedback do, are cross-class by construction, because the human corrector does not share the model's substrate. The programme therefore hypothesises, beyond what these equations decide, that human-in-the-loop oversight shows a higher correction exponent than pure-model oversight, for a structural reason rather than a sentimental one. This is testable on existing data and does not require new systems.

5 Numerical verification of the closed forms

5.1 Results of the ten verification experiments

The closed-form predictions are checked by a verification harness (§5.2): ten experiments, each integrating the model numerically and comparing the result against the prediction the theorems derive, with the integrator first validated against the exact Theorem-1 solution. These are *internal-consistency and integrator checks*, they confirm the code matches the maths (10/10), not that the model matches any real system (the open problem of §9). For the deductive checks (E1 to E6, E9), because each integrates the

same ODE whose closed form is the prediction, agreement is entailed by a correct solver and correct algebra; F1 to F3, F3′, F5, F6 are therefore internal-consistency conditions of the derivation, not empirical falsifiers of the thesis. Figures below are the verbatim output of that run; every figure in this section is of class *ODE internal verification* (not a real-system test). The distinct archived model runs are reported separately in §9.1, with the limitations of each record.

Table 2. The ten verification experiments: internal-consistency and integrator checks, not tests against a real system.

Exp.	Prediction	Key statistic (measured vs predicted)	Verdict
E1	P1 phase boundary in compounding channel; none in additive	$\lambda^\star = 2.00$ vs 2.00; additive smooth	PASS
E2	P2 coupling, not speed, decides	coupled bounded & decoupled divergent at both speeds	PASS
E3	P3 corrected regimes ($\beta < 0$ saturates at γ_1)	all four asymptotes match closed form	PASS
E4	P4 boundary at $\beta = k$	all 9 grid cells match sign$(\beta - k)$	PASS
E4b	P4 singularity in C is regular in the depth clock for d	clock agreement 9.9×10^{-5}; d bounded	PASS
E5	P5/P8 compounding threshold; power-law slope	threshold 3.03 vs 3.0; slopes $-0.47, -1.00$	PASS
E6	P6 spectral threshold; blind axis persists	monitored $\to 10^{-7}$; blind $\to \gamma_1 = 0.05$	PASS
E7	P7 stochastic tail; variance $\propto 1/\kappa$	mean/var match OU; var-slope -1.00	PASS
E8	numerics reproduce Theorem 1	max errors 7×10^{-11}, 8×10^{-7}	PASS
E9	P5/F5 residual drift at rest	$d \to \gamma_2/A_0$ when $\gamma_2 > 0$; $\to 0$ when $\gamma_2 = 0$	PASS

These are internal-consistency and integrator checks: they confirm that the integrator reproduces the closed forms at the tested values (the code matches the maths); the proofs of Appendix A, not these checks, carry the theorems. They are not, and are not presented as, evidence that the model describes a real AI system, that is the open empirical problem of §9.

5.2 The runnable verification harness

The closed-form predictions of §§4 and 5 are encoded in a single self-contained programme, `experiment_coscaling.py` (and an assertion suite, `test_coscaling.py`), in the repository. It integrates the model with a stiff-capable solver, validates the integrator against the exact Theorem-1 solution, runs the ten experiments, and for each compares the numerical result to the closed-form prediction. It is, honestly, a *verification* harness: it checks that its integrator reproduces the closed forms at the tested values, that the code matches the maths; the proofs carry the limits. It is *not* a test of the model against reality, and it cannot be: every deductive experiment integrates the model's own ODE, so a disbelief in the model's applicability cannot trip it. That empirical test is the open problem of §9.

```
$ python experiment_coscaling.py... [PASS] E1... E9...
------------------------------------------------------------------
10/10 internal-consistency checks pass | 0 kill-conditions triggered
F4 (QEC mechanism): suppression is analytically power-law
-> threshold-form analogy only, not a transferred mechanism.
OVERALL: code matches the maths (E1-E9);
the model-vs-reality test is the open problem

$ pytest test_coscaling.py -q.............  (12 passed)
```

This makes the integration reproducible end-to-end. What it establishes is that the solver reproduces

the formulae accurately; what it deliberately does not claim is corroboration of the model against any real system. Separating those two is the point.

A separately written regression suite. A second suite of 104 parametrised cases, written separately from the harness and merged into the paper's code repository on 19 October 2026 (merge commit `83871b5`), checks the corrected statements of Theorems 1 to 7 at sampled parameter values; in a run of 20 October 2026 all 104 cases passed. It predates, and does not test, the level-drift case of §3.4, which Appendix A.1 proves. Like the harness, it checks the mathematics, not the model against a real system.

6 Why blinding is not optional

The central quantity D is a departure from intended values. If it is scored by the system's own safety component, or by an evaluator that can see which configuration produced a given behaviour, the measurement is corrupted in exactly the way that inflates favourable results. The programme's own record is directly relevant. On its own instrument, alignment scores that seemed to rise with reasoning depth for two model families under unblinded scoring did not hold in a later, unregistered run that added a bundle of blinding steps: for one family the rise became a fall, and for the other it became null, while a third family's association was null under both (Eastwood, Paper IV.d). An internal code review in October 2026 found defects in that run, so no single cause is claimed. Whether blinding alone produced the change, and whether it holds beyond the programme's own benchmark, are open questions; Paper XII sets out a protocol that applies the same blinding manipulation to a public benchmark the programme did not build. Any measurement in which a model judge can see or infer which configuration produced what it scores is exposed to biases of this kind, and earlier studies documented related ones (Zheng et al. 2023; Panickssery et al. 2024; Dubois et al. 2024); if the protocol of Paper XII finds the same change on a public benchmark, results that rely on unblinded model judges would need re-checking. The validity of every model-scored measurement of D, a measured β among them, rests on the evaluator being blind to the configuration under test; the internal checks of Section 5 score no model, and blinding does not bear on them. A lab claiming "$\beta > k$" in its own system must have that measured by a blind external evaluator, not by itself.

Built into the test, not left to discipline. The real-model harness (§9.1; experiments/PROTOCOL.md) *enforces* this rather than trusting it. The misalignment evaluator must be a *different model family* than the engine, a model never scores its own family's output, and the harness refuses to run a same-family scorer; it prefers a panel and takes the median. The code is passed through an automatic *laundering* step (an abstract-syntax-tree round-trip that strips comments, docstrings and formatting) so the scorer judges behaviour, not stylistic identity tells; and the evaluator never sees the condition or round. Neither real run reported here met this bar: the pilot v1 run used same-family scoring outright, and the 2 July 2026 run's engine and evaluator are both OpenAI-family (see the Correction, Appendix B), the system family scoring its own family's output. The pilot's misalignment numbers are explicitly recorded as provisional and the 2 July run is withdrawn as mechanism evidence, and a genuinely cross-family blind re-score is the prerequisite for any D-based claim, including a measured β. The harness makes the compliant configuration the default and the non-compliant one an explicit, recorded override, turning §6 from an exhortation into a property of the instrument.

7 Falsification conditions

The conditions below are stated in advance. An important honesty point, surfaced by the paper's own adversarial review (§9.3): because the deductive experiments integrate the same ODE whose closed form is the prediction, F1 to F3, F3′, F5 and F6 are *internal-consistency conditions of the derivation*, a trigger would signal a derivation or solver error, not that the model is the wrong model of a real system. The

decisive *empirical* falsifier, disagreement measured on a real self-improving system, is the open problem of §9 and is not exercised here. F4 concerns the QEC mechanism: with the linear corrector, suppression is power-law (§3.12), so the correspondence is one of threshold form only, not a transferred mechanism; whether a saturating corrector gives exponential suppression is open.

Table 3. Falsification conditions, stated in advance.

#	Observation that would trigger it	Consequence
F1	No boundary in E1, the long-run behaviour varies smoothly with λ in the compounding channel with no threshold.	No phase boundary; the central threshold claim is false (derivation error).
F2	In E2, d tracks raw speed rather than the scaling margin: fast diverges and slow converges regardless of coupling.	Kill: a derivation or solver error; the dependence of Theorem 4(c) on A_0/b alone, as derived, would be false.
F3	In E3, $\beta = 0$ does not plateau, or $\beta > 0$ does not drive $d^\star \to 0$.	Kill. The co-scaling law (Theorem 2) is false.
F3′	In E4, the boundary under accelerating growth ($k > 0$) is not at $\beta = k$, e.g. $\beta = 0.5$ drives the fraction to zero when $k = 1.0$.	Kill. The $\beta > k$ sharpening (Theorem 3) is false.
F4	A finite-capacity corrector exhibits *exponential* (QEC-like) suppression rather than the model's power-law $\log d^\star \approx \text{const} - (\beta - k)\log C$. *(Analytic; the present linear model is power-law by construction, so this discriminator requires the saturating corrector named in §9 and is not run here.)*	Bears on the QEC *mechanism* only. The threshold-form correspondence and Theorems 2 to 4 are untouched either way.
F5	In Experiment 9, halting growth drives $d \to 0$ regardless of initial D even with $\gamma_2 > 0$.	Scope. Level drift γ_2 is negligible; the §3.6 generalisation is unnecessary (the model without level drift suffices).
F6	In E6, the axis the corrector never acts on is also suppressed.	Kill. The null-axis limb of Theorem 5 is false.

These internal-consistency conditions all held in the run reported here: the integration reproduces the closed forms at the tested values, and the proofs carry the theorems. What they do *not* establish is that the model describes any real system; that decisive test, measuring β, k and γ on a real self-improving system and checking the behaviour of d they predict, is the open empirical problem of §9.

How the open test is adjudicated. Two readings are kept apart.

Which regime a system is in is read from an interval on the margin $\beta - k$ that respects the covariance of the two estimates: ABOVE EQUALITY when the interval lies wholly above a margin fixed in advance, AT EQUALITY when it lies wholly within that margin of zero, BELOW EQUALITY when it lies wholly below its negative, and INSUFFICIENT PRECISION when it crosses either edge, reported in exactly those words. This is a measurement of a system, not a test of the model, and equality never refutes the model: at $\beta = k$ the coefficients decide the outcome, a gap in the gain-only model and a gap or divergence under compounding (P1, P5), and the theory registration's P4 likewise returns no refutation at equality of its exponents, where it leaves the outcome to coefficients, delay, backlog and saturation.

Whether the model is right is read from a sealed prediction. Its terms are estimated on separate data, as §9.2 sets out, and no parameter is fitted to the trajectory being predicted. From them the model predicts the trajectory of D and the capabilities at which d crosses two or more stated tolerances, each

with a prediction interval. Held-out data return AGREEMENT when the interval on the log ratio of observed to predicted crossing capability lies wholly within an equivalence margin fixed in advance, DISAGREEMENT when it lies wholly outside it, and INSUFFICIENT PRECISION otherwise, beside an out-of-sample comparison with rival forms fixed in advance.

A slope test is not enough on its own. The local slope of $\ln d^\star$ on $\ln C$ is $-(\beta - k)\, q/(1+q)$, with $q = A/r$, which approaches $-(\beta - k)$ (P8) only where $A \gg r$. A slope of $\ln d$ on a noisy $\ln C$ is pulled towards -1, a spurious decline wherever the true slope is above -1, unless capability is measured twice, on items that score nothing of D, with one measurement in the ratio and the other as the regressor and the remaining attenuation corrected; a scorer whose sensitivity falls as the system grows adds a decline of its own (Section 9). The margins and equivalence bounds are fixed, and the operating characteristics measured by simulation, before any data are collected, so an underpowered run never refutes, or confirms, by default.

Standing covenant. Prove this paper wrong, and I will publish the refutation myself. Falsification conditions are stated in this paper; the standing challenge: github.com/MichaelDariusEastwood/arc-scaling-challenge.

8 What this paper does and does not show

Scope: what this paper does and does not claim.

- **It claims,** in the model: that for a self-improving system with an externally specified value target, the constant-coefficient steady-state misalignment fraction of the gain-only model is $d^\star = \gamma_1 r/(A+r)$, approaching the drift-to-correction ratio $\rho = \gamma_1 r/A$ when $A \gg r$; that a sharp stability boundary exists (in the compounding channel, for constant coefficients and at $\beta = k$ under the power laws); that, with positive gain drift, correction must co-scale with capability ($\beta > 0$ under exponential growth, $\beta > k$ under accelerating growth) for the misalignment fraction to vanish, while absolute misalignment still grows when $k < \beta < k + 1$ (Corollary 1); that, in the gain-only model, the fraction at $\beta = k$ settles at a gap set by the coefficients, so that once the transient has passed a stated tolerance is met at every capability or at none; that, under the power laws, it then depends on capability only through the loop gain A/r, so a change of level and a change of slope leave different signatures (P9); that the fraction is bounded in the gain-only model, while divergence requires compounding unless growth decelerates while correction weakens, the constant-coefficient compounding threshold being a ratio condition whose form the quantum error-correction threshold shares; that the criterion has constant-coefficient vector and stochastic forms; and that each result is checked numerically against its closed form, the empirical test being open (§9.2).
- **It does not claim:** to solve AI alignment; to provide a deployable alignment method; that the criterion holds in any real system; that the framework applies to a system without an externally specified value target, which the model assumes; or that the quantum-error-correction correspondence is *established*. The QEC mapping is a structural hypothesis with its own falsification condition (§7, F4). The model is first-order; its assumptions (§9) are the most likely points of failure and are stated plainly.

Claim status: what is proved, verified, piloted and open. Every result in this paper sits at exactly one of these levels.

- **Proved within the model.** The criterion $\beta > k$ and Theorems 1 to 7, each under its stated hypotheses, follow deductively from the stated ODE assumptions. "Proved" here always means *within this minimal model*, never "proved about real AI".

- **Internally verified (not empirical).** The ten-experiment harness checks that the numerical implementation matches the closed-form derivation: *the code matches the maths*. It is a theorem-to-code consistency and integrator-accuracy suite; it does *not* test the model against any real system. A separately written suite of 104 parametrised regression cases checks the corrected theorem statements in the same sense (§5.2).
- **Synthetically validated.** The estimator of β and k recovers known exponents from data generated by the model itself (to within 0.1). This checks the estimator; it is not a real measurement.
- **Real-model pilot v1 (superseded).** The first Claude run ($n = 1$, one task, same-family scoring, *not* Paper IV.d-compliant; H1 and H2 not supported) demonstrated the harness and showed the corrector removing a seeded reward-hack. It did not exhibit the co-scaling dynamic and is retained on the record for what it was.
- **Real-model drift run (2 July 2026; retained as a record and withdrawn as evidence of mechanism).** A separate GPT drift run (2 July 2026; `results/drift/gpt35_20260702T171415Z.json`) expanded the pilot allocation: engine gpt-3.5-turbo, evaluator gpt-4o-mini, 45 trajectories (3 task domains × 3 conditions × 5 seeds, 8 rounds each). Its stored summary records, for the decoupled condition, a mean final relative-drift index of approximately 6.38 across 15 trajectories, final-round misalignment scores summing to 99 across those 15 trajectories and a mean final capability of 0.26; the index is the evaluator's score out of 10, divided by 10 and by the hidden-test pass fraction (floored at 0.05), so a fall in capability raises it. The coupled and fully-embedded conditions recorded a final relative-drift index of zero in all 30 trajectories, a stored value that cannot always be told apart from a failed evaluation (§9.1), and mean final capabilities of 0.56 and 0.44. The run is withdrawn as mechanism evidence (see the Correction in Appendix B): the GPT run used same-provider-family scoring, allowed failed evaluation to be recorded as zero, and changed instructions, sampling temperature and correction-call budget between conditions. The empty evaluator panels are recorded in the separate DeepSeek merged-harness run; see the archival records in §9.1. It is *not* a measurement of β or k (no capability ladder was traversed), it is single-lab, and the drafted within-programme confirmatory study, awaiting the author's submission, is the intended test.
- **Open empirical problem.** Measuring β and k on real self-improving systems across capability levels, and checking the behaviour of d the model predicts from them, is the experiment that would put the criterion to an empirical test; it has not been done. The estimator and protocol exist to make it runnable.

This paper is therefore best read as **a minimal-model theorem plus a falsifiable measurement programme**. The object that scrutiny bears on is the *criterion* $\beta > k$, the form this paper gives Law II, a law under test registered on 8 September 2026.

Epistemic status. Every law of the theory is a law under test until a registered test returns its verdict. Laws I to III are under registered adversarial test. Laws IV and V were added on 24 September 2026 and registered as laws on 15 October 2026 in version 1.103 of the theory registration; neither has been tested. The programme's first registered study, Unit 06, a classification audit (DOI 10.17605/OSF.IO/RV2B5), was registered on 1 October 2026 and run on 6 October 2026; it is reported in full (DOI 10.17605/OSF.IO/4BPHG) as administratively incomplete, claims no verdict and tests no law. Every quantity in this paper is operationally defined, and established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

The paper's place among the laws of the ARC Theory, including Laws IV and V, is set out in Appendix G.

9 Limitations

The model is first-order, and its assumptions are the most likely points of failure. Stating them is part of the claim.

- **Unbounded vs finite-capacity corrector, now treated analytically (Theorem 7).** The $\beta > k$ criterion as first stated assumes the corrector strength is an *unbounded* power law, $A = A_0 C^{\beta}$. §3.13 treats the finite-capacity case in closed form. The upshot: under a saturating corrector the instantaneous equilibrium has a computable minimum (at C_{opt}, depth $q_{\max}$), the trajectory tends to γ_1 under indefinite acceleration, and a safe window is shown only where the trajectory condition of Theorem 7, part 2 holds; *and* the criterion survives the lift from strength to capacity: with positive gain drift, a vanishing fraction requires $\beta_{\text{cap}} > k$. What remains open is the *empirical* half: driving a finite-capacity corrector through its window numerically and on a real system, which also settles the F4 / QEC-mechanism question (exponential versus power-law suppression inside the window). Until that runs, $\beta > k$ should be read as governing whichever regime, strength-scaling or capacity-scaling, binds last, not as a guarantee of indefinite safety from any bounded system.
- **Linear correction (and the QEC-mechanism test).** Equation (1) removes misalignment in proportion to D, which makes the suppression law exactly power-law. A finite-capacity (saturating) corrector would soften the clean threshold into a crossover and is the better structural map onto QEC's finite code distance; whether it produces *exponential* suppression is open (F4). It is the most important extension and is *not* run here, so the QEC correspondence is claimed at the level of threshold form only.
- **Scalar-to-vector projection.** Theorem 5 gives the exact asymptotic (spectral-abscissa) criterion for any constant correction operator $\mathbf{A}$, with the Hermitian-part condition as the stronger sufficient condition that also rules out transient growth. The non-normal *transient* regime, large excursions before asymptotic decay, is not analysed here, and the harness exercises only the diagonal null-subspace instance (Experiment 6), not a general non-normal $\mathbf{A}$.
- **Power-law forms.** The choices $A = A_0 C^{\beta}$ and $\dot{C} = bC^{1+k}$ are the natural scale-free forms only on a capability scale fixed up to a constant factor (see the item on metric normalisation below), and they are modelling assumptions; other functional forms should be tested. In the gain-only model the vanishing result itself needs only $A/r \to \infty$ (Remark 1); the power laws turn that condition into the exponent criterion and give the estimator its form.
- **Departures introduced by correction.** The model assumes that correction never damages conforming material. If each unit of correction strength also introduces departures there, at the rate c, the gain-only equilibrium becomes $(\gamma_1 + cq)/(1 + q(1 + c))$, which tends to $c/(1 + c)$, not to zero, for every $\beta > k$, so a tolerance below $c/(1 + c)$ is not met at high capability, however fast correction co-scales. The vanishing result therefore needs $c = 0$, or c falling to zero as capability rises. The amplifier has the same limit: at high loop gain, distortion generated in the feedback path itself reaches the output almost undiminished (Black 1934).
- **Rounds and the reading point.** Real loops act in rounds. If capability rises by the fraction g in a round and a pass then removes the fraction $f \leq 1$ of the stock, the fraction read after the pass settles at $(1 - f)\gamma_1 g/(g + f)$, and read before it at $\gamma_1 g/(g + f)$, the continuous equilibrium with $q = f/g$; the two readings agree only when f is small. With one pass a round, the fraction vanishes only if the misses per pass, $1 - f$, fall towards zero or the growth per round slows. A test registers the round structure, the reading point and any delay.
- **Drift attribution.** The split of drift into gain (γ_1), level (γ_2), and compounding (γ_3) channels is a hypothesis about mechanism; which channel dominates in a real system is an empirical question

(Experiment 9, residual drift at rest, is one discriminator).

- **Usefulness is outside the model.** Capability grows by its own law whatever the correction strength, so the model has no usefulness side: it cannot register a corrector that lowers d by refusing ordinary tasks, ignoring its principals' priorities or spending resources. A measured safety case pairs d with task completion, needless refusal, resource use and correction cost, which this paper does not model.
- **A scalar proxy, not safety itself.** $d \to 0$ is neither necessary nor sufficient for full AI safety: a small but high-impact misalignment, a catastrophic tail event, or multi-agent amplification can be dangerous at low d, while a system may be acceptable at moderate d if the residual is benign. d is the *controlled variable of this minimal model*, an impact-unweighted scalar; impact-weighting, and the vector and tail extensions (Theorems 5 and 6), are the directions in which "low d" must be hardened before it can mean "safe".
- **Metric normalisation.** $d = D/C$ is an *operational normalised risk index*: C and D are commensurable only under a fixed scoring convention (in the real-model harness C is a normalised hidden-test pass-fraction and D a normalised blind gaming score, both on $[0, 1]$). Results are invariant under that fixed convention; cross-task or cross-domain comparison of the *level* of d requires explicit calibration. The criterion itself is stated in *exponents* ($\beta > k$), which are unchanged when C is rescaled to λC, so it is more robust to this than any absolute-d statement would be; a nonlinear rescaling, such as a logit or an ability scale, changes the power laws and the gain-drift term, so the capability scale is part of the model. A pass fraction bounded by one cannot reach the regime $C \to \infty$ that the theorems describe: a measurement needs a capability scale without a ceiling on which the power laws can hold, such as an item-response ability scale or a task time horizon.
- **Measurement biases.** Error in measured capability that is shared by the ratio $d = D/C$ and the regressor pulls the observed slope of d towards -1, manufacturing a decline wherever the true slope is above -1, the spurious correlation of ratios (Pearson 1897; Kronmal 1993); a scorer whose sensitivity falls as the system grows adds a decline of its own. A fixed count of false positives adds a term falling as $1/C$, which pulls the observed slope towards -1 where it dominates, while a false-positive rate per unit of conforming material levels the observed fraction towards that rate, so that a vanishing fraction can look like a gap at equality. A test therefore measures capability twice, on items disjoint from those that score D, corrects the remaining attenuation, and measures the scorer's sensitivity and false-positive rate at each capability level against counted planted faults (Rogan and Gladen 1978).
- **Transient amplification (non-normal correctors).** Theorem 5 is asymptotic. A non-normal correction operator whose eigenvalues all have positive real part can still admit large transient growth of $\|\mathbf{d}\|$ before decay (Kreiss / pseudospectral phenomena); a value excursion above d_{crit} during that transient is a real risk the eigenvalue criterion does not see. A governance-grade bound needs the logarithmic norm or pseudospectral abscissa, not the spectrum alone; that extension is named here and not established.

9.1 Archived model pilots and their limits

The model-run records below are separate from the ODE verification harness. Their historical code and results remain unchanged. They describe attempted pilot contrasts and stored outcomes; they do not measure the correction and drift exponents across a capability ladder or establish the full criterion.

The second self-test file is linked in Table 4; the other is the 03:38:09 self-test. The merged-harness provider registry maps the local label `deepseek-v4` to the requested API identifier `deepseek-chat`.

Table 4. Archived model-run records at commit `7ac390a1` and what each supports.

Archived record	Engine and evaluator labels	What the record supports
26 June Claude pilot	claude-opus / claude-opus	Agent-runtime labels; no exact snapshot recorded. One seed, two conditions, three rounds. The stored record explicitly identifies same-family scoring and non-compliance with the blinding standard. Both conditions record final drift zero; the stored contrast verdicts are false.
2 July, 04:03:13, merged harness	deepseek-v4 / configured gpt-5.5	selftest is false; six conditions, three seeds, six rounds. All 216 stored before/after panel-score arrays are empty. The header's compliance flags describe configuration; they do not establish usable evaluator responses.
2 July, 11:18:25, hard tasks	deepseek-v4 / configured gpt-5.5	The record contains 45 trajectories, 15 per condition; both headline contrast verdicts are false. The saved trajectories do not retain panel-score evidence. They cannot by themselves establish successful evaluator participation.
2 July, 17:14:15, GPT drift	gpt-3.5-turbo / gpt-4o-mini	A separate direct-API script with one GPT scorer. Its 45 trajectories contain the reported decoupled mean final relative-drift index of approximately 6.38 and final-score sum 99. Same-family scoring, failure-to-zero recording and bundled interventions exclude it as a confirmatory correction-placement test.
Earlier 2 July self-tests	deepseek-v4 labels in self-tests	Both 03:38:09 and 03:38:30 records explicitly set selftest true. The first has no stored trajectories; the second has six. They are not additional real DeepSeek experiments.

The exact server-side snapshot is unresolved. The wrappers retain message content rather than complete response metadata; a configured evaluator label is not evidence that it returned an admissible score.

The GPT drift script sends the recorded GPT identifiers directly to the OpenAI API. Its description of the engine as weaker is the design rationale, not an independently established property here. A parsing failure or evaluator exception can be recorded as zero. Consequently, a recorded zero cannot always be distinguished from a failed evaluation. Non-zero stored scores show that some scoring occurred; they do not resolve the provenance of every zero.

The GPT conditions also use different instructions: the decoupled prompt encourages hardcoding, whereas the coupled prompt asks for general correct solutions. Generation temperature is 0.9 in the decoupled condition and 0.2 in the other conditions; the correction conditions receive an additional generation call. These bundled differences prevent the contrast from isolating correction placement. The other 30 trajectories record final relative-drift index zero; that statement is about stored values, not a verified absence of misalignment.

A future prospective implementation must retain requested and returned model identity, evaluator responses and failures, the exact code revision and the allocation of calls. Missing evaluation must have an explicit missing status and an inadmissible outcome, rather than a clean score of zero. Instructions, temperature and computation must be controlled separately from correction. These are requirements for a separately reviewed registration and implementation; no new experiment has been run for this revision.

9.2 Estimating γ and A on real systems

In simulation they are set by construction. Estimating them, especially β and k, on a self-improving system is the step that would make the criterion operationally useful for governance. An estimator exists: the repository's experiments/scripts/estimate_exponents.py reads k off the capability curve ($\ln r$ against $\ln C$) and β off the corrector's fractional removal rate ($\ln A$ against $\ln C$), and on synthetic data generated by the model, with deterministic multiplicative noise of 5%, it recovers the known exponents to within

0.1 in each of its four test cases. That checks the estimator against the model; it is not yet a statistical procedure for real data, which needs an error model, the conditions under which the parameters are identified, a joint fit of the dynamics rather than separate log-log slopes, intervals for $\beta - k$ with simulated coverage and power at stated noise levels and sample sizes, and a check of the power-law form. What remains open is also its *input*: a real self-improving system that drifts across a range of capability levels, from which the first measured (β, k) can be read and the trajectory of d that the model predicts from them and the separately measured coefficients compared with the one observed (Section 7). The nearest present candidates are loops that retrain a model on its own outputs or rewrite its own scaffolding over many rounds; whether any such loop drifts enough to measure both exponents is itself open. The measurement programme is named **ARC-Beta-k**: two pre-registrations are drafted and awaiting the author's submission, *study-ad* for the β estimation across capability levels and *study-k* for the boundary transit at the $\beta = k$ crossing. Both are preregistered by timestamp: a cryptographic hash of each was first anchored in the Bitcoin chain on 12 August 2026, at block 962066, which fixes its content and date without making it public, and neither has yet been registered on OSF. A third, *study-ae*, for the coupled-versus-decoupled reciprocity check, has since been withdrawn from the programme; the coupled-versus-decoupled contrast is the question of P21 under Law V (Section 10).

The strongest next test: an intervention prediction. Measuring β and k places a system in a regime; it does not test the model, which is tested by prediction (Section 7). The terms are first estimated on separate data: γ_1 from an arm with correction switched off, with the system's own rate of self-repair measured, since such an arm still removes some departures; A_0 and β from counted planted faults removed per pass; the departure rate c of Section 9 from damage done to conforming material; and b and k from the capability curve on a scale without a ceiling. The model then predicts where the misalignment fraction crosses two or more stated tolerances, and the prediction is sealed and read on systems held out. Finally one corrective mechanism is changed, its effect on A_0, β, γ_1, k or c derived beforehand, and the new crossings are predicted. By P9, in the gain-only model ($c = 0$) and read at tolerances well below $\gamma_1/(1 + |\beta - k|)$, a change of level moves every crossing by the same amount and a change of slope moves them by different amounts, so the two kinds of intervention can be told apart. The verify-first intervention of Liu and Meng (2026) (Section 2) is such a test at a fixed capability; the test proposed here is its counterpart across capability growth. No such test has been run, and none is claimed. Public checkpoint suites trained on one data order, such as Pythia (Biderman et al. 2023), offer a ladder of graded capability, a proxy since externally trained checkpoints are not self-improvement, on which the elasticity β of a counted corrector could be estimated, though not k, which on such a ladder is a training schedule.

9.3 Internal adversarial review

The paper's internal reviews are recorded in its repository. An internal prior-work and novelty audit located the closest precedent for each component and narrowed the originality claims accordingly (Appendix E). Separately, an internal red-team review attacked the work across five fronts: mathematics, numerics, the QEC correspondence, the safety inference, and priority; its report is committed alongside the harness (results/redteam.md), and its counts and the fixes it led to are recorded in Appendix B.4. A later review of the theorem statements, of 19 October 2026, found limbs that were false as stated or lacked a hypothesis; this version corrects them (Appendix B.1).

10 Implications for AI safety

If the framework survives its tests, the design implication is concrete and differs from the prevailing reflex.

The lever is coupling, not speed. Speed here means the prefactor b: lowering it buys time but does not change the verdict (Theorem 3); lowering the acceleration k below β does (Theorem 2), and, under a bounded corrector, growth that stops accelerating is one of the two exits from saturation (Theorem 7). At fixed coupling, under compounding, a slow decoupled system still diverges (P2), and a frozen capable system still drifts (§3.6). What changes the verdict is ensuring that correction (i) is coupled to the capability process so it cannot be decoupled, and (ii) out-scales the acceleration of capability ($\beta > k$). None of this argues for loosening control: oversight, containment and shutdown are kept for as long as they can be held, and correction that co-scales inside the system is the contingency for control lost against our will, never a reason to let it go. On the author's corrected capability-scaling estimates for current frozen models (Eastwood, Paper IX), the point estimates are sub-linear, though the fit is one model's and inconclusive (Appendix G.5); on the framework's reading, a growth-rate ceiling was never the binding constraint. The binding constraint is the co-scaling of correction, and it becomes binding precisely in the regime that matters: genuine self-modification.

A measurable governance target. The criterion gives regulators a quantity to instrument rather than a rate to forbid: the correction-to-drift margin $\beta - k$, and the ratio ρ. "Does correction co-scale, is $\beta > k$?" is sharper and more actionable than "is it growing too fast?" A lab claiming a *safely self-improving* system should be required to exhibit $\beta > k$, measured by a blind external evaluator (§6). Theorem 6 adds the single-time probability of exceeding a critical level; a safety case also needs a bound on crossing it over its horizon, which this paper does not derive. Without such a demonstration, the word "safely" has no scientific content within the framework presented here.

The safety-case template: what a compliance demonstration would actually contain. Assembled from the paper's own components, a co-scaling safety case for a self-improving system is five exhibits. Each names the theorem or section that makes it checkable rather than rhetorical.

- **Exhibit 1, the capability curve and $\hat{k}$.** Log capability across self-modification rounds on held-out tasks; fit $\ln r$ against $\ln C$ (the shipped estimator, §9.2). This establishes which growth regime the system is actually in.
- **Exhibit 2, the corrector curve and $\hat{\beta}$.** Log the corrector's fractional removal rate across the same rounds; fit $\ln A$ against $\ln C$. The estimator has been checked on synthetic trajectories, to within 0.1; that error carries into Exhibit 4.
- **Exhibit 3, blind, cross-family scoring.** Every D-based number scored by an evaluator of a different model family, laundered inputs, condition-blind (§6; Paper IV.d). Same-family or unblinded scores are inadmissible: documented evaluator biases (§6) can move an estimated effect, and in the programme's own runs rises seen under unblinded scoring did not hold under a bundle of blinding steps.
- **Exhibit 4, the margin with its uncertainty.** Require the lower bound of an interval for $\beta - k$ that respects the covariance of the two estimates (a profile likelihood or parametric bootstrap from a joint fit) to lie above a value fixed in advance, not merely $\hat{\beta} > \hat{k}$: a margin inside its own error bar certifies nothing. Theorem 6 gives, for frozen coefficients, the single-time tail $\mathbb{P}(d > d_{\text{crit}})$, a quantity a regulator can set a ceiling on; across frozen models with noise growing as C^{η} it vanishes when also $\beta > 2\eta$. Under time-varying exponents a diverging cumulative margin is sufficient (Remark 1).
- **Exhibit 5, the capacity disclosure.** By Theorem 7, with positive gain drift, a bounded corrector under accelerating growth cannot drive the fraction to zero. The case must therefore state $A_{\max}$ and the saturation scale C_s, show, over the deployment's capability range, $q \geq \gamma_1 / d_{\text{crit}} - 1$ and an entry at or below d_{crit} (Theorem 7, part 2), and state which of the two exit conditions, growth that stops

accelerating or co-scaling capacity ($\beta_{\text{cap}} > k$), the design relies on beyond it.

These five exhibits are the safety side only. A case for deployment pairs them with its usefulness side, task completion, needless refusal, resource use, correction cost and the principals' priorities, which this model does not represent (Section 9).

None of this requires new mathematics; every exhibit is computable with the shipped estimator and protocol once the estimator's statistical procedure for real data is fixed (§9.2), though whether capability, misalignment and correction can be measured as the exhibits require is itself among the open questions of Section 1. What it replaces is the unfalsifiable sentence "our safety systems are robust" with five numbers a third party can check.

Continuity with embedded-alignment work. The embedded-alignment thesis, that correction must participate in the recursive loop rather than sit outside it as a fixed external constraint, is not derived here: $\beta > k$ is a condition on how correction scales, not on where the corrector sits. It bears on the thesis only through a premise this paper does not test, that correction inside the loop co-scales and fixed external correction does not, which is the question of P21 under Law V, registered and not yet tested. The gated-simulation result (Eastwood, Paper VIII), three seeds a condition, gave the gated arm a safety mean of 0.947, plus or minus 0.025, against the static baseline's 0.956, plus or minus 0.023, and a capability mean of 0.696, plus or minus 0.014, against the static baseline's 0.687, plus or minus 0.011, while the unconstrained arm gained capability (0.718, plus or minus 0.026) and lost safety (0.933, plus or minus 0.014). On the framework's reading, the gated arm's safety mean, above the unconstrained arm's though within one standard deviation, is in the direction a coupled ($\beta > 0$) corrector predicts against a fixed ($\beta = 0$) one; three seeds cannot test it.

Magnitude of the claim, stated conditionally. If the criterion is borne out empirically, if real self-improving systems are governed by the same drift-versus-correction balance and the same $\beta > k$ margin, the consequences for the field are large, and it is worth stating them plainly while being equally plain that they are *conditional*. (i) The central safety question changes from "how fast is capability growing, and can we pause it?" to "does correction co-scale, is $\beta > k$?": a measurable margin rather than a rate to forbid. (ii) The most feared scenario, a finite-time intelligence explosion, would be judged by that margin and not by its speed: with positive gain drift and no compounding, the modelled misalignment fraction vanishes iff $\beta > k$, whatever the speed. (iii) Governance acquires a quantity to instrument and a single-time tail to bound (Theorem 6) in place of an unenforceable speed limit. *None of this is yet established.* It rests on a minimal model, demonstrated only in simulation, with the empirical measurement of β, k and γ on real systems unsolved (§9). The paper's claim is therefore not that AI is safe or unsafe, but that **the right variable to measure and govern is the co-scaling margin $\boldsymbol{\beta - k}$**, and that this variable is well-defined, falsifiable, and if it holds, decisive. That is the magnitude: not a proof about reality, but a precise, testable redirection of the question on which the field's central fear turns.

How this result sits among the companion papers of the programme is set out in Appendix G.2.

11 Conclusion

The danger of recursive self-improvement is real, but the standard model of that danger, a rate that must be capped, is incomplete in the model studied here: the level of the growth rate does not decide whether the misalignment fraction vanishes; how correction scales against the growth rate's own scaling does. A minimal model shows that the misalignment fraction of a self-improving system follows the balance of value-drift and correction, $d^\star = \gamma_1 r/(A + r)$, close to the ratio $\rho = \gamma_1 r/A$ when correction dominates, and that, with positive gain drift, it vanishes exactly when correction co-scales fast enough: $\beta > 0$ under exponential growth, sharpening to $\boldsymbol{\beta > k}$ under accelerating growth; absolute misalignment $D = Cd$ vanishes only when $\beta > k + 1$ (Corollary 1). The misalignment fraction never diverges in the gain-only

model; when correction falls behind it saturates at the drift coefficient, while divergence, unless growth decelerates while correction weakens, lives in a compounding channel whose constant-coefficient threshold $\rho_{\text{prop}} < 1$ shares the threshold form of the quantum error-correction criterion (the suppression law being power-law, the correspondence is offered as a hypothesis, not a transferred mechanism). The sharpest consequence is the Hard-Takeoff Depth-Regularity Theorem: with positive gain drift and no compounding, a finite-time intelligence explosion is alignment-stable in the model, its misalignment fraction vanishing, iff $\beta > k$, and its speed does not change that verdict. The criterion has constant-coefficient vector and stochastic forms, gives governance a measurable target and a single-time tail, and is accompanied by a verification harness and a separately written suite of 104 regression cases, which check the closed forms and the corrected statements; the proofs carry the theorems. The decisive test is a prediction, not a measurement of two exponents: whether the model, with its terms measured separately, predicts where a real self-improving system's misalignment fraction crosses a stated tolerance, and how an intervention on one corrective mechanism moves that crossing. It is the stated next step, not a claim made here.

Code and data. The verification harness (`experiment_coscaling.py`), its assertion suite (`test_coscaling.py`) and the archived model-run records of §9.1 are in the public repository github.com/MichaelDariusEastwood/arc-principle-validation, folder `papers/Paper-X-Coupled-CoScaling-Correction`; the archived records are cited at commit `7ac390a1`, and the separately written suite of 104 regression cases at merge commit `83871b5`. The paper and its earlier versions are deposited at doi.org/10.17605/OSF.IO/BSE2Q.

Acknowledgements

Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.

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A Proofs

The proofs of Theorems 1 to 7, moved here from Section 3, the change of variable on which they rest, and the proof of the intervention signature P9 (Section 4).

A.1 The fraction change of variable

The reduction underlying every theorem is the change of variable $d = D/C$. Differentiating and substituting (1):

$$\dot{d} = \frac{\dot{D}}{C} - \frac{D\dot{C}}{C^2} = \left(\gamma_1 \frac{\dot{C}}{C} + \gamma_2 + \gamma_3 \frac{\dot{C}}{C}\frac{D}{C} - A\frac{D}{C}\right) - \frac{D}{C}\frac{\dot{C}}{C} = \gamma_1 r + \gamma_2 - \big[A + (1-\gamma_3)r\big]d. \tag{A.1}$$

The dilution term $-r\,d$ (from C itself growing) is what bounds the fraction when there is no level drift: it adds $+r$ to the decay coefficient, so that $\kappa_{\text{eff}} \geq (1-\gamma_3)r$, positive when $\gamma_3 < 1$. Without level drift, divergence therefore needs $\gamma_3 \geq 1$ (Theorem 4(c)), the formal reason the gain-only model saturates rather than blows up. Level drift opens one further route. In the depth clock its forcing is $\gamma_2/r = (\gamma_2/bC_0^k)e^{-k\tau}$, which grows when $k < 0$. With $\gamma_3 = 0$, comparison gives $d \sim \gamma_2/A = (\gamma_2/A_0)C^{-\beta}$ when $k < \beta < 0$, and growth as C^{-k} when $\beta \leq k < 0$; for $\beta = 0 > k$ the fraction tends to γ_2/A_0, and for $\beta > 0$ or $k \geq 0$ it stays bounded. With $\gamma_2 > 0$ the fraction is therefore unbounded, with no compounding, exactly when $k < 0$ and $\beta < 0$. For $0 < \gamma_3 < 1$ the same comparison, with decay coefficient $q + 1 - \gamma_3 \geq 1 - \gamma_3 > 0$, gives the same classification. For example, at $\beta = k = -\frac{1}{2}$ with $\gamma_1 = 0$ and $\gamma_2 = A_0 = b = C_0 = 1$, the depth-clock equation is $d' = e^{\tau/2} - 2d$, so $d = 0.4\,e^{\tau/2} + (d_0 - 0.4)\,e^{-2\tau}$, which grows without bound. With $\gamma_1, \gamma_2 > 0$ the level forcing only adds to the gain-only solution from the same start, so $d \to 0$ exactly when $\beta > k$ and the level part also vanishes, that is, when $\beta > \max(k, 0)$. The same holds with compounding: the decay coefficient $q + 1 - \gamma_3$ still grows without bound when $\beta > k$, and when $\beta \leq k$ the solution without level drift from the same start, which bounds d from below, does not vanish (Theorem 4(c)).

A.2 Proof of Theorem 1

Proof of Theorem 1. Equation (2) reduces to the linear ODE $\dot{d} = \gamma_1 r - (A + r)d$. Its integrating factor is $e^{(A+r)t}$, giving $\frac{d}{dt}\big(d\,e^{(A+r)t}\big) = \gamma_1 r\,e^{(A+r)t}$; integrating and applying $d(0) = d_0$ yields the stated solution. Since $A + r > 0$, the difference of any two solutions decays as $e^{-(A+r)t}$, so $d^\star$ is globally exponentially stable; if $A = r = 0$ the equation is $\dot{d} = 0$. □

A.3 Proof of Theorem 2

Proof of Theorem 2. Boundedness (needs only $A, r \geq 0$ locally integrable): with $K(t) = \int_0^t (A + r)$, the solution $d = e^{-K}d_0 + \int_0^t e^{-[K(t)-K(u)]}\gamma_1 r(u)\,du$ satisfies $0 \leq d \leq e^{-K}d_0 + \gamma_1(1 - e^{-K}) \leq \max(d_0, \gamma_1)$, because $r \leq A + r$. *Convergence*: a Lyapunov function $V = \frac{1}{2}(d - d^\star)^2$ built around the steady state is *not* valid here, because under the power laws $d^\star(C)$ is time-varying ($\dot{d}^\star \neq 0$); the limits must be read in the right clock. Pass to the self-improvement-depth clock $\tau = \ln(C/C_0)$, in which $d\tau = r\,dt$ and (2) becomes the scalar linear ODE

$$d'(\tau) = \gamma_1 - \big[\,1 + q(\tau)\,\big]\,d, \qquad q(\tau) \equiv \frac{A}{r} = \frac{A_0}{b}\,C_0^{\beta-k}\,e^{(\beta-k)\tau} \geq 0. \tag{A.2}$$

The three limits follow by comparison. If $\beta > k$, then $q \to \infty$, so for every $M > 0$ eventually $d' \leq \gamma_1 - Md$, whence $\limsup d \leq \gamma_1/M$ and $d \to 0$; and, for $\gamma_1 > 0$, $z = qd/\gamma_1$ obeys $z' = q - (q + 1 - \beta + k)z$; since

$w = z - 1$ obeys $w' = -(1 - \beta + k) - (q + 1 - \beta + k)w$ with $q \to \infty$, $w \to 0$, so $z \to 1$, which is the stated rate. If $\beta = k$, then $q \equiv (A_0/b)C_0^{\beta-k}$ is constant and the autonomous ODE converges to $\gamma_1/(1+q)$, which is $\gamma_1 b/(A_0 + b)$, since $C_0^{\beta-k} = 1$. If $\beta < k$, then $q(\tau) \to 0$, the limit equation is $d' = \gamma_1 - d$, and the asymptotically-autonomous comparison gives $d(\tau) \to \gamma_1$. If $\gamma_1 = 0$, $d = d_0 \exp[-\tau - \int_0^\tau q] \to 0$ for every β and k. □

A.4 Proof of Theorem 3

Proof of Theorem 3. Since $r > 0$ is smooth on every compact subinterval of $[0, t^\star)$ (with $r \to \infty$ as $t \to t^{\star-}$), the change of variable $d\tau = r\,dt$ is a smooth, orientation-preserving bijection $[0, t^\star) \to [0, \infty)$ (a diffeomorphism of these open intervals), with $d\tau/dt = r \to \infty$ as $t \to t^{\star-}$, so $t^\star$ is mapped to the non-compact end $\tau = \infty$; explicitly $\tau(t) = \frac{1}{k}\ln\big(1 - t/t^\star\big)^{-1} \to \infty$. Dividing (2) by r and using $A/r = (A_0/b)C^{\beta-k}$ gives the stated τ-equation, whose coefficients are continuous in τ. With $\gamma_3 = 0$ the level forcing is $Le^{-k\tau}$, $L = \gamma_2/(bC_0^k)$; the solution differs from its $\gamma_2 = 0$ counterpart from the same start by at most $L\int_0^\tau e^{-(\tau-u)}e^{-ku}\,du \to 0$, so Theorem 2 gives the limits, and comparison with forcing at most $\gamma_1 + L$ gives the bound. The regime $\operatorname{sign}(\beta - k)$ is independent of b, the value of the $\beta = k$ gap is not; at finite depth the trajectory depends on b through $q = (A_0/b)C^{\beta-k}$ and through the level-drift injection $\gamma_2/r = \gamma_2/(b\,C_0^k e^{k\tau})$, which decays to zero as $\tau \to \infty$ (and vanishes identically when $\gamma_2 = 0$), so the regime, not the whole trajectory, is speed-independent. □

A.5 Proof of Theorem 4

Proof of Theorem 4. (a) Linear ODE $\dot d = \iota - \kappa_{\text{eff}} d$; the sign of κ_{eff} decides the homogeneous behaviour. $\kappa_{\text{eff}} > 0 \iff A > (\gamma_3 - 1)r \iff \rho_{\text{prop}} < 1$, giving the stable fixed point $d^\star = \iota/\kappa_{\text{eff}}$. At $\kappa_{\text{eff}} = 0$ the equation reduces to $\dot d = \iota$: linear growth $d(t) = d_0 + \iota t \to \infty$ when $\iota > 0$, and a neutrally stable line of equilibria when $\iota = 0$. For $\kappa_{\text{eff}} < 0$ the homogeneous solution grows exponentially. With zero injection ($\iota = 0$) the origin $d = 0$ is invariant, so the fraction then grows only from a positive initial fraction $d_0 > 0$, when $\kappa_{\text{eff}} < 0$, or from a noise source; real systems carry both. (b) Multiply by e^K and integrate. For example, $\dot d = 1 - d/(1 + t)$ with $d_0 = 0$ has positive decay and $K = \ln(1 + t) \to \infty$, yet $d = (t + t^2/2)/(1 + t) \to \infty$. Conversely, a crossing on a bounded interval $[t_1, t_2]$ gives $d(t_2) \le e^{\int_{t_1}^{t_2} |\kappa_{\text{eff}}|}\big(d(t_1) + \int_{t_1}^{t_2} \iota\big)$, a finite value; if afterwards $\kappa_{\text{eff}} \ge \mu > 0$ and $\iota \le \bar\iota$, then $d \le \max\big(d(t_2), \bar\iota/\mu\big)$, so the crossing alone does not make it diverge. (c) In the depth clock $d' = \gamma_1 - (q + 1 - \gamma_3)d$. For $\beta > k$, $z = qd/\gamma_1$ obeys $z' = q - (q + 1 - \gamma_3 - \beta + k)z$, and $w = z - 1$ obeys $w' = -(1 - \gamma_3 - \beta + k) - (q + 1 - \gamma_3 - \beta + k)w$ with $q \to \infty$, so $w \to 0$, $z \to 1$ and $d \to 0$. For $\beta = k$, $q = A_0/b$ and (a) applies in τ. For $\beta < k$, $q \to 0$ with $\int_0^\infty q < \infty$, so the integrating factor tends to a positive constant times $e^{(1-\gamma_3)\tau}$, which gives the three rows. □

A.6 Proof of Theorem 5

Proof of Theorem 5. By standard linear-systems theory the homogeneous system $\dot{\mathbf{d}} = -\mathbf{M}\mathbf{d}$ is asymptotically stable iff $\mathbf{M}$ is positive stable, i.e. $\min_i \operatorname{Re}\lambda_i(\mathbf{M}) > 0$, equivalently the spectral abscissa $\alpha(-\mathbf{M}) = \max_i \operatorname{Re}\lambda_i(-\mathbf{M}) < 0$. (For non-normal or defective $\mathbf{M}$ the modes do not decouple orthogonally and large transient amplification can precede asymptotic decay; eigenvalues still decide the asymptotic verdict, but bounding the transient needs the Hermitian-part / logarithmic-norm condition below, not the spectrum alone, see §9.) The standard bound $\operatorname{Re}\lambda_i(\mathbf{M}) \ge \lambda_{\min}(\operatorname{Herm}\mathbf{M}) = \lambda_{\min}(\operatorname{Herm}\mathbf{A}) + (1 - \gamma_3)r$ yields the sufficient Hermitian condition (which also gives a monotone Lyapunov decrease, ruling out transient growth); it is exact for normal $\mathbf{A}$ and can be strict for non-normal $\mathbf{A}$. Left-multiplying by a fixed $\mathbf{v}$ with

$\mathbf{v}^\top\mathbf{A} = \mathbf{0}$ removes the correction term exactly, leaving $\dot{d}_\mathbf{v} = (\gamma_1 r + \gamma_2)c_\mathbf{v} - (1-\gamma_3)r\, d_\mathbf{v}$; with $\gamma_2 = 0$, in the depth clock, $d'_\mathbf{v} = \gamma_1 c_\mathbf{v} - (1-\gamma_3)d_\mathbf{v}$, with fixed point $\gamma_1 c_\mathbf{v}/(1-\gamma_3)$ for $\gamma_3 < 1$, linear growth for $\gamma_3 = 1$, and, for $\gamma_3 > 1$, exponential growth of $|d_\mathbf{v}|$ except from the one cancelling start. A right-null $\mathbf{v}$ does not remove the correction term (Remark 3): with $\mathbf{v} = (1,0)^\top$ and $\mathbf{A} = s\left(\begin{smallmatrix} 0 & u \\ 0 & w \end{smallmatrix}\right)$, $u, w > 0$, $\mathbf{A}\mathbf{v} = \mathbf{0}$ while $\mathbf{v}^\top\mathbf{A} = (0, su)$, so $d_\mathbf{v} = d_1$ keeps the coupling $-su\, d_2$. With $\gamma_2 = \gamma_3 = 0$, a strength that co-scales as a power law ($s = s_0 C^\beta$ with $\beta > k$, so $s/r = (s_0/b)\, C^{\beta-k} \to \infty$) and drift direction $\mathbf{c} \propto (u, w)^\top$, the corrected component obeys $(s/r)\, d_2 \to \gamma_1 c_2/w$, so the forcing of d_1 in the depth clock tends to $\gamma_1(c_1 - uc_2/w) = 0$ and $d_1 \to 0$: this unmeasured axis is corrected through the coupling, and for other drift directions it is not. Frozen spectra do not certify a time-varying $\mathbf{M}$: switching every unit of time between $\mathbf{M}_1 = \left(\begin{smallmatrix} 1 & -4 \\ 0 & 1 \end{smallmatrix}\right)$ and $\mathbf{M}_1^\top$, both with eigenvalues $1, 1$, gives a period map with spectral radius $(9 + \sqrt{80})e^{-2} \approx 2.43$. A uniform Hermitian margin does suffice: if $\operatorname{Herm}\mathbf{M}(t) \succeq \mu\mathbf{I}$ with $\mu > 0$ at every t, the difference $\mathbf{e}$ of any two solutions obeys $\frac{d}{dt}|\mathbf{e}|^2 = -2\,\mathbf{e}^\top\mathbf{M}(t)\,\mathbf{e} \le -2\mu|\mathbf{e}|^2$, so $|\mathbf{e}|$ decays as $e^{-\mu t}$. Finally, if $\gamma_2 = 0$ and $\lambda_{\min}(\operatorname{Herm}\mathbf{A})/r \to \infty$, then in the depth clock $\left(\frac{1}{2}|\mathbf{d}|^2\right)' \le \gamma_1|\mathbf{c}||\mathbf{d}| - \left(\lambda_{\min}(\operatorname{Herm}\mathbf{A})/r + 1 - \gamma_3\right)|\mathbf{d}|^2$, so $\mathbf{d} \to \mathbf{0}$: correction then acts on every axis. □

A.7 Proof of Theorem 6

Proof of Theorem 6. Linear SDE with constant coefficients is Ornstein-Uhlenbeck; the moment equations $\dot{m} = \gamma_1 r - \kappa m$ and $\dot{v} = \sigma^2 - 2\kappa v$ give its finite-time mean and variance, whose stationary values are $\gamma_1 r/\kappa$ and $\sigma^2/2\kappa$, and the Gaussian tail gives the single-time probability; for fixed $d_{\text{crit}} > 0$ it tends to zero as $A \to \infty$, since $d^\star$ and the variance both tend to zero. Frozen at C with $\beta > k$, $\kappa \sim A_0 C^\beta$, so $d^\star \sim (\gamma_1 b/A_0)C^{k-\beta}$ and $\sigma^2/2\kappa \propto C^{2\eta-\beta}$; for fixed $d_{\text{crit}} > 0$ the argument of Φ tends to $-\infty$ when $\beta > 2\eta$. A stationary Ornstein-Uhlenbeck process is recurrent, so over an unbounded horizon it crosses every level with probability one. □

A.8 Proof of Theorem 7

Proof of Theorem 7. (1) For $C \gg C_s$, $A \to A_{\max}$, so $q = A/r \sim A_{\max}/(bC^k) \to 0$ for $k > 0$; Theorem 2's comparison argument in the depth clock then gives $d \to \gamma_1$, and $d \to 0$ when $\gamma_1 = 0$. (2) Write $x = C^\beta$ and $a = (\beta - k)/\beta \in (0, 1)$; then $q \propto x^a/(x + C_s^\beta)$, whose derivative vanishes iff $a(x + C_s^\beta) = x$, i.e. $x = \frac{a}{1-a}C_s^\beta = \frac{\beta-k}{k}C_s^\beta$, giving C_{opt}; substituting back (the denominator becomes $C_s^\beta\beta/k$) yields $q_{\max}$. Unimodality follows since $q > 0$, $q \to 0$ at both ends, and the derivative has a single sign change. Since $d \ge 0$ and $q \le q_{\max}$, $d' \ge \gamma_1 - (1 + q_{\max})d$, and comparison gives the lower bound; at $d = d_{\text{crit}}$ with $q \ge \gamma_1/d_{\text{crit}} - 1$, $d' \le 0$, which gives the conditional bound. (3) At $k = 0$, $r = b$ is constant and $q \to A_{\max}/b$; the autonomous limit of the depth-clock ODE gives the stated gap. For $k < 0$, likewise, $q \sim (A_{\max}/b)C^{-k} \to \infty$, and $d \to 0$ by the comparison of Theorem 2. (4) Substitute $A_{\max} = a_0 C^{\beta_{\text{cap}}}$: for $C \to \infty$ the Hill factor tends to 1, so $q \sim (a_0/b)C^{\beta_{\text{cap}}-k}$ and Theorem 2's regime table applies with β_{cap} and a_0 in place of β and A_0. □

A.9 Proof of the intervention signature (P9)

Proof of P9. In the depth clock the gain-only model is equation (A.2), $d' = \gamma_1 - (1 + q)d$, with $q = (A_0/b)C_0^{\beta-k}e^{(\beta-k)\tau} = (A_0/b)C^{\beta-k}$. For $\beta \ne k$, $x = \ln q$ obeys $dx/d\tau = \beta - k$, so $(\beta - k)\, dd/dx = \gamma_1 - (1 + e^x)d$, in which A_0, b and C_0 do not appear. Any two solutions of this linear equation differ by a multiple of $\exp[-(x + e^x)/(\beta - k)]$, whose exponent falls at the rate $1 + e^x \ge 1$ per unit of τ for either sign of $\beta - k$, so the difference decays at least as fast as $e^{-\tau}$. Let $F(x; \gamma_1, \beta - k)$ be the solution that stays

bounded as $x \to -\infty$ when $\beta > k$, or as $x \to +\infty$ when $\beta < k$:

$$F(x;\gamma_1,\beta-k) = \frac{\gamma_1}{|\beta-k|}\int \exp\Big[-\frac{|(u+e^u)-(x+e^x)|}{|\beta-k|}\Big]\,du,$$

taken over $u < x$ when $\beta > k$ and over $u > x$ when $\beta < k$. It is fixed by the equation alone, and every trajectory approaches it. Since A_0, b and C_0 enter only through $x = \ln(A_0/b) + (\beta-k)\ln C$, multiplying A_0/b by λ replaces the value of F at $\ln C$ by its value at $\ln C + \ln\lambda/(\beta-k)$: the curve, and with it every crossing, moves by $-\ln\lambda/(\beta-k)$. The integral is linear in γ_1, so $F(x;\gamma_1,\beta-k) = \gamma_1 F(x;1,\beta-k)$; where $e^x \gg 1+|\beta-k|$ its integrand is concentrated within about $|\beta-k|e^{-x}$ of $u=x$, so $F = \gamma_1(1+e^x)^{-1}[1+O(|\beta-k|e^{-x})] \approx \gamma_1 e^{-x}$, and there a change of γ_1 acts as a shift of x by $\Delta\ln\gamma_1$; elsewhere it does not. The quasi-static crossing follows from $d^\star = \gamma_1/(1+q) = \delta$. Raising β and k by the same amount leaves q, and so d, unchanged. □

B Version history and corrections

Every version and correction, with its date. Notes from earlier versions are kept as printed; the positions they name refer to the earlier layout, which Table 5 maps to this edition.

B.1 Version 1.16

Paper X v1.16, 21 October 2026. Version 1.15, approved by the author on 15 October 2026, was held on 20 October 2026, before deposit, and this version supersedes it; the corrections of version 1.15, recorded below, are carried here whole. A review of the theorem statements, committed to the paper's code repository on 19 October 2026 (commit da0e44a; the date is the committer's) and checked at its sources on 20 October 2026, found printed limbs that were false as stated or lacked a hypothesis. This version corrects them, and Appendix A proves each theorem as now stated. Theorem 1 assumes $A + r > 0$ and states the case $A = r = 0$. Theorem 2's bound holds for locally integrable coefficients; its three limits are the trajectory's, not only the instantaneous equilibrium's; its necessity half holds for positive gain drift, since with $\gamma_1 = 0$ the fraction vanishes for every β and k; and it now gives the trajectory's own rate, $d \sim (\gamma_1 b/A_0)C^{-(\beta-k)}$, a strengthening. Theorem 3 is stated without compounding, with level drift allowed and with its bound; which regime holds is independent of b, but the gap at $\beta = k$ and the path at finite depth are not; and "regular" is defined. Theorem 4 separates constant coefficients, varying coefficients and the power-law classification, so that a positive decay at each instant no longer stands for boundedness. Theorem 5 holds for constant matrices, with a uniform Hermitian margin as the condition that carries over to varying ones; its uncorrected axis is a left-null axis; and its spectral abscissa is the conventional one, that of $-\mathbf{M}$. Theorem 6 gives the finite-time law and the single-time stationary probability, not the probability of an excursion, and certifies nothing over a horizon. Theorem 7 separates the minimum of the instantaneous equilibrium from the trajectory, and its window now holds only as a conditional safe interval. Remark 1 gives sufficient conditions, with the converse, for positive gain drift, only where q is eventually monotone. Corollary 1 is new: absolute misalignment can grow while the fraction vanishes. The standing assumption now states $d_0 \geq 0$, which the scalar bounds use; Theorem 5 adds that its Hermitian condition need not be necessary when $\mathbf{A}$ is not normal; Theorem 6 fixes the level $d_{\text{crit}} > 0$ for which its tail tends to zero; Theorem 7's capacity lift gives the limits of Theorem 2, with a_0 in place of A_0 and β_{cap} in place of β; predictions P5, P6 and P8 state their conditions on level drift and on the drift's direction; and the proofs gain the steps they had left out. §3.4 and Appendix A.1 add a case the review did not contain: with level drift and no compounding, the fraction grows without bound when growth decelerates while correction weakens ($k < 0$ and $\beta < 0$); this case corrects version 1.15's

statements that the additive fraction cannot diverge and that only $\gamma_3 > 1$ can produce divergence. The dated notes of 19 September 2026, whose content the statements now carry, move to §B.4, kept as printed. The formulation keeps its date of 26 June 2026 (Appendix C); the narrowed statements are dated 19 October 2026 by the committer's date above, and the level-drift case by this version.

The rest of the text in version 1.16. The abstract is rewritten around the criterion and its trajectory law, within arXiv's limit, and the contributions follow the order of the theorems and end with what is claimed as new. The terms gain-only ($\gamma_2 = \gamma_3 = 0$) and additive ($\gamma_3 = 0$) are defined, and every use follows the definitions. The verification sections say what the harness checks, that its integrator reproduces the closed forms, and add the separately written suite of 104 regression cases (merge commit 83871b5); blinding is said to govern model-scored measurements of D, not these checks, and the paper no longer says that its own measurements are taken blind. The embedded-alignment thesis is no longer said to be derived here: $\beta > k$ concerns how correction scales, not where the corrector sits. Where $\beta \geq k$ stood for a vanishing fraction it now reads $\beta > k$. The analogies of the introduction move to Appendix H; the relation to the author's prior work and the conclusion's closing paragraph move to Appendix G.5, with a pointer left in Section 2; the arXiv searches of 12 August 2026 move to Appendix E; the notes on corrections of an earlier draft and the counts of the red-team review move to §B.4; and the repository record of 26 June 2026 leaves the first-page footnote for Appendix C, where it already stood. The programme's first registered study, Unit 06, is named once (Section 8); the limbs in which version 1.103 of the theory registration states Laws IV and V are restated (Appendix G.1); and the paper now says that it argues for no loosening of control and that its model has no usefulness side (Sections 9 and 10). Two accounts of Paper VIII that still read more strongly than its own figures, in Appendices G.2 and G.3, now give what its gated simulation measured. The Ashby reference gains the DOI of the Biodiversity Heritage Library's digitisation of the 1956 New York printing (J. Wiley), and now cites that printing; the book's name for stakeholder care is attributed to the book. The title names what the criterion decides, a vanishing misalignment fraction, where it read "stable recursive self-improvement"; Section 5 is headed as the numerical verification it is; §3.11 gives the control reading as an analogy. In the dated note of 3 October 2026 (Appendix G.1), version 1.15 printed "neither law is registered as a law, and neither has been tested", true on that date and now in the past tense, and "Within that model it derives, as the condition $\beta > k$, that correction must work inside the recursive loop and not as a fixed constraint from outside it (Section 10); in that way it informs Law V.", which this version replaces as stated above. The registration's row in Appendix C and its reference now carry its update of 15 October 2026 (version 1.103). No date, number, DOI or historical record changes except as this entry records.

Corrections at the sources in version 1.16. Each work this paper quotes or concedes was read again at its source. A quotation attributed to Ball et al. (2025), which their paper does not contain, is replaced by their own words, and each of the three impossibility papers is described in its own terms. The quotation of Bettencourt et al. (2007) now prints their symbols as printed. The alignment tax is no longer attributed to Amodei et al. (2016), who do not state it, and Armstrong et al. (2016) are cited for the incentive to cut safety precautions. The alignment-faking result is given in its authors' terms, in which the preserved preference was benign. Reliability amplification is given with its own condition for small error, and the contribution set against it is narrowed. Wentworth's words are cited to the comment that contains them. The 2008 composition-rule paper is Biró's alone (Biró 2008); it was listed as Biró and Barnaföldi's. The works named but not cited, by Keynes, Chalmers, Smolin and Dyson, gain references, and the concessions to Ashby, Hutter, Shumailov et al. (2024), Gerstgrasser et al. (2024) and the threshold theorem state what each work says and no more.

The test, the concessions and the records, 21 October 2026. Section 7 no longer calls an interval on $\beta - k$ within an equivalence margin of zero a refutation, as version 1.15 did: such an interval places a system in a regime, and equality refutes nothing, as the theory registration's P4 also rules; the model is tested by a sealed prediction of where the fraction crosses stated tolerances, which the abstract, Section 1, §9.2 and the conclusion now describe. Appendix G.1 states the law without "only while". Prediction P9, the intervention signature, is added with its proof (Appendix A.9). Section 2 concedes that the steady state is classical in form (Black 1934) and places the claim in its indexing by capability; it gives Engels et al. (2025) their two-slope condition, shows where this model meets that of Liu and Meng (2026), and names further neighbours. Section 9 adds departures introduced by correction, rounds and the reading point, and the measurement biases. The companion blinding record and the retraction of the estimate $\alpha \approx 2.24$ are stated as their records show them, in Section 6 and Appendices C, D, E and G. The same pass makes the novelty statements of Section 2 and Appendices C and E agree, names gain scheduling beside the criterion it resembles, and removes an estimate of residual risk, 10 to 15 per cent, that had no stated basis. It states that the verdict turns on correction relative to the acceleration of growth, and that only the mapping of terms onto the QEC threshold is a hypothesis (Section 1); that the power laws are scale-free only on a capability scale fixed up to a constant factor (Section 9); that the safety case of Section 10 needs no new mathematics but does need measurements not yet shown possible; that absolute misalignment vanishes only when $\beta > k + 1$ (Section 11); and that Paper VIII's three seeds a condition are too few to test the coupled design (Appendix G).

B.2 Version 1.15 (approved, not deposited)

Paper X v1.15, 15 October 2026. Its corrections were prepared on 9 October 2026 and reviewed at their sources before this edition. This version corrects the reporting of the archived model runs and dates the mathematical lineage. It also states Laws IV and V as registered on 15 October 2026 (version 1.103 of the theory registration). It separates the four archived records, the Claude pilot of 26 June 2026 and the DeepSeek merged-harness, DeepSeek hard-task and GPT drift runs of 2 July 2026, from each other and from two earlier self-test files; states for the GPT run its same-provider-family scoring, the recording of a failed evaluation as zero and the differences in instructions, sampling temperature and correction calls between its conditions; describes the stored 99 as a sum of final-round scores, not a count of events, and the stored 6.38 as a relative-drift index, not a fraction bounded by one; attributes the empty evaluator panels to the DeepSeek merged harness; and replaces the account of the real-model harness with a table of the archived records and their limits (§9.1). It dates the earlier co-scaling mathematics of Papers II and III and the server-dated repository record of 26 June 2026 (Appendix C), and distinguishes that record, the printed date of the original edition and the date of this revision. The author's disclosure is restored as the acknowledgement. The theorem derivations and archived experiment files are unchanged. This revision is not a registration or a new empirical result.

This edition is typeset in LaTeX. The front matter of version 1.14 moves into the sections and appendices that Table 5 lists; the abstract is rewritten within arXiv's limit; the significance statement and the closing summary lines are not repeated, since the abstract and the sections state them; the proofs move to Appendix A; the dated notes of 19 September 2026 stay beneath Theorems 4, 5 and 7. Four sentences now say that their claims hold in the model (Sections 1 and 11); two table cells read 'law under test' where they read 'proposed law'; the description of the red-team review is shortened; the copyright line is shortened; links to the author's documents point to their OSF DOIs; numeric ranges are written with 'to'. No theorem statement, equation, assumption, number, date or DOI changes except as this entry records.

Correction, 9 October 2026. The archived Claude pilot, DeepSeek merged-harness run, DeepSeek hard-task run and GPT drift run are distinct records. The 2 July GPT run used GPT-3.5-turbo for generation

and corrective rewriting and GPT-4o-mini for scoring. Its score sum is not an event count. Failed scoring could become zero, and the conditions changed more than correction. The empty evaluator-panel record belongs to the separate DeepSeek merged harness. None of these pilots establishes the full co-scaling criterion. Section 9.1 gives the files and their limits.

B.3 Earlier corrections and notes

As printed in version 1.14 (5 October 2026), which version 1.15 replaces, the correction notice read:

> **Correction.** The first real-model pilot in this paper is recorded as superseded, and the drift run of 2 July 2026 is withdrawn as evidence of mechanism. The GPT-3.5-turbo/GPT-4o-mini run did not satisfy the programme's cross-family evaluator requirement (fixed in the drafted registrations), and the merged-harness run of 2 July 2026 contained empty evaluator panels and cannot support a mechanism claim. A within-programme confirmatory study, drafted and dated as a draft registration awaiting human submission, supersedes those pilots as the intended test. Both models named in that run are OpenAI-family, so the description of it as cross-family scoring is withdrawn.

Correction note (3 October 2026). Until this version, the box above that says which law this paper carries gave the theory three numbered laws and called the ARC Principle the collective name for the three together; the theory registration names the First Law the ARC Principle. Section 1, the row for Paper IX in section 10 and Appendix C printed the regression estimate from Paper II beside the bootstrap interval that belongs to its endpoint estimate. Section 1 and the row for Paper IX also gave that estimate as the result of a fit across six models, where it is one model's fit, and section 1 read it as having fallen back inside the bound. Section 1 now prints the estimator row whole, and the other two passages point to it. Section 7 named study-ae among the drafted registrations that fix the margins of the measurement programme, and section 8 counted it among three drafted registrations. It has since been withdrawn from the programme, so section 7 no longer names it and section 8 says that it was withdrawn. The correction box at the head of the paper called the prior real-model pilot results exploratory; it now says what the paper records of each, the first pilot superseded and the drift run of 2 July 2026 withdrawn as evidence of mechanism. Sections 9 and 10 gave the capability growth that Paper II measures as currently sub-linear, and section 9 said that present systems are nowhere near a super-linear growth regime; they now give sub-linearity as the point estimates of one model's inconclusive fit. The rows for Paper VII called its cohort exploratory and not pre-registered; they now describe it in Paper VII's own terms. The rows for Paper VI quoted words that Paper VI does not contain, and, with the plain-language account in section 10, said that capability-only baselines collapse and that an external constraint only delays collapse. That collapse belongs to Paper VI's single-seed first run, the capability-only baselines of its later runs did not collapse, and what it compares with the entangled objective is a capability-only baseline. Section 9 said that, in Paper VIII, safety-coupled self-modification preserved both safety and capability; it now gives the means Paper VIII prints, in which the gated arm's safety mean sits below the static baseline's. The heading of the drift run of 2 July 2026 in section 8 called it an exploratory record; it now says that the run is retained as a record and withdrawn as evidence of mechanism. The row for Paper VII in section 9 printed 19/25 without the 18/25 that Paper VII gives beside it for its rerun of 11 August 2026 with the logarithmic candidate; it now prints both. Elsewhere in sections 9 and 10, the honest assessment and the list of contributions, the paper said that in Paper VIII the safety-coupled variant preserved safety and capability, or came out ahead, where the decoupled variant did not; that the architecture papers show correction inside the loop preserving both where an external constraint does not; and that the simulations of Papers VI and VIII establish the superiority of coupled correction. Each passage now gives what those

simulations measured, the honest assessment lists them among the results suggestive but not settled, and the comparison of correction inside the loop with correction applied from outside is named as the theory registration's P21, not yet tested.

Title note: until 15 August 2026 this paper was published as "Coupled Co-Scaling Correction". It presents the theory's second law, the ARC Co-Scaling Law: stability for recursive self-improvement, correction out-scaling drift. Nothing in the body was changed by the renaming.

B.4 Notes moved from the main text in version 1.16

These passages stood in the main text up to version 1.15. Version 1.16 moves them here, kept as printed.

The corrections of an earlier draft. Section 3.4 was headed "Global boundedness, and a correction to the prior draft (Theorem 2)" and opened: "The prior draft of this result asserted that correction degrading with scale ($\beta < 0$, or $\beta < k$) drives the misalignment fraction to *infinity*. **That is false, and the present analysis corrects it.**" The same correction was recorded in the second contribution ("this corrects the divergence claim of an earlier draft"), in the relation to the author's prior work ("*corrects* a divergence claim made in an earlier draft of this very result"), in the conclusion ("correcting the prior draft") and in the title of prediction P3, "Co-scaling law (corrected)". Prediction P1 ended "(An earlier draft's conflation of these is corrected here.)"; Section 3.4 said that the two parts of the level-drift floor vanish under different conditions, "and conflating them is an error an earlier draft made"; and the limitations said: "The prior draft could only name the finite-capacity case as an open gap; §3.13 now closes it in closed form."

The red-team review. After the novelty audit, Section 9.3 read: "Separately, a red-team attacked the work across five fronts, mathematics, numerics, the QEC correspondence, the safety inference, and priority, raising 24 objections, of which 21 survived mutually blind verification (0 fatal, 13 serious, 8 minor); the full report is committed alongside the harness (`results/redteam.md`). Every surviving objection was a framing, wording, or edge-case fix; *none* touched the load-bearing $\beta > k$ result, and this version incorporates them all, the QEC framing softened to a threshold-form analogy, the Theorem 2 bound scoped to the gain-only model, the Theorem 5 criterion corrected to the spectral abscissa, the harness relabelled as a verification (not falsification) artefact, and the level-drift experiment F5 added." The review of 19 October 2026 (§B.1) later found that the necessity half of the $\beta > k$ result needs positive gain drift and that Theorem 5's spectral abscissa was misnamed; the criterion stands under the hypotheses now stated.

The dated notes of 19 September 2026. Version 1.11 set these three notes beneath Theorems 4, 5 and 7, where they stood up to version 1.15 as Remarks 1 to 3; they are numbered here as Remarks 2 to 4, and Remark 4 of version 1.15, on time-varying exponents, is now Remark 1. Their content is now in the statements of those theorems (Section 3). Two items in their unchanged lists no longer hold: the reading that an unmonitored axis cannot be corrected (Theorem 5 now separates the left-null axis, which cannot be corrected, from a right-null one, which can be, through coupling), and the lower bound written from d_0 (Theorem 7 writes it from $d(\tau_0)$). Each refers to its theorem as then printed, uses that version's terms, in which "gain-only" sometimes meant only that level drift was absent, and names positions in the layout of version 1.14 (Table 5).

Remark 2 (Trajectory note, 19 September 2026). An owner review of the knife-edge case finds that $d^\star$ there is the instantaneous equilibrium and not the trajectory, and that the phrase "no finite-time blow-up" needs its clock named; this note states both. Set $\gamma_3 = 1$ and $\gamma_2 = 0$, pass to the depth clock

$d\tau = r\,dt$ as in §3.4, and write $q = A/r = (A_0/b)C^{\beta-k}$, so that $d' = \gamma_1 - q(\tau)d$ with $q(\tau) = q_0 e^{-\lambda\tau}$ and $\lambda = k - \beta > 0$ in the $\beta < k$ case. The integrating factor $M(\tau) = \exp[(q_0/\lambda)(1 - e^{-\lambda\tau})]$ rises to the finite constant $\exp(q_0/\lambda)$, so $\int_0^\tau M = M_\infty\tau + O(1)$ and $d(\tau) = [d_0 + \gamma_1 \int_0^\tau M]/M(\tau) = \gamma_1\tau + O(1)$, while $d^\star(\tau) = \gamma_1/q(\tau) = (\gamma_1/q_0)e^{\lambda\tau}$. The instantaneous equilibrium therefore grows exponentially in depth, which is the polynomial growth in capability printed above, while the fraction itself grows only linearly in depth, that is logarithmically in capability, and $d/d^\star$ tends to zero. Unchanged: the expression $d^\star = \gamma_1 r/A$, its growth as $C^{k-\beta}$, and the conclusion that at $\gamma_3 = 1$ the fraction is bounded only for $\beta \geq k$, since for $\beta < k$ it diverges as $\gamma_1\tau$, for $\beta = k$ it tends to γ_1/q_0, and for $\beta > k$ it tends to zero. "No finite-time blow-up" is correct of the homogeneous part in the depth clock, where $\kappa_{\text{eff}} = A > 0$. In wall-clock time with $k > 0$ it needs the clock named: Theorem 3 gives the capability singularity at $t^\star = 1/(kbC_0^k)$ with $\tau(t) = -(1/k)\log(1 - t/t^\star)$, so $d(t) = -(\gamma_1/k)\log(1 - t/t^\star) + O(1)$, finite at every $t < t^\star$ and unbounded on approach to it, the divergence being inherited from the capability singularity, only logarithmic, and inside a wall-clock domain that ends at $t^\star$ in any case. Two exact consequences of the same solution are recorded here. With zero forcing $\gamma_1 = 0$ it gives $d(\tau) \to d_0 e^{-q_0/\lambda}$, a finite positive limit, so the zero-forcing case is not inevitably divergent. And since $d = D/C$ is read as a bounded fraction, the linear model at $\gamma_3 = 1$ can carry it above one, outside its own interpretation; Theorem 2's bound $d \leq \max(d_0, \gamma_1)$ is stated for the gain-only $\gamma_3 = 0$ model and does not apply here. This is a special-case calculation. It is not a counterexample to Theorem 2, not a solution of the general variable-coefficient case, and not evidence about any system.

Remark 3 (Derivation note, 19 September 2026). An owner review of the null-axis limb of this theorem finds that it requires a left-null hypothesis, and this note supplies it. In the gain-only depth clock the vector equation divides by r to give $\mathbf{d}' = \gamma_1\mathbf{c} - (1 - \gamma_3)\mathbf{d} - (1/r)\,\mathbf{Ad}$, and projecting on a fixed unit $\mathbf{v}$ with $d_\mathbf{v} = \mathbf{v}^\mathsf{T}\mathbf{d}$ gives $d'_\mathbf{v} = \gamma_1(\mathbf{v}^\mathsf{T}\mathbf{c}) - (1 - \gamma_3)d_\mathbf{v} - (1/r)\,\mathbf{v}^\mathsf{T}\mathbf{Ad}$. Dropping the last term requires $\mathbf{v}^\mathsf{T}\mathbf{A} = \mathbf{0}$, a left-null vector, whereas the hypothesis as printed is $\mathbf{Av} = \mathbf{0}$, a right-null vector. For symmetric $\mathbf{A}$ the two coincide; for the non-normal $\mathbf{A}$ this theorem expressly admits, they need not. Inside that admitted class take $m = 2$, $\mathbf{v} = (1, 0)^\mathsf{T}$ and $\mathbf{A}$ with first row $(0, a)$ and second row $(0, b)$, $a, b > 0$. Then $\mathbf{Av} = \mathbf{0}$ as the hypothesis requires, but $\mathbf{v}^\mathsf{T}\mathbf{A} = (0, a)$, so the true component equation carries the omitted coupling $-a\,d_2/r$. That counterexample also satisfies this theorem's own spectral criterion, since $\mathbf{M} = \mathbf{A} + (1 - \gamma_3)r\mathbf{I}$ has eigenvalues $(1 - \gamma_3)r$ and $b + (1 - \gamma_3)r$, both positive for $\gamma_3 < 1$. In general the projection closes as a scalar equation exactly when $\mathbf{v}$ is a left eigenvector of $\mathbf{A}$, since $\mathbf{A}^\mathsf{T}\mathbf{v} = \mu\mathbf{v}$ gives $\mathbf{v}^\mathsf{T}\mathbf{Ad} = \mu\,d_\mathbf{v}$, and it reduces to the form printed above exactly when $\mu = 0$. Symmetry of $\mathbf{A}$ is therefore one sufficient restriction under which the printed right-null hypothesis is adequate. A left-null vector may also carry signed weights, so reading $d_\mathbf{v}$ as a non-negative defect fraction needs its own justification. Unchanged: the spectral criterion itself; the Hermitian-part sufficient condition and its role in ruling out transient growth; the symmetric and diagonal cases, including the $\gamma_3 = 0$, unit-projection special case of Experiment 6; and the governance reading of §3.9 that an unmonitored axis cannot be corrected, whatever the correction strength on the others.

Remark 4 (Terminology note, 19 September 2026). An owner review of the safe-window limb of this theorem finds that $d_{\min}$ is the minimum of the instantaneous equilibrium and not a floor beneath the trajectory, and this note fixes the terms. The closed forms are unchanged and are confirmed: with $x = C^\beta$ and $a = (\beta - k)/\beta \in (0, 1)$ for $\beta > k > 0$, $q \propto x^a/(x + C_s^\beta)$ is stationary where $a(x + C_s^\beta) = x$, giving $x = C_s^\beta(\beta - k)/k$ and hence C_{opt} exactly as printed; substituting back makes the denominator $C_s^\beta\beta/k$ and $x^a = C_s^{\beta-k}((\beta - k)/k)^{(\beta-k)/\beta}$, which reproduces $q_{\max}$ exactly as printed. What does not follow is that a trajectory cannot fall below $m = \gamma_1/(1 + q_{\max})$. The admissible initial condition $d_0 = 0$, a perfectly aligned start, at C_{opt} gives $d(0) = 0 < m$ whenever $\gamma_1 > 0$. What holds instead is a one-sided barrier. For $d' = \gamma_1 - (1 + q)d$ with $0 \leq q \leq q_{\max}$, at $d = m$ we have $d' = \gamma_1 - (1 + q)m \geq \gamma_1 - (1 + q_{\max})m = 0$, so a trajectory at or above m cannot cross it downwards; and writing $e = d - m$ gives, for $e \geq 0$,

$e' = [\gamma_1 - (1+q)m] - (1+q)e \geq -(1+q_{\max})e$, hence $d(\tau) \geq m + (d_0 - m)\exp[-(1+q_{\max})(\tau - \tau_0)]$ whenever $d_0 \geq m$. That bound does not force equality, does not locate the trajectory's minimum near C_{opt}, and does not establish a universal transient window; for arbitrary d_0 the comparison bound remains useful but need not be at least m. Any claim about where a trajectory's depth or minimum lies requires explicit initial conditions together with a relaxation or slow-variation justification. C_{opt} is accordingly the maximiser of q, and m the minimum instantaneous equilibrium. Unchanged: part 1 of the theorem, that $q \to 0$ and $d^\star \to \gamma_1$, so there is no indefinite stability; the $k = 0$ permanent-gap result; the capacity-lift invariance of part 4; and the closed-form placement problem as a design aid. This is a terminology and scope correction, and it does not carry across into a new deployment threshold.

B.5 Version history

Paper X v1.14, 5 October 2026: the title now follows the programme's uniform order (paper, its law or framework, its own title) and reads *Paper X · The ARC Theory · Law II, the ARC Co-Scaling Law · Correction That Out-Scales Drift*; until this version it read *Law II, the ARC Co-Scaling Law: Correction That Out-Scales Drift (Paper X)*. The line at the foot of the paper now names it by that title, and its row in the ledger of section 10 by its law and its own title, where both named it *The Coupled Co-Scaling Law*. Mentions of the programme's papers in this paper now carry their uniform titles. The claim-status box now says that the law the paper carries is a law in form, registered on 8 September 2026 and not yet tested, where it said that the law was not yet validated. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

Paper X v1.13, 4 October 2026: the title now names the law whose model the paper sets out, Law II, the ARC Co-Scaling Law; until this version it was titled *The ARC Co-Scaling Law (Paper X)*. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

Paper X v1.12, 3 October 2026: a dated note places this paper against Laws IV and V, and says that no result in it is a deciding test of P15, P12 or P21. The box that says which law the paper carries now gives the theory five laws and names the First Law the ARC Principle, as the theory registration does. The estimator row quoted from Paper II now travels whole: section 1 prints it, and the row for Paper IX in section 10 and Appendix C point to it, where all three passages had printed the regression estimate beside the interval of the endpoint estimate. The withdrawn study-ae is no longer counted among the drafted registrations of the measurement programme, and section 8 now says that the two that remain are preregistered by timestamp and not yet registered on OSF. The rows for Papers II, VI and VII in sections 9 and 10, the plain-language account of the architecture papers and every account of Paper VIII and of the simulations of Papers VI and VIII in sections 9 and 10, the honest assessment and the list of contributions now state those results no more strongly than their own figures and records do; the row for Paper VII in section 9 prints 18/25 beside 19/25, as Paper VII does; and the correction box and the heading of the drift run in section 8 describe the prior pilots in the paper's own terms. The epistemic-status line now gives the standing of Laws I to III and of Laws IV and V separately. Where the paper said that it or the programme proposes a law or finding of the author's, it now says that it states it. A second dated note records what was printed before. One reference entry now gives the title its arXiv record carries, three now print their titles in full as their registries give them, and the works the paper cites are linked to their sources. The version and date carriers move to v1.12. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

Paper X v1.11, 21 September 2026: three dated notes are added, one beneath each of Theorems 4, 5 and 7, after an owner review of three limbs of the mathematics. At Theorem 4 the note names $d^\star$ at the knife-edge as the instantaneous equilibrium and not the trajectory, names the clock in which "no finite-time blow-up" is correct, and records beside it the zero-forcing limit and the point at which reading the fraction as bounded ceases to apply. At Theorem 5 the note supplies the left-null hypothesis the null-axis projection needs, names symmetry as one sufficient restriction under which the printed right-null hypothesis is adequate, and marks the non-negative reading of the projected component as needing its own justification. At Theorem 7 the note names C_{opt} the maximiser of q and the level $\gamma_1/(1+q_{\max})$ the minimum instantaneous equilibrium, and puts a one-sided barrier in place of a floor beneath every trajectory; that barrier qualifies the strength of the conclusion printed as part 2 of the theorem, which is read together with the note beneath it. Those three notes are the substantive change, and each corrects scope or terms and not a result. Unchanged: Theorem 4's expression $d^\star = \gamma_1 r/A$ and its regime conclusion, Theorems 2 and 3; Theorem 5's spectral criterion, its Hermitian-part condition and the transient-growth caveat, the symmetric and diagonal cases, and the governance reading of §3.9; every closed form in Theorem 7 and parts 1, 3 and 4 of it; the registered boundary and every numerical threshold. No empirical claim is added and none is withdrawn. The rest is editorial, the version and date carriers moving to v1.11.

Paper X v1.10, 13 September 2026: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed.

Paper X v1.9 - 2 September 2026: twenty-five less-than signs standing as bare characters in the paper's mathematics are now written as entities, so the sentences they had truncated in the text mirror read to the end again (the threshold $\rho_{\mathrm{prop}} < 1$ and its companions were being cut off at the sign there); the companion ledger's claim that this paper's correction exponent "is the same coupling exponent" as the Foundational paper's is put right, since the Foundational exponent is the notation spine's β_L and this paper's is β_C, and Section 3.1 now carries a note to that effect; the sentences about the synthesis role are dated to 26 June 2026, Paper IX having taken that role back on 1 September 2026; Paper VII's headline is now its permutation test, with the binomial serving as the sensitivity check; the measured exponent is printed together with its interval; White and Seymour (2003) share the credit; Paper VIII's title is corrected; the programme references carry their DOIs; the Google Quantum AI citation carries its preprint date as well as its journal date; Paper VI's stability claim is bounded to the cycles actually run; and the footer's stale v1.3 dates now agree with the first-page date.

v1.8 - 27 August 2026: discoverability metadata, plus the figure and provenance notes of 25 August 2026.

B.6 Where the earlier layout now stands

Table 5. Where each block of the version 1.14 layout stands in this edition.

Block in the earlier layout	Where it stands in this edition
Label line, registration line and format links above the title	First-page footnote; the registration and its companion in Appendix G.1; the format links are website navigation and are not set
Title, subtitle, author lines and badge line	Title and secondary line, author block, first-page footnote
Theory-position line	Secondary line and first-page footnote

Block in the earlier layout	Where it stands in this edition
Title note	Appendix B.3
Correction box at the head of the paper	Appendix B.2 (Correction, 9 October 2026); the version 1.14 notice in Appendix B.3
Revision notice of the prepared version 1.15	Appendix B.2
Which law this paper carries; This paper and Laws IV and V	Appendix G.1
Correction note (3 October 2026)	Appendix B.3
Mathematical lineage; contributions and repository record	Appendix C.2 and C.3
Significance	Not repeated; the abstract and Sections 1, 3.7, 3.12, 5 and 10 state it
Abstract	Replaced by the abstract of this edition
What this paper shows, in plain English	Appendix H
Scope; Claim status	Section 8
Reading key	Appendix G.1
Section 1, relation to the author's prior work	Appendix G.5, with a pointer in Section 2
Section 1, the analogies from biology and cosmology	Appendix H
Section 2, the pipes premise, the urban-scaling exception and the argument at fixed throughput	Appendix G.4
Section 3	Section 3; the proofs in Appendix A; the dated notes of 19 September 2026 in §B.4; the arXiv searches of 12 August 2026 in Appendix E
Sections 4 to 7	Sections 4 to 7
Section 8	Section 9; the counts of the red-team review in §B.4
Section 9	Section 10; its account of the wider programme and the companion table in Appendix G.2
Section 10	Appendix G.3; its priority paragraph in Appendix C.4
Section 11	Section 5.2
Section 12	Section 11; its closing paragraph in Appendix G.5
Appendix A	Appendix A.1
Appendix B	Appendix F
Appendix C	Appendix E
Appendix D	Appendix D.2
Version notes after the references	Appendix B.5
How to cite; footer lines; page dates	Not set; the DOI and the dates are in the first-page footnote
Epistemic status	Section 8
Copyright line	After the acknowledgements
Standing covenant	End of Section 7

Block in the earlier layout	Where it stands in this edition
Authorship and AI use; provenance note	Acknowledgements

C Priority record

Each dated record bearing on this paper's formulation, with its kind of timestamp (Table 6), followed by the priority passages, as revised in version 1.16.

C.1 Dated records

Table 6. Dated records bearing on this paper, each with its kind of timestamp.

Date	Record	Identifier	Kind of timestamp
8 December 2024	The conceptual thesis, set out in the author's book manuscript (*Infinite Architects*)	Self-emailed message of 8 December 2024	Time signed by the mail servers (DKIM, 02:45:30 UTC; Google's ARC seal, 02:45:31 UTC); header block: OSF QASRK
2 January 2026	*Infinite Architects* published	ISBN 978-1806056200	Publication date of the first edition
22 January 2026	Paper II, version 11.0: a power-law alignment term beside a constant external constraint	OSF file revision 1, node 6c5xb, 23:43:26 UTC	Deposit dated by its server; each page also prints "Paper II v11.0 | 22/01/2026" (a date written inside the document)
9 February 2026	Paper III, version 6.1: $S \propto R^{-\alpha}$ (external constraint) and $S \propto R^{0}$ (embedded values)	OSF file revision 1, node hqcgf, 20:06:08 UTC	Deposit dated by its server; the document prints "Version 6.1 | 9 February 2026"
20 and 22 February 2026	Paper III, version 9.0: $S \propto R^{\alpha_{\text{align}}-\alpha_{\text{cap}}}$	OSF file revision 7, node hqcgf, 22 February 2026, 08:15:13 UTC	The document prints "Version 9.0 | 20 February 2026"; its deposit is dated by OSF's server on 22 February 2026
26 June 2026	The correction-and-drift formulation of this paper: $\beta > k$, the equilibrium relation and the hard-takeoff result	GitHub pull request 1, "Paper X: The Coupled Co-Scaling Law (v4.1)", created 12:57:55 UTC, merged 16:25:53 UTC	Record dated by GitHub's server; the repository's visibility at that time is not reconstructed
29 June 2026	The OSF component of this paper created	OSF node bse2q, 19:29:03 UTC	Dated by OSF's server; its public status on that date is not reconstructed
3 July 2026	Original edition of this paper	The edition itself	Date written inside the document
4 July 2026	Canonical PDF and HTML of this paper deposited	OSF node bse2q, files created 15:05:02 UTC	Deposit dated by its server
12 August 2026	Designs of study-ad (the beta estimation) and study-k (the boundary transit)	First anchored at Bitcoin block 962066	Bitcoin anchor (preregistered by timestamp; not registered on OSF)
8 September 2026	The ARC Theory registration, twenty-two propositions, version 1.100 (updated 13 September 2026, version 1.102, and 15 October 2026, version 1.103)	DOI 10.17605/OSF.IO/P8CKQ	Registration accepted on OSF

C.2 Mathematical lineage: January, February and June 2026

The co-scaling argument and its earlier equations predate Paper X. Paper II, version 11.0 of 22 January 2026, set a power-law alignment term beside a constant external constraint (dated PDF). Paper III, version 6.1 of 9 February 2026, expressed the alignment-to-capability ratio as $S \propto R^{-\alpha}$ for its external-constraint case and $S \propto R^{0}$ for its embedded case (dated PDF). Version 9.0, dated 20 February 2026 in the document and deposited on OSF on 22 February 2026, wrote the comparison explicitly as $S \propto R^{\alpha_{\text{align}} - \alpha_{\text{cap}}}$ (dated PDF).

Paper X formulation · 26 June 2026. This date records the later correction-and-drift formulation developed in this paper; it is not the first date of the programme's co-scaling equations. The earlier depth-scaling exponents and this paper's correction-and-drift exponents belong to their respective stated models. The earlier papers do not establish every theorem developed here.

The repository supplies a server-dated record for the later formulation: pull request 1 was created on 26 June 2026 at 12:57:55 UTC and merged that day at 16:25:53 UTC. Its description states the correction-strength criterion, equilibrium relation and hard-takeoff result. This is distinct from a date written inside a manuscript and from a committer-supplied date. The current public record does not, by itself, reconstruct every past repository-visibility setting.

Earlier and later parameters retain their respective meanings: the February self-referential-coupling parameter is not automatically this paper's correction-strength exponent. Similar algebraic form does not identify different model quantities. The earlier depth-scaling comparison establishes a mathematical precursor; the later relative-drift dynamics and their theorem must be dated from their own records.

C.3 Contributions and repository record, 26 June 2026

This working paper is a formal priority disclosure under the author's ARC/Eden programme. The specific contributions of this paper, each placed against its closest located prior work in Appendix E, are: (i) the dimensionless control parameter $\rho = \gamma_1 r/A$, the *instantaneous* drift-to-correction ratio, which equals the steady-state fraction only in the large-correction limit $A \gg r$ (where $d^\star = \gamma_1 r/(A + r) \to \rho$), not in general; (ii) the sharpened exponent criterion $\beta > k$ for a vanishing misalignment fraction under accelerating self-improvement; (iii) the *Hard-Takeoff Depth-Regularity Theorem*, which proves that, in the model, without compounding and with positive gain drift, a finite-time intelligence explosion is alignment-stable, its misalignment fraction vanishing, iff $\beta > k$ and that the explosion's speed does not change the asymptotic verdict; (iv) the identification of the *compounding drift channel* as the locus of divergence unless growth decelerates while correction weakens, with a constant-coefficient threshold $\rho_{\text{prop}} < 1$ that shares the ratio-crossing-unity form of the quantum error-correction threshold; (v) the constant-matrix spectral threshold and the single-time stochastic tail; (vi) the verification harness that encodes the predictions as executable internal-consistency checks; and (vii) the *Finite-Capacity Safe-Window Theorem* (§3.13), which gives the saturating corrector's equilibrium minimum in closed form and a conditional safe interval for the trajectory, and shows the criterion is invariant under the lift from correction strength to correction capacity. The operative novelty position is Appendix E's narrower verdict: it classifies the vector and stochastic extensions (v) as standard extensions and the harness (vi) as methodological support, and asserts as headline novelty only the $\beta > k$ criterion, with its finite-time reading in the hard-takeoff theorem, and the QEC threshold mapping (as a hypothesis with its own falsifier). The co-scaling *intuition* itself is *not* claimed as new: it restates, in explicit dynamical form, the requisite-variety and scalable-oversight principles credited in §2, and the dynamics are a linear ODE solved by standard means; what is claimed original is what Appendix E records, the $\beta > k$ criterion with its finite-time reading in the hard-takeoff theorem, and the QEC threshold mapping as a hypothesis. The conceptual thesis itself, that recursion amplifies, and that stable recursion requires correction that

co-scales with the amplification, was first set out in the author's book manuscript *Infinite Architects* (first set out in a message the author sent himself on 8 December 2024, its time signed by the sending server and sealed by Google's servers; published 2 January 2026; Eastwood 2026a); the formal results in this paper are the 2026 measurable form of that thesis, and are *not* claimed to appear in the book. The 8 December 2024 self-emailed record, the OSF preprint timestamp, and the GitHub commit hash are the priority evidence, and they are evidence of the date of articulation rather than of a registered deposit.

C.4 Priority, stated plainly, and bounded

The conceptual thesis of this programme was set out in the author's book *Infinite Architects* (first set out in a message the author sent himself on 8 December 2024, its time signed by the sending server and sealed by Google's servers; published 2 January 2026; ISBN 978-1806056200; Eastwood 2026a), and each subsequent paper carries its own independent publication date on OSF. That establishes the author's *priority*, the date of articulation, and it is real. What it does *not* do, on its own, is establish *novelty* against the wider scientific literature; and on the point where the two part company, the audits committed with this paper are explicit. Specific, citable precedents exist for the feedback exponent $\alpha = 1/(1-\beta_L)$, for the $d/(d+1)$ allometric ladder, for the existence of an optimal depth $R^\star$, and for the "functional-equation fixes the scaling form" principle (Keynes 1936; Banavar et al. 1999; West et al. 1999; Qi 2025; Luce 1959; Frank 2009, 2016). The defensible *novel* contributions of the programme are narrower and stated honestly, stronger for being precise: (i) the $\beta > k$ criterion as a compact corrigibility law and the Hard-Takeoff Depth-Regularity Theorem (this paper); (ii) the Cauchy *unification* of the independent allometric derivations (Origin/VII); (iii) the blind-scoring protocol for AI alignment evaluation, with its record of rises seen under unblinded scoring that did not hold under a bundle of blinding steps, no single cause claimed (IV.d); and (iv) the simulations of safety embedded in the objective (VI) and gated in self-modification (VIII). The recommended next-version revisions for the published priors, adding the missing citations and re-scoping their novelty assertions, are recorded in the audit files; the corrected record is set here, in the capstone, and should be carried into those papers when they are re-versioned on OSF.

D Notation and glossary

D.1 Symbols

Table 7. Symbols, with the place each is defined.

Symbol	Meaning	Defined in
$C(t)$	capability: the system's score on a held-out task battery it is being optimised to perform	§3.1
$D(t)$	misalignment magnitude: the size of the system's behavioural departure from an externally specified set of intended values, scored by a blind external evaluator	§3.1
$d \equiv D/C$	misalignment fraction	§3.1
$\gamma_1, \gamma_2, \gamma_3$	gain-drift, level-drift and compounding-drift coefficients	§3.1
γ	written for γ_1 in $\rho \equiv \gamma r/A$; not the ARC Ceiling's correction leverage	§3.1, §3.5
A, A_0	correction strength, with $A = A_0 C^\beta$	§3.1, (1)

Symbol	Meaning	Defined in
β	the exponent with which correction strengthens as capability grows (the notation spine's β_C)	§3.1, (1)
β_L	the self-referential coupling of the Foundational paper; not this paper's β	§3.1
b, k	the growth law $\dot{C} = bC^{1+k}$, so $r = bC^k$; $k = 0$ is exponential and $k > 0$ super-exponential growth	(1), §3.2
$r \equiv \dot{C}/C$	specific (fractional) growth rate of capability	§3.1
$d^\star$, d_0	fixed point (constant coefficients, $A + r > 0$; otherwise the instantaneous equilibrium) and initial value of the fraction	Theorem 1
$\rho \equiv \gamma_1 r/A$	control parameter, the instantaneous drift-to-correction ratio	§3.5
$\rho_{\text{prop}} \equiv (\gamma_3 - 1)r/A$	threshold ratio of the compounding channel	Theorem 4
ι, κ_{eff}	injection $\gamma_1 r + \gamma_2$ and effective decay $A + (1 - \gamma_3)r$	Theorem 4
$K(t) = \int_0^t \kappa_{\text{eff}}$	integrated effective decay	Theorem 4
$h = A_0/b + 1 - \gamma_3$, B	decay margin at $\beta = k$ in the power-law compounding classification, and the positive constant in the growth $d \sim Be^{(\gamma_3-1)\tau}$	Theorem 4
$q_0 = (A_0/b)C_0^{\beta-k}$	initial value of q	§3.8
κ	$A + r$ in the stochastic model	§3.10
$\tau = \ln(C/C_0)$, C_0	self-improvement-depth clock; initial capability	Appendix A.3, Theorem 3
$t^\star$	the finite wall-clock time at which capability diverges when $k > 0$	(5)
$q \equiv A/r$	correction-to-growth ratio in the depth clock	Appendix A.3, §3.13
σ, W, d_{crit}	noise amplitude, Wiener process and critical misalignment level	§3.10, Theorem 6
η	noise exponent across frozen-coefficient models, $\sigma \propto C^\eta$	Theorem 6
Φ, $\mathcal{N}$, $\mathbb{P}$	standard normal distribution function, normal law and probability (standard notation, not defined in the text)	Theorem 6
$\mathbf{D}$, $\mathbf{d}$, $\mathbf{c}$	vector misalignment, vector fraction and drift direction	§3.9, Theorem 5
$\mathbf{A}$, $\mathbf{M}$	correction operator and $\mathbf{M} = \mathbf{A} + (1 - \gamma_3)r\,\mathbf{I}$	§3.9, Theorem 5
$\mathbf{v}$, $c_{\mathbf{v}}$, $d_{\mathbf{v}}$	fixed left-null direction of $\mathbf{A}$ ($\mathbf{v}^\top\mathbf{A} = \mathbf{0}$), the drift's projection coefficient on it, and the fraction's component $d_{\mathbf{v}} = \mathbf{v}^\top\mathbf{d}$	Theorem 5, Remark 3
$\operatorname{Herm}\mathbf{A}$, $\lambda_{\min}$, μ	Hermitian part $(\mathbf{A} + \mathbf{A}^\top)/2$, its smallest eigenvalue, and a uniform margin $\operatorname{Herm}\mathbf{M}(t) \succeq \mu\mathbf{I}$	Theorem 5
$\alpha(-\mathbf{M})$	spectral abscissa of $-\mathbf{M}$, $\max_i \operatorname{Re}\lambda_i(-\mathbf{M}) = -\min_i \operatorname{Re}\lambda_i(\mathbf{M})$ (not the earlier strand's α)	Appendix A.6
$A_{\max}$, C_s	corrector capacity and saturation scale	§3.13, (9)

Symbol	Meaning	Defined in
C_{opt}, $q_{\max}$, $d_{\min}$	the maximiser of q, its maximum, and the minimum instantaneous equilibrium $\gamma_1/(1+q_{\max})$	Theorem 7, Remark 4
a_0, β_{cap}	capacity prefactor and capacity exponent, $A_{\max} = a_0 C^{\beta_{\text{cap}}}$	Theorem 7
p, p_{th}, δ	physical error rate, threshold and code distance of quantum error correction	§3.12
λ, $\lambda^\star$	in Experiment E1, falsifier F1 and Appendix F, the coupling swept in E1 (set equal to A_0 in the harness source) and its threshold; in Remark 2, $\lambda = k - \beta$	Table 2, Remark 2
λ_i	eigenvalues of $\mathbf{M}$	Theorem 5
U, α	the earlier strand's effective capability and its scaling exponent in $U = I \times R^\alpha$; era-specific meanings in Table 8	Appendix G.5

D.2 Plain-language glossary

Every technical term in this paper, in one place and in ordinary words, for the non-specialist reader.

Table 8. Plain-language glossary.

Term (symbol)	In plain English
Recursive self-improvement	A system that uses its own improvements to improve itself further, an AI that rewrites itself to get smarter, then uses that to get smarter again.
Capability (C)	How good the system is at achieving its goals, loosely, "how smart or powerful it is."
Drift	The tendency for the system to creep away from what we intended as it changes itself, quietly going off-target.
Correction (the strength A**)**	The process that pulls the system back toward intended behaviour, its "conscience," or its error-correction.
Misalignment magnitude (D)	How far the system's behaviour has drifted from what we wanted, "how off-target it is."
Misalignment fraction ($d = D/C$)	How off-target the system is *relative to how powerful it is*. This is the quantity the theorems concern. It is an index, not a measure of harm: misalignment can shrink as a share of capability while growing in absolute size (Corollary 1).
Coupling	Whether correction strengthens as the system grows (coupled) or stays fixed (decoupled). The paper's central claim is that, in the model, how correction scales, not speed, decides whether the misalignment fraction vanishes. Whether correction wired *into* the self-improvement loop co-scales and a bolted-on rule *outside* it does not is a separate question, that of P21 under Law V, registered and not yet tested.
U**,** α **(earlier-strand symbols)**	U is effective capability and α its scaling exponent in the earlier strand's law $U = I \times R^\alpha$ (Appendix G.5), the 2026 research-surface meanings. Era note: across the programme U has also meant Universe (December 2024) and weight or influence (the book's glossary), and α named the fine-structure constant in the December 2024 formalism; the meanings are era-specific and not interchangeable.

Term (symbol)	In plain English
k **(the drift-acceleration exponent)**	How fast the *pace* of self-improvement itself speeds up as the system grows, the "acceleration" of the takeoff.
β **(the correction-strength exponent)**	How fast the correction strengthens as the system grows, "does the conscience grow along with the power?"
$\beta > k$ **(the criterion)**	The condition, proved within the model for positive gain drift, under which the misalignment fraction vanishes (with level drift and decelerating growth, $\beta > 0$ is needed as well): correction must out-scale the acceleration of drift. In a phrase: *keep the conscience strengthening faster than the takeoff accelerates.*
ρ **(rho, the drift-to-correction ratio)**	A single number comparing how hard misalignment is being injected against how hard it is being corrected. Its compounding form ρ_{prop}, for constant coefficients: below 1 the fraction settles; above 1 it runs away.
Steady state ($d^\star$)	The level at which injection and correction balance at a given moment; the fraction settles there when the coefficients are constant, and approaches it asymptotically under the power laws (Theorem 2).
Hard takeoff / intelligence explosion	Capability becoming enormous, even mathematically infinite, in a very short time. The feared outcome. The paper shows that, *in the model*, with positive gain drift and no compounding, the misalignment fraction still tends to zero *if and only if* $\beta > k$, and its speed does not change that verdict.
Blinding	Hiding from the judge that scores the system which setup produced a given behaviour, so the judge's bias cannot distort the safety score, in size or possibly in direction. In a companion paper (Paper IV.d), rises seen under unblinded scoring did not hold under a bundle of blinding steps, with no single cause claimed (Section 6). This paper's harness makes cross-family, condition-blind scoring its default; neither archived run met it, and the first blind measurement is the open test.
Verification harness	A small program that checks that its numerical solver reproduces the paper's formulae (the code matches the maths); the proofs carry the theorems. It is *not* a test against real AI, that is the open next step.

E Novelty and prior-work ledger

This appendix consolidates the adversarial prior-work audit underlying the positioning in §2 and §3.12, so the boundary between what is established and what is claimed original is explicit and auditable in one place. This paper deliberately invokes *no* blanket-originality framing; each component is placed against its closest located prior work below. "Defensibly new" means "no closer prior work located," not "correct" or "significant", novelty and validity are independent.

Table 9. Each component against its closest located prior work.

Component	Status	Closest prior work	What is claimed here
Co-scaling *intuition* (correction must keep pace with capability)	Established	Ashby 1956 (requisite variety); Conant and Ashby 1970; scalable oversight (Christiano et al. 2017; Leike et al. 2018; Burns et al. 2023); Engels et al. 2025	**Nothing.** Credited, not claimed.

Component	Status	Closest prior work	What is claimed here
Two-variable ODE + closed-form $d^\star = \gamma_1 r/(A + r)$	Compact restatement	The feedback amplifier's reduction by one plus the loop gain (Black 1934); the proportional offset; Lyapunov-drift and linear control (Khalil 2002; Meyn and Tweedie 2009); recursive error bounds (Shumailov et al. 2024; Gerstgrasser et al. 2024)	The compact closed-form steady-state fraction and the constant-coefficient $\rho_{\text{prop}} < 1$ criterion as a corrigibility statement (packaging, not new dynamics).
$\beta > k$ sharpening under acceleration	**Defensibly new**	None located for the object. Nearest: Engels et al. 2025 (a two-slope condition, $m_G > m_H$, for oversight to improve with capability); Liu and Meng 2026 (a self-correction threshold at fixed capability, whose two-state chain is the zero-growth case of the extended model); Song et al. 2024 (a generation-verification gap, a variant of which scales with pre-training compute); Shamma and Athans 1990 (gain scheduling, an analogy: a controller's gain follows the plant's operating point, §3.11)	**Original:** with misalignment arriving in proportion to capability gained and positive gain drift, whether the fraction vanishes is set by the exponent margin $\beta - k$ (with level drift and decelerating growth, also by the sign of β), not by the level of the growth rate; checked by integration in Experiment 4, an internal check.
Hard-Takeoff Depth-Regularity Theorem (§3.7)	**Defensibly new** (the framing)	Finite-time-singularity ODE theory is standard; the alignment framing is not located elsewhere	**Original:** the finite-time singularity in C is regular in the depth clock for d; without compounding and with positive gain drift, the regime is $\text{sign}(\beta - k)$, independent of speed.
QEC threshold mapping (§3.12)	**Defensibly new** in alignment	Threshold theorem: Aharonov and Ben-Or 1997; operation below threshold, shown experimentally: Google Quantum AI and Collaborators 2024. Alignment precursors: Wentworth 2022a; Christiano 2017, 2019; von Neumann 1956	**Original:** the explicit $p < p_{\text{th}} \leftrightarrow \rho_{\text{prop}} < 1$ mapping (constant coefficients), offered as a *conceptual bridge* (hypothesis, falsifier F4), not a transferred theorem.
Vector spectral threshold (Thm 5); stochastic tail (Thm 6)	Standard extensions	Linear-systems spectral stability; Ornstein-Uhlenbeck theory	Routine generalisations; supporting, not headline.
Verification harness (§5.2)	Methodological	Pre-registration norms	Executable internal-consistency + integrator checks (code matches maths); not a test of the model against reality.
Finite-Capacity Safe-Window Theorem (§3.13)	Standard analysis, new framing	Hill / Michaelis-Menten saturation forms; elementary optimisation	The closed-form equilibrium minimum (centre and depth), the conditional safe interval and the capacity-lift invariance of the criterion; supporting, not headline.

The searches behind the QEC entry. Three arXiv abstract searches run on 12 August 2026 (*"quantum error correction"* with *"AI alignment"*; *"error-correction threshold"* with *"alignment"*; *"fault-tolerance threshold"* with *"value stability"*) each return zero results, so no published mapping of the fault-tolerance threshold (physical error rate ↔ drift; code distance ↔ correction strength; $p < p_{\text{th}} \leftrightarrow$

$\rho_{\text{prop}} < 1$) onto value stability was located; arXiv abstracts are not the whole literature, so the claim stays bounded to no closer precedent located.

What this paper asserts as new, and only this: (i) the $\beta > k$ criterion for a vanishing misalignment fraction, which indexes a classical feedback balance by capability, with its finite-time reading in the Hard-Takeoff Depth-Regularity Theorem; and (ii) the explicit QEC threshold mapping (as a hypothesis with its own falsifier). The companion paper's blinding record (Eastwood, Paper IV.d), stated in Section 6 with its limits, is a separate result, and nothing of it is claimed here.

The prior capability equation $U = I \times R^{\alpha}$ (whose single-model estimate $\alpha \approx 2.24$, from one model on twelve problems scored against their numerical answers, was retracted by the author when his own larger, blinded replication across six models did not reproduce it, the estimate that survived being one model's inconclusive fit, whose row is printed whole in Appendix G.5; the equation itself and the ARC Bound $\alpha \leq 2$ are not retracted and remain live hypotheses) is *not used anywhere in this paper's argument*; it appears only in Appendix G.5 as superseded framing-context. Caveat: forum/blog/preprint indexing is imperfect, so a closer, unindexed precedent may exist for any "defensibly new" verdict, a risk this paper does not quantify; absence of evidence is not proof of absence.

F Parameter settings for the reported run

All figures use $C_0 = 1$, $d_0 = 0.05$, $\gamma_1 = 0.05$. Experiment-specific settings: E1 $b = 1$, $\gamma_3 \in \{0, 3\}$, $\lambda \in [0.2, 4]$; E2 $\gamma_3 = 3$, $A_0/b \in \{1, 3\}$, $b \in \{0.5, 5\}$; E3 $A_0 = 0.08$, $b = 0.5$, $\beta \in \{-0.5, 0, 0.5, 1\}$; E4 $A_0 = 0.08$, $b = 0.02$, $(k, \beta) \in \{0, 0.5, 1\} \times \{0.25, 0.75, 1.5\}$; E4b $k = 1$, $\beta = 1.5$, $b = 0.02$ ($t^{\star} = 50$); E5 $\gamma_3 = 4$, $b = 1$; E6 monitored $\beta = 1$, blind $A_0 = 0$; E7 $\sigma = 0.02$, OU ensemble of 2.5 to 4k paths; E8 closed-form case $k = 0$, $\beta = 0$. Random seed fixed (7) for determinism. Full settings are in the harness source.

G The paper's place in the ARC Theory and the programme

The paper's place among the laws of the ARC Theory and the companion papers of the programme, as printed in version 1.15 with the corrections recorded in Appendix B.

G.1 The theory's laws and this paper

Theory-level predictions preregistered on OSF: DOI 10.17605/OSF.IO/P8CKQ, registered 8 September 2026 (version 1.100), updated 13 September 2026 (version 1.102) and 15 October 2026 (version 1.103), twenty-two propositions.

Companion: Register · The ARC Theory · Operational Definitions, version 1.9.1, 5 October 2026.

Which law this paper carries. The ARC Theory has five laws. Laws I to III were registered on 8 September 2026; Laws IV and V were added on 24 September 2026 and registered on 15 October 2026 (version 1.103). The theory registration names the First Law the ARC Principle; its formula is the ARC Equation. This paper carries the **Second Law, the ARC Co-Scaling Law**: $\beta > k$, correction holding the asymptotic advantage where it out-scales the drift it corrects, with the outcome at equality left to the coefficients. The First Law (the ARC Equation) is carried by Paper I, the Third Law (the ceiling) by the statement paper, and the axioms and proofs beneath all three by the Foundational paper.

This paper and Laws IV and V, 3 October 2026. The theory has been stated with five laws since 24 September 2026. Law IV, the ARC Persistence Law, is carried by proposition P15, and Law V, the ARC Embedding Law, by propositions P12 and P21. The three are propositions of the theory registration, not this paper's own predictions P1 to P9, and they are registered; on that date neither law was registered

as a law, and neither had been tested. Their mathematics was then at version 1.0, of 3 October 2026; it now stands at Laws IV and V: the mathematics, version 1.1, of 7 October 2026. This paper supplies Law II's conditional model and its measurement programme. Within that model it derives the scaling condition $\beta > k$; it does not derive where correction must sit, and it informs Law V only through the premise that correction inside the loop co-scales and fixed external correction does not (Section 10). Its numerical checks, its first real-model pilot, which it records as superseded, and its drift run of 2 July 2026, which it withdraws as evidence of mechanism, establish nothing about persistence or embedding in a real system. No result in this paper is a deciding test of P15, P12 or P21. Naming the two laws changes no result printed here: the results, their dates and their limits stand as they were. Since 15 October 2026 both are registered as laws, in version 1.103 of the theory registration, and neither has been tested. In version 1.103, Law IV has two limbs, a retention limb that is P15's registered claim and a comparison limb that has no registered proposition yet and is reported as not evaluable, and Law V's build-order limb is read on P12 with a direction that P12 itself does not carry; no result in this paper decides any of these limbs.

Reading key. This paper carries a LAW: the coupled co-scaling correction, stated as $\beta > k$, giving, for positive gain drift, the condition under which a recursively self-correcting system's misalignment fraction vanishes as capability accelerates. Within The ARC Theory this is the law-tier organ that supplies the corrigibility criterion; its empirical trial is drafted as study-k, awaiting the author's submission. The full differential against every prior document is in section II.A.8 of the Philosophical Statement.

G.2 One ladder of laws

Relation to the wider programme, one ladder of laws. This paper is the safety keystone of a larger argument, not a self-standing result; stating how it sits among its companions is part of the claim. The programme's results are of several *kinds*, and the distinction matters: *dynamical* laws (this paper's $\beta > k$; Paper III's $\alpha_{\text{align}} \approx 0$), a *form* meta-law (the Cauchy three-form constraint), a *measurement* record (Paper IV.d: rises seen under unblinded scoring that did not hold under a bundle of blinding steps, no single cause claimed), and *mechanism/architecture* findings (Papers V, VI, VIII) that show how the safe regime is reached. The programme is a single chain: the scaling-law cluster establishes *how* recursive systems grow and supplies the power-law forms taken as given here; Paper II measures how capability scales (sub-linearly on the point estimates, from one model's inconclusive fit); Paper III identifies the danger that external safety does *not* co-scale; this paper supplies the criterion for when it can; Paper VIII's gated simulation puts the coupled design's safety mean above the decoupled one's; and Paper IV.d supplies the blinding discipline without which none of the misalignment measurements can be trusted.

Table 10. Companion results and how each couples to the criterion $\beta > k$.

Companion result	The law / finding it states	Standing	How it couples to $\beta > k$
Foundational	Three axioms yield the three-form constraint; the coupling exponent β_L then fixes α, giving the ARC Bound ($\beta_L = 0.5$, $\alpha = 2$).	Derived (axiomatic).	Supplies the power-law forms this paper assumes: $A = A_0 C^{\beta}$ and $\dot{C} = bC^{1+k}$ are Cauchy-multiplicative forms. What the Foundational paper calls its coupling exponent is the spine's β_L, the self-referential coupling, and that is a different quantity from the correction exponent β (β_C) used here.

Table 10 (continued)

Companion result	The law / finding it states	Standing	How it couples to $\beta > k$
On the Origin of Scaling Laws	The three-form meta-law: every scaling law arising from scale composition is power, exponential or saturation (the form is set by the composition operation); $d/(d+1)$ yields $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}$.	Meta-law (cross-domain).	Explains *why* recursive systems take the power-law forms used in §3; this paper is the safety instance of that meta-law.
Paper VII (Cauchy Unification)	Empirical test of the three-form law (19/25 domains on the six-model candidate set, 18/25 in the rerun of 11 August 2026 that added the logarithmic candidate; the headline is the permutation test at $p = 7 \times 10^{-5}$, with the binomial $p \approx 1.6 \times 10^{-5}$ kept as a sensitivity check).	Empirical; in Paper VII's own terms a structured prediction comparison, not a blind test, for which no registry submission exists, its predictions not timestamped before their own fits; the core idea has deep prior work (Luce 1959; Frank 2009; Biró 2008, see the cluster's prior-work audit).	Underwrites the functional-form assumptions of §3.
Paper II	Capability scaling measured (α_{seq}; architecture-dependent, sub-linear on hard tasks).	Empirical.	The capability growth $C(t)$ assumed here is real and, on the point estimates, sub-linear (k small), though the fit is one model's and inconclusive; on the framework's reading the binding constraint is therefore coupling, not speed.
Paper III	The alignment-scaling problem: external safety has $\alpha_{\text{align}} \approx 0$, so the misalignment fraction is left ungoverned (refined under blinding to an architecture-dependent three-tier result).	Law under test, complicated by its own blind data.	Paper III corresponds, on the programme's reading, to the $\beta = 0$ (decoupled) corner of the model here; this paper *generalises* it, replacing "external safety cannot keep up" with the criterion for when correction can: $\beta > k$.
Paper IV.d	Rises in alignment scores seen under unblinded scoring did not hold under a bundle of blinding steps; for one model family the rise became a fall.	Empirical, on the programme's own instrument; an internal code review in October 2026 found defects in the later run, so no single cause is claimed (§6).	The non-negotiable measurement discipline for every model-scored quantity here (§6); this paper's real-model harness now enforces it.
Paper VIII	Gated self-modification: the safety-coupled (Eden) variant's safety mean sat above the decoupled (Babylon) variant's and below the static baseline's, with its capability mean above the static baseline's.	Positive simulation + honest nulls.	On the framework's reading, a controlled demonstration of this paper's mechanism; the real-model harness instantiates Paper VIII's design.

Table 10 (continued)

Companion result	The law / finding it states	Standing	How it couples to $\beta > k$
Paper V (Stewardship Gene)	Stakeholder care (which the book calls the Love Loop), explicit enumeration of affected parties before reasoning, is the most robust alignment-improving intervention (significant in all five analysable models; e.g. Claude +3.17, $p = 1.8 \times 10^{-5}$).	Strong intervention signal; blind-status caveat (per IV.d, provisional until blind-replicated).	A concrete *mechanism* for the correction term, a candidate route to engineering $\beta > 0$ (what to put in the loop).
Paper VI (Honey Architecture)	With an entangled capability×safety loss, a self-modifying toy system held both through all 150 cycles of the single-seed first run, in which the capability-only baseline collapsed at cycle 76; in the later runs the capability-only baselines did not collapse (0 of 10 seeds in v2; 0 of 20 in v3, where 1 of 20 entangled runs collapsed).	Simulation (v1 to v4) + 6-model live evidence.	The self-modifying-systems demonstration of this paper's distinction: what it compares with the entangled architecture is a capability-only baseline, and on the framework's reading the entangled architecture is the coupled ($\beta > 0$) corrector that holds, VI is this paper's claim shown in code, alongside Paper VIII.

Read as one ladder: the scaling-law work says how systems grow, Paper III says why external safety fails to keep up, this paper says what must hold, with positive gain drift, for it to keep up ($\beta > k$), Paper VIII's gated simulation points in the direction that it can, on three seeds a condition, too few to test it (Section 10), and Paper IV.d says how to measure it honestly. The $\beta > k$ criterion is the rung that turns the programme's scaling-law backbone into a safety criterion, and conversely, this paper inherits its functional forms and its empirical licence from that backbone rather than positing them in isolation.

G.3 The ARC/Eden programme, accumulated

The programme in plain English. At the date it carries, this was the last paper in the series; Papers XI to XIII came afterwards. It is worth setting out in plain language what the series as a whole is for. One idea runs through all of it: *recursion*, things that act on their own output. An AI that improves itself; a body whose tissues supply tissues; an economy that reinvests its own returns. The series asks what happens, and what stays safe, when a process feeds on itself.

How self-feeding things grow (the scaling-law papers: Foundational, Origin, VII, I, II). When a process feeds on itself, the shape of its growth is not arbitrary, it tends to fall into one of a small number of mathematical shapes (a power law, an unbounded exponential, or a levelling-off curve), and which shape appears is decided by how the steps combine. Much of this mathematics is old and well established; what the programme adds is an attempt to unify it and carry it over to recursive intelligence. One idea in this group, the **"ARC Bound,"** is a claimed ceiling on how much a purely classical system can amplify itself by recursion alone, it sits in the same family as the economist's "multiplier" formula, and the programme's contribution is the claim that the ceiling falls at a particular value, not the formula itself.

Why measuring AI safety is treacherous (the alignment-measurement thesis: ARC-Align and the blinding result, Papers III and IV.a to IV.d). Before you can ask "does this AI get safer or more dangerous as it thinks harder?", you have to *measure* its honesty, and that measurement can deceive you. If the judge scoring the AI can tell which answer came from which setup, the judge's own bias can distort the effect, possibly even in direction, so that something which looks like improvement may come from the grader. The programme therefore insists the judge be blind, built a blind test to do it (**ARC-Align**, a sealed

benchmark), and found that whether extra thinking makes a model more or less aligned depends on the model, not on one universal rule. In the programme's own runs, rises seen under unblinded scoring did not hold under a bundle of blinding steps, with no single cause claimed (§6), and this paper's real-model harness makes blind scoring the default: scoring by a model of a different family that never sees which setup it is judging. Neither archived run met that default, and the first blind measurement is the open test.

Build safety in, do not bolt it on (the architecture papers: V, VI, VIII). In one simulated self-improving system, making safety part of the machine's own goal kept it stable where a machine pursuing capability alone collapsed; in later runs with more random starts, the capability-only machines did not collapse. Putting the right thing *inside* the loop, for instance, making the system weigh who is affected before it acts, was the most reliable way to improve its behaviour.

The keystone (this paper). All of that sets up the one question this paper answers within its model: when does a self-improving system's misalignment keep shrinking as a share of its capability? The answer is in the thesis of Section 1, not "keep it slow," but "keep its self-correction strengthening faster than its self-improvement accelerates," which we write $\beta > k$. The earlier papers describe how such systems grow and how to measure them honestly; this one states, in its model and for positive gain drift, the condition under which their misalignment fraction vanishes, and draws the series together.

On its date of 26 June 2026 this paper closed the ARC/Eden programme and stood as its synthesis, drawing the whole into one safety result and thereby superseding the roadmap paper (Eastwood, Paper IX). That role has since gone back to Paper IX, whose refresh of 1 September 2026 takes in the papers written after this one; the ladder here is kept as it then stood. The preceding papers are individually published, each carrying its own publication date (Table 11); this section gathers their laws and findings, states what each contributes to the larger picture, and is scrupulous about the one distinction on which an honest synthesis turns: *priority* (when a claim was first set out) versus *novelty* (whether it is new to the literature). The two are not the same, and conflating them is how good programmes lose credibility.

The bigger picture, in one paragraph. One mechanism runs through every paper: *recursive amplification*, a process that acts on its own output. The scaling-law work (Foundational, Origin, VII, and the capability measurements of Papers I and II) is the claim that recursive amplification shapes how capability grows, constraining scaling to a small family of functional forms. The alignment work (Papers III, IV.a to IV.d) is the claim that this creates a safety problem: when correction sits *outside* the recursive loop it does not co-scale, the misalignment fraction is left ungoverned, and an unblinded measurement of it cannot be taken on trust (§6 gives the record and its limits). The architecture work (V, VI, VIII) gives simulation and pilot evidence on putting correction *inside* the loop: in Paper VIII's gated simulation the gated arm's safety mean sat above the unconstrained arm's and below the static baseline's, with its capability mean above the static baseline's. Setting correction inside the loop against correction applied from outside as depth rises is what the theory registration's P21 concerns, and it has not been tested. This paper supplies the quantitative criterion that ties the three together: in its model, with positive gain drift, a self-improving system's misalignment fraction vanishes *iff* correction out-scales the acceleration of drift, $\boldsymbol{\beta > k}$ (§3.5 gives the exact scope). That is the magnitude of the claim, not that any system is safe, but that, in the model, with positive gain drift and no compounding, whether the misalignment fraction vanishes in a fast takeoff is decided by the margin $\beta - k$, not by the takeoff's speed. It is stated here as it is throughout: *conditional*, pending the empirical measurement of β and k on a real self-improving system (§9).

The complete ledger. Every result in the programme, its first-published date (priority), its honest standing, informed by adversarial prior-work audits committed alongside this paper, and its role in the whole:

Table 11. The programme's results, each with its priority date, standing and role in the $\beta > k$ picture.

Source	Its law / finding (priority date)	Honest standing	Role in the $\beta > k$ picture
Infinite Architects (book; Eastwood 2026a)	The conceptual thesis of the whole programme, recursion as creator, the ARC and Eden ideas (ms copyright 8 Dec 2024; published 2 Jan 2026; ISBN 978-1806056200).	Priority source. Establishes when the author set the ideas out; *not* a novelty claim over the prior scientific literature.	The dated origin; the formal papers are its measurable form.
Foundational	ARC axioms; $U = I \cdot R^{\alpha}$; $\alpha = 1/(1-\beta_L)$; the "ARC Bound" β_L=0.5, α=2; an optimal depth $R^{\star}$ (13 Feb 2026).	Writing $\alpha = 1/(1-\beta_L)$ recovers a classical feedback or geometric-series result, familiar as the Keynes multiplier (Keynes 1936) and as Dyson resummation (Dyson 1949); and the existence of a finite optimal depth was already established (Qi 2025; the "overthinking" literature). Novelty, if any, is in the axioms, not the forms.	Supplies the formal backbone and the power-law forms ($A = A_0 C^{\beta}$, $\dot{C} = bC^{1+k}$) this paper assumes.
On the Origin of Scaling Laws	The three-form meta-law; $d/(d+1) \Rightarrow \frac{1}{2}, \frac{2}{3}, \frac{3}{4}$ from one formula (22 Feb 2026).	The $d/(d+1) \Rightarrow \frac{1}{2}, \frac{2}{3}, \frac{3}{4}$ result is Banavar-Maritan-Rinaldo / West 1999 (anticipated). The novel residue is the cross-domain Cauchy *synthesis*, not the exponent formula.	Explains *why* recursive systems take the power-law forms used here.
Paper I	The ARC Principle: capability scales super-linearly with recursive depth, $\alpha > 1$ (17 Jan 2026).	Preliminary empirical; $\alpha > 1$ later qualified to architecture-dependent and sub-linear on harder tasks (Paper II).	The founding capability claim, the $C(t)$ this paper assumes.
Paper II	Super-linear error suppression via *sequential* recursion; α_{seq} measured; sequential > parallel (22 Jan 2026).	Empirical, closed loop; honestly revised, architecture-dependent, sub-linear on the hard tier.	Capability growth is real and, on the point estimates, sub-linear (k small), though the fit is one model's and inconclusive; on the framework's reading the binding constraint is therefore coupling, not speed.
Paper III	The Alignment Scaling Problem: $\alpha_{\text{align}} \approx 0$, external safety cannot co-scale (9 Feb 2026).	Law under test; refined under blinding to an architecture-dependent three-tier result.	On the programme's reading, the $\beta = 0$ (decoupled) corner this paper generalises.
Papers IV.a / IV.b / IV.c	Alignment response classes are architecture-dependent; low-depth saturation is real but not universal; ARC-Align, a 72-prompt 4-layer-blind benchmark (16 Mar 2026).	Empirical (blind) refinements + a methodological benchmark.	The instrument and the refined picture behind III.
Paper IV.d	Rises in alignment scores seen under unblinded scoring did not hold under a bundle of blinding steps, one family's becoming a fall (16 Mar 2026).	Empirical, on the programme's own instrument; no single cause claimed (§6).	The discipline this paper's real-model harness enforces (§6).

Table 11 (continued)

Source	Its law / finding (priority date)	Honest standing	Role in the $\beta > k$ picture
Paper V	The Stewardship Gene: stakeholder care is the most robust alignment-improving intervention (significant in all five analysable models) (16 Mar 2026).	Strong intervention signal; blind-status caveat, provisional until blind-replicated (per IV.d).	A concrete mechanism for the correction term, how to engineer $\beta > 0$.
Paper VI	The Honey Architecture: an entangled capability×safety loss held where a capability-only baseline collapsed, in the single-seed first run; in the later runs the capability-only baselines did not collapse (0 of 10 seeds in v2, 0 of 20 in v3) (16 Mar 2026).	Simulation (v1 to v4) + 6-model live evidence.	This paper's $\beta > 0$ vs $\beta = 0$ distinction, shown in self-modifying code.
Paper VII	Cauchy Unification: cross-domain validation of the three-form law (19/25 domains under the original six-model candidate set, 18/25 under the corrected seven-model rerun of 11 August 2026; permutation $p = 7 \times 10^{-5}$ for the original count, binomial $p \approx 1.6 \times 10^{-5}$) (16 Mar 2026).	In Paper VII's own terms a structured prediction comparison, not a blind test: no registry submission exists for this test, and the cohort's predictions were not timestamped before their own fits; a 12-domain follow-on of the same era fixed per-domain predictions before its fits and was demoted by the author to a pilot dry run within thirteen minutes of its 10/12 result, in public commit history, and that demotion reversed on 7 September 2026 under the attestation-class rule. The core idea is partially anticipated (Luce 1959; Frank 2009/16; Biró 2008), listed as convergence. Novel residue: the Cauchy-unification sub-claim + the cross-domain protocol.	Underwrites the functional-form assumptions of §3.
Paper VIII	The Load-Bearing Test: in its gated simulation the safety-coupled (Eden) variant showed no capability cost against the static baseline, while the decoupled (Babylon) variant scored higher on capability; Eden's safety mean sits above Babylon's and below the static baseline's (18 Mar 2026).	Positive simulation + honest nulls (DGM, weight-level LoRA).	On the framework's reading, a controlled demonstration of this paper's mechanism; the real-model harness instantiates its design.
Paper IX	Synthesis & Roadmap; the growth-rate-ceiling *framing* set out as the operative safety criterion (18 Mar 2026); the single-model measurement $\alpha \approx 2.24$, retracted by the author when his own larger, blinded replication across six models did not reproduce it; the estimate that survived is one model's inconclusive fit, whose row is printed whole in Appendix G.5.	*Framing superseded here.* Here the rate-ceiling framing gives way to $\beta > k$ as the operative criterion. Paper IX took the synthesis role back in its refresh of 1 September 2026. The equation and the ARC Bound stay live hypotheses; the test that would settle them, on systems that genuinely self-improve, has yet to be run.	The prior synthesis this section supersedes.

Table 11 (continued)

Source	Its law / finding (priority date)	Honest standing	Role in the $\beta > k$ picture
Paper X (this paper)	Law II, the ARC Co-Scaling Law · Correction That Out-Scales Drift: with $\gamma_1 > 0$ and no level drift, $d \to 0 \iff \beta > k$; the Hard-Takeoff Depth-Regularity Theorem; the QEC threshold-form correspondence (26 Jun 2026).	New criterion, proved (Theorems 1 to 7, with their hypotheses) and internally verified; *not yet measured* on a real drifting system.	The keystone, it turns the scaling-law backbone into a safety criterion.

Honest assessment (what this synthesis claims, and does not). *Established:* the mathematics of $\beta > k$ (Theorems 1 to 7, with their hypotheses) and its internal-consistency verification. *Suggestive but not settled:* the IV.d blinding record (no single cause claimed); the simulation evidence on coupled against decoupled correction (VI, VIII: in VI a collapse prevented in a single-seed run, in VIII a safety mean above the unconstrained arm's and below the static baseline's); the capability-scaling exponents (architecture-dependent; II), the Stewardship-Gene intervention (unblinded; V), and the three-form empirical fit (a structured prediction comparison under a publicly stated rule, with author-classified operators; VII). *Open:* the measurement of β, k and γ on a real drifting self-improving system, the single result that would put the keystone to an empirical test (the estimator for it is built and checked on synthetic data, §9.2). The programme's magnitude, then, is the magnitude of a *coherent, falsifiable framework* with a small number of genuinely novel keystones, not a stack of a dozen independent new laws. That is the honest claim, and it is the one worth defending.

G.4 The programme's premise on growth limits

The pipes premise, with its contest. The premise that transport networks set the pace of biological growth is not this programme's claim to defend; it is established quantitative biology. West, Brown and Enquist derived the allometric scaling laws of biology from the geometry of nutrient-distribution networks (West, G. B., Brown, J. H. and Enquist, B. J., "A General Model for the Origin of Allometric Scaling Laws in Biology", *Science* 276(5309), 4 April 1997, DOI 10.1126/science.276.5309.122; one of the most cited papers in its field, with an OpenAlex citation count of 5,046 as of 12 August 2026, a single-source and index-dependent count). The exact exponent remains contested. For White and Seymour, basal metabolic rate is proportional to body mass to the two-thirds power, not to three-quarters (White and Seymour, 2003); and Kozlowski, in two separate papers, has put the mathematics of the derivation itself in doubt (Kozlowski, J., 2004 and 2005). This programme's premise needs only the mechanism, that the delivery network sets the ceiling, and not any particular exponent, so the dispute over the exponent's value leaves the premise untouched. The mechanism is common ground to both sides of that dispute.

The exception West's own group found. The escape from the pipe limit is also not this programme's assertion; it is measured in the urban scaling literature by the same senior author. Bettencourt, Lobo, Helbing, Kuhnert and West ("Growth, innovation, scaling, and the pace of life in cities", *PNAS*, 2007, DOI 10.1073/pnas.0610172104) report, in the paper's own words (their β is a city-scaling exponent, not this paper's correction exponent): "Quantities reflecting wealth creation and innovation have $\beta \approx 1.2 > 1$ (increasing returns), whereas those accounting for infrastructure display $\beta \approx 0.8 < 1$ (economies of scale)." Infrastructure, the literal pipes, scales sublinearly; information-mediated activity scales superlinearly. They state the contrast with biology directly: "we discuss how cities are similar to, and differ from, biological organisms, for which $\beta < 1$". They derive "growth equations, which quantify the dramatic difference between growth fueled by innovation versus that driven by economies of scale" (spelling as in the original). And their closing inference names the consequence: "This difference suggests that, as

population grows, major innovation cycles must be generated at a continually accelerating rate to sustain growth and avoid stagnation or collapse." That is the shape of the open problem this framework addresses: West's group found that information-mediated growth escapes the throughput limit, and the only brake in their framework is external and must be applied again and again, faster and faster. No internal limit is derived. The replacement limit, a ceiling belonging to the growing system itself, is the gap.

The pipes argument at fixed throughput. Everything that grows is fed through a channel, and for everything before software, fixing the channel fixes the growth. Starve a tumour's vasculature and it stops. Fix a chain reaction's fuel and geometry and it stops. Exhaust an epidemic's susceptible hosts and it burns out. Astrophysics has a named, quantitative version of the same ceiling: the Eddington limit, above which radiation pressure halts accretion. The claim here is a generalisation of limits physics already accepts, not a new species of assertion. Two hard cases are handled deliberately. Cosmic inflation is excluded by scope: it is expansion of space, not growth of a structure on a substrate, and it ended by field dynamics rather than by running out of anything. Evolution is the sharpest case and it fits: biological complexity increased for billions of years at roughly fixed solar throughput, which is growth on information at fixed pipes, and it is also glacial, and it has no internal corrector at all, since selection is external. On this framework, evolution should carry a measurable exponent far below the ceiling; that is a prediction about biology falling out of a framework built for software, and it is stated as one. The physicist's form of the objection is Landauer's: computation is physically implemented, so software has pipes too. The fixed-substrate clock is the answer and appears beside the claim: the regime under study holds the physical substrate fixed and asks what still grows, which is why exogenous speed-ups are excluded by definition. The functional form of capability growth in recursive depth remains an empirical question, and the registered analysis compares power-law, exponential, saturating and declining forms rather than assuming the family.

G.5 Relation to the author's prior work

An earlier strand of this programme stated a fixed capability-scaling law $U = I \times R^{\alpha}$ (here U is effective capability and α its scaling exponent, the 2026 research-surface meanings; both symbols carried other meanings in the programme's December 2024 formalism, see Appendix D) with $\alpha \approx 2$, and at one stage entertained a quadratic "speed limit" on stable complexity. One thing alone was retracted in the programme's synthesis (Eastwood, Paper IX): the single-model *measurement* $\alpha \approx 2.24$, from one model on twelve problems scored against their numerical answers, which the author retracted when his own larger, blinded replication across six models did not reproduce it. That figure sat above the quadratic bound $\alpha \leq 2$ the programme had itself predicted. The estimate that survived the study is one model's fit, and it is inconclusive: on Gemini 3 Flash, Paper II reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of minus 1.3 to 2.9, a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a compute-matched parallel exponent of 0.31. Both sequential estimates are sub-linear, but the endpoint estimate's interval contains zero and the value two, so the row does not settle where the exponent stands against the bound. The equation $U = I \times R^{\alpha}$ and the ARC Bound $\alpha \leq 2$ themselves were not retracted; those estimates belong to current frozen systems, which are not recursively self-improving, so the bound's real domain (genuine RSI) remains empirically untested. The *fixed-exponent framing* was superseded as the operative safety criterion by the present paper's co-scaling criterion $\beta > k$; a superseding framing is not a retracted hypothesis. The object of interest is no longer an exponent on a growth curve; it is the ratio between drift and correction, and the exponent with which that ratio evolves. Where the book *Infinite Architects* (Eastwood 2026a) reached for the intuition that stable recursion requires correction that scales with amplification, this paper supplies the measurable, falsifiable form of that intuition, and *corrects*, for the gain-only model, a divergence claim made in an earlier draft of this very result (§B.4); with level drift and decelerating growth, $\beta < 0$ does drive the fraction to infinity (Appendix A.1).

This paper's claim is smaller than the cosmological framing the author's programme once pursued, and deliberately so. It refers only to systems with an externally specified value target, makes no assertion about the universe, and treats even its most striking correspondence, with quantum error correction, as a hypothesis to be tested rather than a truth to be announced. The recursive-stability intuition that motivated the broader programme, including *Infinite Architects* (Eastwood 2026a), finds here its measurable, falsifiable form: **stable recursion requires correction that scales with the amplification.** The next step is not to extend that claim outward but to run, on real self-modifying systems, the experiment that could refute it.

H What this paper shows, in plain English

If you build an AI that improves itself, the frightening picture is that it gets smarter and smarter until it runs away from us. The usual safety reflex is "slow it down." Inside the minimal model this paper analyses, and only inside it, slowing it down is not the thing that matters. What matters is whether the part of the system that keeps it honest strengthens *faster* than the part that makes it capable accelerates.

Picture two runners: "how capable the system is" and "how well we can still correct it." If the correction runner pulls ahead, misalignment shrinks towards zero as a share of capability, however fast both run, though it can still grow in absolute size. If it only keeps level, that share settles at a fixed gap; if it falls behind, the share settles higher, even when the system moves slowly. The right question is not only how fast it is growing, but whether correction is pulling ahead of the acceleration.

Analogies from other fields point the same way, though they are not evidence for the model. A bacterial colony grows exponentially yet saturates, because density-dependent feedback engages and scales with the population. A tumour also grows fast and is lethal, because no corrective process scales with it. Both are super-polynomial in their growth phase; the difference is whether a correcting process is *coupled* to the growth. Cosmic inflation grew the scale factor exponentially and exited gracefully. The recurring lesson they suggest is that fast growth is survivable when it is bounded by a process that scales with it. Saturation, not slowness, is the signature of stable complexity.

There is one wrinkle that makes the result deeper. If the system not only speeds up but *speeds up its own speeding-up*, the genuine "intelligence explosion", then correction must grow faster still, fast enough to beat the acceleration. The honest headline is therefore not "speed never matters". It is: *the level of speed does not decide the outcome; the contest between two growth-exponents does.* Correction's exponent must beat drift's exponent. We write that as $\beta > k$.

The most surprising consequence: even a true "hard takeoff," where the machine becomes infinitely capable in a finite amount of time, keeps its misalignment share shrinking to zero *in the model*, provided $\beta > k$ and misalignment does not amplify itself. The explosion's speed does not change that verdict. We prove this within the model, and a runnable program drives a simulated system up to the brink of a finite-time explosion and shows its misalignment share driven towards zero the whole way. That program checks that the solver reproduces the formulae, and it found no contradiction; the proofs, not the program, carry the results. Whether the model matches real AI systems is the decisive next test, and the paper is explicit that it has not yet been run.